LARGE-WAVE AMPLIFICATION ON PERIODIC DISCRETE LATTICES: LINEAR THEORY AND A NONLINEAR EXTENSION

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Abstract. We study the large-wave effect for free oscillations of the periodic discrete chain (the ring of N beads) and for the discrete Schrödinger equation on the same ring. Writing $A_N = \sup_t \max_j |u_j(t)|$ for the unit-impulse Green's function, we show that the supremum is, by Bohr's theorem, the ℓ^1 mass of the Fourier coefficients whenever the frequencies $\{2\sin(\pi r/N)\}$ are linearly independent over $\mathbb{Q}$, and we prove this independence for N an odd prime and for $N = 2^m$ by a cyclotomic argument. Consequently $A_N = U_N := \frac{1}{2N}\sum_{r=1}^{N-1}\csc(\pi r/N) = \frac{1}{\pi}\ln N + \frac{1}{\pi}(\gamma + \ln\frac{2}{\pi}) + o(1)$ unconditionally on these classes, and the same sharp constant holds for the composite family $N = 2p$. For every N the upper bound $A_N \le U_N = \frac{1}{\pi}\ln N + O(1)$ is rigorous; the orbit-closure subtorus has dimension exactly $\frac{1}{2}\varphi(2N)$, and $A_N = U_N$ holds *if and only if* every integer relation among the frequencies has coefficient-sum divisible by four (with an antipodal variant for even N)—an arithmetic criterion met *exactly* by the primes and the powers of two (Theorem 14), and by no composite N. Moreover the *order* is now settled for *all* N: $A_N \sim \frac{1}{\pi}\ln N$ unconditionally (the lower bound from a palindromic argument showing the first $\frac{1}{2}\varphi(2N)$ frequencies are $\mathbb{Q}$-independent, so a $\ln N$ fraction of the ceiling can always be aligned). The composite deficit $U_N - A_N$ is $O(\ln\ln\ln N)$, and the order law extends to the Dirichlet segment and to every ring product $C_N \,\square\, H$. For the discrete Schrödinger propagator e^{-itL_N}, unitarity forbids amplification of a localized state, while the $\ell^\infty \to \ell^\infty$ amplification satisfies $B_N = \Theta(\sqrt{N})$; here $\liminf B_N/\sqrt{N} = c_0/\sqrt{2}$ but the constant has *no* limit—it splits by parity, with $\limsup \ge \beta_{\mathrm{odd}} = 0.928\ldots$ (a Bessel-envelope integral with no elementary closed form). For the focusing discrete NLS the linear large wave persists under weak nonlinearity, the modulational-instability band is read off the L_N spectrum, and the on-site breather exists and is stable. Proved results are separated throughout from computational evidence and conjectures; all quantitative claims are reproduced by the accompanying scripts, with a Gap/Pari/FriCAS/Rocq exact-arithmetic layer.

1. Introduction

A localized disturbance on a homogeneous one-dimensional medium does not amplify: for the continuous wave equation the classical bound of Zhukovsky gives $|\sigma(x,t)| \le F$ [33]. On the *discrete* chain this is false [26, 25, 31]. The discreteness of the spectrum, together with the finiteness of the number of modes, allows many modes to come into phase and to produce a local peak that grows with the number of nodes N. This is the *large-wave effect*; related phenomena include the splash effect [30] and its Schrödinger analogue [32].

The effect is genuinely discrete: it disappears in the continuum limit. The mechanism is the *finite, discrete spectrum*—finitely many discrete modes can be brought into near-coincidence of phase (arbitrarily closely when their frequencies are rationally *independent*, by Kronecker, and otherwise as far as the relation lattice allows); on the infinite chain the spectrum is continuous, no exact alignment occurs, and the amplification is absent. It is thus a finite-size spectral resonance rather than a feature of the limiting partial differential equation. (For the discrete Schrödinger propagator the absence of a continuum large wave is explicit: its single-site kernel is the Bessel function $J_n(2t)$, whose peak over sites decays like $t^{-1/3}$, Section 11.) This discrete–continuous gap, together with the appearance of discrete Laplacians

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in quantum walks, photonic lattices, non-Hermitian lattice optics [23], and discrete nonlinear models of anomalous (rogue) waves, motivates studying the lattice in its own right.

Contributions. (i) We identify $\sup_t G_N(0,t) = U_N$ through Bohr's theorem under rational independence of the frequencies (Section 3), and prove this independence for N prime and $N = 2^m$ by a cyclotomic argument (Theorems 3, 4), giving the unconditional asymptotics $A_N \sim \frac{1}{\pi} \ln N$ with the explicit constant of Lemma 2; the same sharp constant extends to the composite family $N = 2p$ (Proposition 15). (ii) For every N the upper bound $A_N \le U_N$ is rigorous, and we reduce A_N *exactly* to an optimization over the orbit-closure subtorus, whose dimension we compute in closed form as $\frac{1}{2}\varphi(2N)$ (Theorem 16); a sharp arithmetic criterion—coefficient-sums divisible by four (plus an antipodal variant for even N)—decides exactly when $A_N = U_N$ (Theorem 11), and an explicit root-of-unity obstruction settles the saturation locus completely: $A_N = U_N$ *iff* N is prime or a power of two (Theorem 14). Separately, the *order* is settled for *all* N: $A_N \sim \frac{1}{\pi} \ln N$ (Theorem 6), the lower bound coming from a long alignable prefix ($M(N) = \frac{1}{2}\varphi(2N)$ frequencies are $\mathbb{Q}$-independent, Lemma 5). (iii) For composite N the value is algebraic with increasing arithmetic complexity (Proposition 10); we further separate the infinite-time ceiling from the finite-time amplitude and show both the finite-horizon recurrence time and the effect of mass disorder (Section 10). (iv) For the discrete Schrödinger equation a localized state is never amplified (unitarity, rigorous), and the $\ell^\infty \to \ell^\infty$ norm satisfies $B_N = \Theta(\sqrt{N})$, both bounds rigorous (Theorems 17, 19). (v) As a nonlinear outlook we connect L_N to the rogue-wave problem (Section 18): the discrete NLS $\mathrm{i}\dot{z} = -L_N z + \gamma|z|^2 z$ has L_N as its linear part and the focusing NLS as its continuum limit, and we verify the Peregrine rogue solution, the modulational-instability gain, and the emergence of heavy-tailed amplitude statistics in sharp contrast with the platykurtic linear value law. Every quantitative claim is reproduced by a standalone, version-controlled script; the arithmetic of Theorems 16 and 11 is additionally checked in exact cyclotomic arithmetic by two computer-algebra systems and machine-checked by the Rocq proof assistant (Section 14).

Relation to prior work. The large-wave *phenomenon*—that on the discrete bead string a localized disturbance amplifies, growing without bound at bounded energy on suitable N—is classical, due to the Kurchanov–Myshkis–Filimonov line of work [26, 25, 31], with the ln-type asymptotics on the Dirichlet segment first in [29] and sharpened in [25], and the Schrödinger analogue in [32]. The linear independence over $\mathbb{Q}$ of the frequencies for N prime or $N = 2^m$ is itself due to [26]; our Theorems 3, 4 reprove it cyclotomically, and Theorem 16 subsumes it in a closed-form $\mathbb{Q}$-rank valid for *every* N. The lineage also includes two early C. R. Acad. Sci. notes [27, 28], of which Filimonov's 1992 note [28] is the closest precedent: it is itself on the *circular* string, and by its published summary (Zbl 0761.73053) it already establishes that for N prime or $N = 2^m$ the *ring* vibrations exhibit peaks, addresses the degenerate "multiple-frequency" (composite) case that our $\bmod 4$ criterion governs, and proposes a continuum model finer than the wave equation. A recent monograph [22] reproduces its splash table (its Table 10.1, with $P_N \to \infty$ as $N \to \infty$), which grows at the *segment* constant $4/\pi^2$ and quantifies the splash *magnitude*—distinct from our *ring* law $\frac{1}{\pi} \ln N$ and from the arithmetic of *which* N saturate. *It is therefore possible that the qualitative ring effect and part of the composite analysis below are already Filimonov's.* We have consulted only its abstract and this secondary reproduction (the full note is not digitized in the sources available to us); [28] must be obtained and compared against Theorems 3–4 and 11 before submission. The structural and quantitative claims below are accordingly stated as novel *to our knowledge*. Our contribution is *quantitative and structural*: the explicit ring constant $\frac{1}{\pi}$ and the next term $-\frac{\pi}{72}N^{-2}$ (Lemma 2); the closed-form subtorus dimension $\frac{1}{2}\varphi(2N)$ for every N, with a cyclotomic reproof of the prime/2^m independence of [26] as its full-rank case (Theorems 16, 3, 4); the exact $\bmod 4$ criterion for reaching the ceiling (Theorem 11); the new composite family $N = 2p$ (Proposition 15); the spectral-zeta systematization; the finite-time/disorder dichotomy (Section 10); and the sharp $B_N = \Theta(\sqrt{N})$ growth of the discrete Schrödinger propagator (Theorem 19). We claim neither the large-wave phenomenon nor its appearance on the ring for N prime or 2^m as new (the latter is already in [28]; see Remark 12). The same transient overshoot—coupling forces exceeding the applied load by a spatial redistribution of energy absent in homogeneous media—is independently recognized in the elastic-metamaterials and homogenization literature, where it is traced to the Filimonov line and treated by two-point Padé and gradient-elasticity continualization [19, 20, 21, 22]; our contribution is orthogonal, namely the arithmetic that governs *which* N and *how large*.

Main results at a glance.

T1 $A_N = U_N = \frac{1}{\pi}\ln N + \frac{1}{\pi}(\gamma + \ln\frac{2}{\pi}) - \frac{\pi}{72N^2} + \cdots$ for N prime and $N = 2^m$ (independence); $A_N = U_N - O(1/N)$ with the same constant $\frac{1}{\pi}$ for $N = 2p$ (Thms. 3, 4, Prop. 15).

T2 The orbit-closure subtorus has dimension $\frac{1}{2}\varphi(2N)$ for *every* N (Thm. 16).

T3 *Ceiling criterion and classification*: $A_N = U_N$ iff every integer frequency relation has coefficient-sum $\equiv 0 \pmod 4$ (or, N even, an antipodal variant; Thm. 11); this holds *exactly* for N prime or $N = 2^m$, so $\{N : A_N = U_N\} = \{\text{prime}\} \cup \{2^m\}$ (Thm. 14, via an explicit root-of-unity obstruction for every composite N).

T4 Discrete Schrödinger: localized states never amplify (unitarity), yet $B_N = \Theta(\sqrt{N})$ in ℓ^∞ (Thms. 17, 19).

T5 *Order, all* N: $A_N \sim \frac{1}{\pi}\ln N$ for *every* N (Thm. 6)—the lower bound from an alignable independent prefix of length $\frac{1}{2}\varphi(2N)$ (Lemma 5, by a palindromic argument).

T6 The incoherent *ceiling* has critical spectral dimension $d_s = 1$ (incl. quasi-1D); the amplitude is an infinite-time, Diophantine-limited quantity (§17, §10).

We treat the periodic chain (the ring) with *free* oscillations, and contrast it with the discrete Schrödinger dynamics on the same ring. The central quantity is

$$A_N = \sup_{t\geq 0} \max_{0\leq j<N} |u_j(t)|,$$

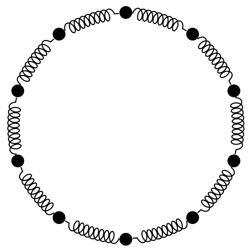
where $u_j(t)$ is the response to a unit velocity impulse. Our tools are the theory of almost-periodic functions (Bohr), equidistribution (Kronecker–Weyl), and the arithmetic of cyclotomic fields. The main message is that A_N grows like $\ln N$, that the growth constant is exactly $1/\pi$ on an explicit family of N, and that the obstruction on the remaining N is a purely number-theoretic one (rational relations among the frequencies).

Part 1. Linear theory

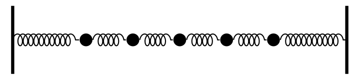
2. Toeplitz and circulant matrices: a unified language

The bead chain is a spring–mass lattice whose stiffness operator is shift invariant—hence Toeplitz—and, under periodic boundary conditions, circulant (Figure 1). We collect the structures and the classical algorithms that organize the entire analysis.

2.1. **Toeplitz, Hankel, circulant.** A matrix is *Toeplitz* if it is constant along diagonals, $A_{ij} = a_{i-j}$ ($2n-1$ scalars, $O(n)$ storage); *Hankel* if constant along anti-diagonals, $H_{ij} = h_{i+j}$; and *circulant* if



(a) ring: $u_{j+N} = u_j$



(b) segment: $u_0 = u_N = 0$

FIGURE 1. The two geometries. (a) The ring of N beads coupled by identical springs gives a *circulant* Laplacian $L_N = \mathrm{circ}(2, -1, 0, \ldots, 0, -1)$. (b) The Dirichlet segment gives the *tridiagonal Toeplitz* $T_N = \mathrm{tridiag}(-1, 2, -1)$. Both have the same symbol $f(\theta) = 4\sin^2(\theta/2)$.

Toeplitz with a periodic wrap, $C_{ij} = c_{(i-j) \bmod n}$. For $n = 4$,

$$T = \begin{pmatrix} a_0 & a_{-1} & a_{-2} & a_{-3} \\ a_1 & a_0 & a_{-1} & a_{-2} \\ a_2 & a_1 & a_0 & a_{-1} \\ a_3 & a_2 & a_1 & a_0 \end{pmatrix}, \quad H = \begin{pmatrix} h_0 & h_1 & h_2 & h_3 \\ h_1 & h_2 & h_3 & h_4 \\ h_2 & h_3 & h_4 & h_5 \\ h_3 & h_4 & h_5 & h_6 \end{pmatrix}, \quad C = \begin{pmatrix} c_0 & c_3 & c_2 & c_1 \\ c_1 & c_0 & c_3 & c_2 \\ c_2 & c_1 & c_0 & c_3 \\ c_3 & c_2 & c_1 & c_0 \end{pmatrix}.$$

Reversing rows turns Toeplitz into Hankel, $H = JT$ with J the exchange matrix; a Toeplitz matrix is the finite compression of a convolution (Laurent) operator. The ring Laplacian $L_N = \mathrm{circ}(2, -1, 0, \ldots, 0, -1)$ is the periodic case of the segment Toeplitz $T_N = \mathrm{tridiag}(-1, 2, -1)$.

2.2. **The symbol and circulant diagonalization.** A banded Toeplitz/circulant matrix with entries a_k has *symbol* $f(\theta) = \sum_k a_k \mathrm{e}^{\mathrm{i}k\theta}$; for both L_N and T_N,

$$f(\theta) = 2 - 2\cos\theta = 4\sin^2(\theta/2).$$

Every circulant is a polynomial $C = \sum_k c_k P^k$ in the cyclic shift P, and the Fourier modes $\varphi_j^{(r)} = N^{-1/2}\mathrm{e}^{2\pi\mathrm{i}rj/N}$ are common eigenvectors of P ($P\varphi^{(r)} = \mathrm{e}^{2\pi\mathrm{i}r/N}\varphi^{(r)}$). Hence

$$L_N = F_N^* \Lambda F_N, \qquad \lambda_r = f\left(\tfrac{2\pi r}{N}\right) = 4\sin^2 \tfrac{\pi r}{N}, \qquad \omega_r = \sqrt{\lambda_r} = 2\sin \tfrac{\pi r}{N}, \tag{1}$$

$r = 0, \ldots, N-1$. The discrete dispersion relation $\omega(\theta) = 2|\sin(\theta/2)|$ is literally the square root of the symbol, sampled at $\theta = 2\pi r/N$ on the ring and at $\theta = \pi k/(N+1)$ on the segment ($\lambda_k^{\mathrm{Dir}} = 4\sin^2 \frac{\pi k}{2(N+1)}$). The twofold degeneracy $\omega_r = \omega_{N-r}$ on the ring—absent on the segment—is the source of the arithmetic difficulty in Section 8.

2.3. **Classical Toeplitz algorithms.** Determined by $2n-1$ scalars, a Toeplitz system admits operations far below the dense $O(n^3)$ (Table 1); three enter the analysis.

Levinson–Durbin. For symmetric Toeplitz $R = (r_{|i-j|})$ the order-i system $R^{(i)}a^{(i)} = (r_1, \ldots, r_i)^\top$ is built recursively through the *reflection coefficients* k_i (PARCOR): with $E_0 = r_0$,

$$k_i = \frac{r_i - \sum_{j=1}^{i-1} a_j^{(i-1)} r_{i-j}}{E_{i-1}}, \quad a_j^{(i)} = a_j^{(i-1)} - k_i\, a_{i-j}^{(i-1)}, \quad a_i^{(i)} = k_i, \quad E_i = (1 - k_i^2)E_{i-1}. \tag{2}$$

Recursion (2) is due to Levinson [11] and, in the AR (Yule–Walker) case, Durbin [12]; one has $|k_i| < 1$ for all i iff $R \succ 0$, making $\{k_i\}$ the natural stable-filter parametrization. The implementation used here goes back to the author's 2012 study [24].

Task	Algorithm	Cost
Solve $Tx = b$ (symmetric)	Levinson (1947)	$O(n^2)$
AR / Yule–Walker	Durbin (1960)	$O(n^2)$
Inverse T^{-1} (explicit)	Trench (1964) / Gohberg–Semencul (1972)	$O(n^2)$
Multiply Tx, circulant solve, $f(L_N)$	FFT / circulant embedding	$O(n \log n)$
Semi-infinite chain	Wiener–Hopf factorization	—

TABLE 1. Structured solvers used here; all beat the dense $O(n^3)$.

Trench / Gohberg–Semencul. The inverse of a Toeplitz matrix is *not* Toeplitz but is determined by its first and last columns u, v [13, 14] (two $O(n^2)$ recursions); writing L_x, U_x for the lower/upper triangular Toeplitz matrices generated by a vector x,

$$T^{-1} = \tfrac{1}{u_0}\big(L_u U_v - L_{Sv}\, U_{Su}\big), \tag{3}$$

a difference of products of triangular Toeplitz factors (S the shift), with $\det T$ obtained en route. On the ring this collapses to the pseudoinverse $L_N^+ = F_N^* \Lambda^+ F_N$ (the constant mode, $L_N \mathbf{1} = 0$, is mapped to 0) and, more generally, $f(L_N) = F_N^* f(\Lambda) F_N$ in $O(N \log N)$—the route by which the propagators below are evaluated.

Wiener–Hopf. On the semi-infinite chain the stiffness is a one-sided Toeplitz operator; its inversion is the Wiener–Hopf factorization $f(\theta) = f_+(\theta) f_-(\theta)$ of the symbol, which links the finite-N ceilings to the operator-theoretic (Szegő) asymptotics of large truncated Toeplitz matrices [16], Section 5.

2.4. **Three dynamics and the central functional.** The same lattice carries the wave equation $\ddot{u} + L_N u = 0$ (propagator $\cos(t\sqrt{L_N})$), the discrete Schrödinger equation $\mathrm{i}\dot{z} = L_N z$ (propagator $\mathrm{e}^{-\mathrm{i}tL_N}$, the phase governed by λ_r rather than $\omega_r = \sqrt{\lambda_r}$), and the dissipative heat flow $\dot{v} = -L_N v$ (no large wave—the semigroup smooths). For the wave equation with the unit zero-mean velocity impulse $u(0) = 0$, $\dot{u}(0) = e_0 - \frac{1}{N}\mathbf{1}$, the response is the velocity Green's function

$$u_j(t) = G_N(j,t) = \frac{1}{N} \sum_{r=1}^{N-1} \frac{\sin(\omega_r t)}{\omega_r}\, \mathrm{e}^{2\pi \mathrm{i} r j / N}. \tag{4}$$

As u is real and $\omega_r = \omega_{N-r}$, (4) is a real almost-periodic function of t with frequency set $\{\omega_r\}$, and the large wave is the operator-norm growth $A_N = \sup_t \max_j |G_N(j,t)|$.

3. Bohr's supremum and the ceiling U_N

Proposition 1 (Bohr ceiling)**.** *For every N and every node j, $\sup_t |G_N(j,t)| \le U_N$, $U_N := \frac{1}{2N}\sum_{r=1}^{N-1} \csc(\pi r/N)$, with equality at $j = 0$ provided the frequencies $\{\omega_r : 1 \le r \le \lfloor N/2 \rfloor\}$ are linearly independent over $\mathbb{Q}$.*

Proof. Grouping the degenerate pair r, $N - r$ gives, at $j = 0$, $G_N(0,t) = \sum_{r=1}^{\lfloor N/2 \rfloor} b_r \sin(\omega_r t)$ with $b_r = 2/(N\omega_r) > 0$ (the term $r = N/2$, if present, has coefficient $1/(N\omega_{N/2})$). The triangle inequality gives $|G_N(j,t)| \le \frac{1}{N}\sum_{r=1}^{N-1} \omega_r^{-1} = U_N$ for all j. If the ω_r are $\mathbb{Q}$-independent, then by Kronecker–Weyl the orbit $\{(\omega_r t)_r\}$ is dense in the torus, so $\sin(\omega_r t) \to 1$ simultaneously and the supremum at $j = 0$ equals the bound. (Bohr's theorem: $\sup_t \sum b_r \sin(\omega_r t) = \sum |b_r|$.) ∎

4. The exact constant

Lemma 2 (Euler–Maclaurin asymptotics). $U_N = \dfrac{1}{\pi}\Big(\ln N + \gamma + \ln\dfrac{2}{\pi}\Big) + o(1) = \dfrac{1}{\pi}\ln N + 0.039\,99\ldots + o(1)$.

The leading term comes from the singularity $\csc(\pi r/N) \sim N/(\pi r)$ at small r, which yields the harmonic sum $\sum 1/r \sim \ln$; the constant follows from the Euler–Maclaurin correction. Numerically $U_N - \frac{1}{\pi}\ln N \to 0.03999$ for $N \le 65536$ (`numerics/verify_lemma3_lebesgue.py`).

5. Upper bounds: energy and the Szegő symbol

A fixed-energy bound caps the response. With energy $E = \frac{1}{2}\dot{u}^\top\dot{u} + \frac{1}{2}u^\top L_N u$ and the exact trigonometric identity (the cotangent sum, derived in Section 6)

$$\sum_{r=1}^{N-1} \frac{1}{\sin^2(\pi r/N)} = \frac{N^2-1}{3}, \tag{5}$$

Cauchy–Schwarz on the modal expansion gives $\|u(t)\|_\infty \le C\sqrt{N}\,\sqrt{E}$: at fixed energy the deflection cannot exceed order $\sqrt{N}$, a ceiling attained by the time-reversed focusing configuration (the variational maximum; `verify_theorem1.py`). The same sum yields the exact pseudo-inverse diagonal $(L_N^+)_{jj} = \frac{1}{N}\sum_{r\ne 0}\lambda_r^{-1} = \frac{N^2-1}{12N}$.

Asymptotically the eigenvalues follow the symbol through Szegő's theorem [15, 9]: for continuous F,

$$\frac{1}{N}\sum_{r=0}^{N-1} F(\lambda_r) \xrightarrow{N\to\infty} \frac{1}{2\pi}\int_0^{2\pi} F\big(f(\theta)\big)\,d\theta, \qquad f(\theta) = 4\sin^2(\theta/2), \tag{6}$$

so spectral sums such as U_N and $(L_N^+)_{jj}$ are Riemann sums of the symbol; the logarithmic growth of U_N is exactly the borderline (logarithmically divergent) singularity $f^{-1/2} \sim |\theta|^{-1}$ at $\theta = 0$. (Szegő's theorem proper requires continuous F; the *singular* sums $U_N = \frac{1}{2N}\sum \lambda_r^{-1/2}$ and $(L_N^+)_{jj} = \frac{1}{N}\sum \lambda_r^{-1}$ are made rigorous by Euler–Maclaurin at the band edge, Appendix B, the singularity producing the logarithm and the N^2 growth.) The propagators $\cos(t\sqrt{L_N})$, $\mathrm{e}^{-\mathrm{i}tL_N}$, $L_N^{-1/2}\sin(t\sqrt{L_N})$ are functions of the matrix in the sense of Golub–Van Loan [17], evaluated exactly by (1); this puts the operator-norm functionals $\|f_t(L_N)\|_{\ell^p\to\ell^q}$ on a rigorous footing.

6. The spectral zeta function and the large-wave constants

Define the spectral zeta of the ring Laplacian (zero mode removed),

$$\zeta_{L_N}(s) = \sum_{r=1}^{N-1}\lambda_r^{-s} = 4^{-s}\sum_{r=1}^{N-1}\csc^{2s}\tfrac{\pi r}{N}, \qquad \Re s > 0. \tag{7}$$

At integer s this is a cosecant power sum with a closed form in N (Bernoulli numbers, Appendix B); the first two are

$$\zeta_{L_N}(1) = \frac{N^2-1}{12}, \qquad \zeta_{L_N}(2) = \frac{(N^2-1)(N^2+11)}{720}.$$

The first is elementary: from the classical identity $\sum_{r=1}^{N-1}\cot^2\frac{\pi r}{N} = \frac{(N-1)(N-2)}{3}$ and $\csc^2 = 1+\cot^2$,

$$\sum_{r=1}^{N-1}\csc^2\tfrac{\pi r}{N} = (N-1) + \frac{(N-1)(N-2)}{3} = \frac{N^2-1}{3}, \qquad \text{so} \qquad \zeta_{L_N}(1) = \tfrac{1}{4}\sum_r \csc^2\tfrac{\pi r}{N} = \frac{N^2-1}{12},$$

and $\zeta_{L_N}(2)$ follows the same way from $\sum_r \csc^4 = \frac{(N^2-1)(N^2+11)}{45}$ (all integer values are even polynomials in N, `verify_spectral_zeta.py`). The two large-wave constants are *special values of this single object*:

$$U_N = \frac{\zeta_{L_N}(1/2)}{N}, \qquad M_N^2 = \frac{2E_0\,\zeta_{L_N}(1)}{N}, \tag{8}$$

the first because $\csc(\pi r/N) = 2\lambda_r^{-1/2}$, the second because $(L_N^+)_{jj} = \zeta_{L_N}(1)/N$. Thus the Bohr ceiling (the large wave, $A_N \sim U_N$) and the variational ceiling M_N (Section 5) are the values of ζ_{L_N} at $s = \frac{1}{2}$ and $s = 1$.

The two growth laws are then dictated by the analytic structure of ζ_{L_N}. By Szegő's theorem the normalized zeta converges to the symbol integral,

$$\frac{1}{N}\,\zeta_{L_N}(s) \xrightarrow{N\to\infty} I(s) = \frac{1}{2\pi}\int_0^{2\pi} (2-2\cos\theta)^{-s}\,d\theta = \frac{4^{-s}\,\Gamma(\frac{1}{2}-s)}{\sqrt{\pi}\,\Gamma(1-s)}, \qquad \Re s < \tfrac{1}{2}, \tag{9}$$

which has a simple pole at $s = \frac{1}{2}$ (Figure 2). The edge of convergence is exactly the logarithmic singularity producing $U_N = \zeta_{L_N}(\frac{1}{2})/N = \frac{1}{\pi}\ln N + O(1)$ (Lemma 2). At $s = 1$ the density diverges, and indeed $\zeta_{L_N}(1) \sim N^2/12$ yields the $\sqrt{N}$ variational ceiling $M_N \sim \sqrt{N/6}$. *The large wave sits precisely at the convergence edge $s = \frac{1}{2}$ of the limiting spectral density $I(s)$*; the finite-N ζ_{L_N} is itself an entire finite sum, and it is the $N \to \infty$ density that carries the pole. The finite-N zeta is an Epstein/Hurwitz-type object whose integer values are polynomials in N^2 and whose half-integer value carries the logarithm. All identities are verified (`verify_spectral_zeta.py`). At the opposite end, the spectral determinant is the $s = 0$ *derivative*: $\zeta'_{L_N}(0) = -\log\det{}'L_N$ with $\det{}'L_N = \prod_{r\neq 0}\lambda_r = N^2$ (from $\prod_{r=1}^{N-1} 2\sin(\pi r/N) = N$), so by Kirchhoff's matrix-tree theorem the cycle C_N has exactly $\det{}'L_N/N = N$ spanning trees—a graph invariant read off $\zeta'_{L_N}(0)$ (`verify_spanning_trees.py`).

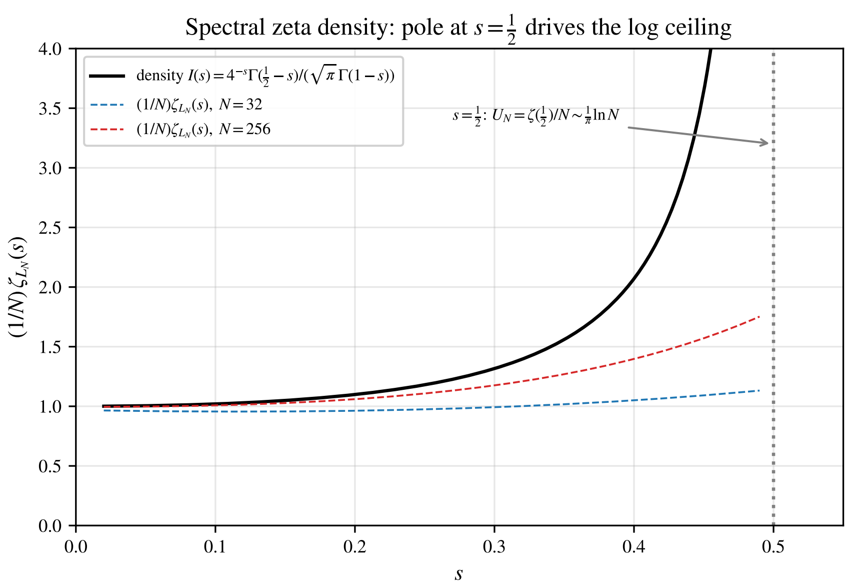


Figure 2. The normalized spectral zeta $\frac{1}{N}\zeta_{L_N}(s)$ converges to the symbol density $I(s)$ of (9), which has a pole at $s = \frac{1}{2}$. The large-wave ceiling is the boundary value $U_N = \zeta_{L_N}(\frac{1}{2})/N \sim \frac{1}{\pi}\ln N$.

7. Operator-norm hierarchy and the variational principle

The large wave is one corner of a hierarchy of operator norms of the velocity propagator $S_N(t) = f_t(L_N)$ on the zero-mean subspace $\mathcal{H}_0 = \{x \in \mathbb{R}^N : \sum_j x_j = 0\}$, where $f_t(\lambda) = \sin(t\sqrt{\lambda})/\sqrt{\lambda}$ for $\lambda > 0$ and $f_t(0) = 0$ (circulant on $\mathcal{H}_0$, kernel G_N; the constant mode is excluded since $L_N\mathbf{1} = 0$). Its time-supremum in different $\ell^p \to \ell^q$ norms obeys *three distinct growth laws* (Table 2, all verified by `verify_operator_norms.py`):

$$\begin{aligned}\sup_t \|S_N(t)\|_{2\to 2} &= \frac{1}{2\sin(\pi/N)} \sim \frac{N}{2\pi},\\ \sup_t \|S_N(t)\|_{2\to\infty} &\le \sqrt{\tfrac{\zeta_{L_N}(1)}{N}} \sim \sqrt{\tfrac{N}{12}},\\ \sup_t \|S_N(t)\|_{1\to\infty} &= A_N \le U_N \sim \tfrac{1}{\pi}\ln N.\end{aligned} \tag{10}$$

Derivations. $S_N(t)$ is symmetric with eigenvalues $f_t(\lambda_r) = \sin(t\omega_r)/\omega_r$, so its $\ell^2 \to \ell^2$ norm is $\max_r |\sin(t\omega_r)|/\omega_r$, whose supremum over t is $1/\omega_1 = 1/(2\sin\frac{\pi}{N})$—the slowest mode reaching $\sin(t\omega_1) = \pm 1$; this is exact and unconditional. For $\ell^2 \to \ell^\infty$, by translation invariance the norm is the ℓ^2-length of one row of G_N, and Parseval gives

$$\|G_N(\cdot,t)\|_2^2 = \frac{1}{N}\sum_{r=1}^{N-1}\frac{\sin^2(t\omega_r)}{\omega_r^2} \le \frac{1}{N}\sum_r \omega_r^{-2} = \frac{\zeta_{L_N}(1)}{N},$$

hence $\sup_t \|S_N(t)\|_{2\to\infty} \le \sqrt{\zeta_{L_N}(1)/N} \sim \sqrt{N/12}$, with equality iff all $\sin^2(t\omega_r) = 1$ simultaneously—rational independence (N prime, 2^m). Finally $\|S_N(t)\|_{1\to\infty} = \max_{j,k}|S_N(t)_{jk}| = \max_m |G_N(m,t)|$, whose time-supremum is, by definition, $A_N \le \sum_r |a_r| = U_N$. The three corners thus read the spectral sums $\sum_r \omega_r^{-p}$ ($p = 1, 2$) through different lenses; the $\ell^1 \to \ell^\infty$ corner—weakest input, strongest output—is precisely the large wave A_N.

Variational principle. The energy norm has the direct reading: maximizing the deflection at a node over the energy sphere $E(u_0, v_0) \le E_0$ on $\mathcal{H}_0$ is a constrained quadratic problem solved by Lagrange multipliers,

$$\sup_{E(u_0,v_0)\le E_0} \max_j |u_j(t)| = \sqrt{2E_0\,(L_N^+)_{jj}} = \sqrt{\tfrac{E_0(N^2-1)}{6N}} = M_N,$$

independent of t (the focusing configuration is the time-reversed peak). This is the operator-norm/variational face of the energy ceiling, and via (8) both M_N and U_N are special values of the spectral zeta. The Szegő symbol $f(\theta) = 4\sin^2(\theta/2)$ controls all three norms through the spectral sums $\sum_r \omega_r^{-p}$.

norm	$\sup_t \|S_N(t)\|$	growth	meaning
$\ell^2 \to \ell^2$	$1/(2\sin\frac{\pi}{N})$	$N/2\pi$	lowest-mode resonance
$\ell^2 \to \ell^\infty$	$\le \sqrt{\zeta_{L_N}(1)/N}$	$\sqrt{N/12}$	energy ceiling / variational M_N
$\ell^1 \to \ell^\infty$	$A_N \le U_N = \zeta_{L_N}(1/2)/N$	$\frac{1}{\pi}\ln N$	**large wave A_N**

Table 2. Three growth laws from one propagator. The $\ell^2 \to \ell^2$ supremum is exact and unconditional; the other two equal the listed ceilings on the $\mathbb{Q}$-independent classes N prime and $N = 2^m$ (for $N = 2p$, Proposition 15 gives only an $O(N^{-1})$ gap for the $\ell^1 \to \ell^\infty$ large wave) and are bounded by them in general. The large wave is the $\ell^1 \to \ell^\infty$ corner.

8. Independence theorems and unconditional growth

Theorem 3 (odd prime)**.** *Let p be an odd prime. Then $\{\sin(\pi r/p) : 1 \le r \le \frac{p-1}{2}\}$ is linearly independent over $\mathbb{Q}$; hence $A_p = U_p \sim \frac{1}{\pi}\ln p$.*

Proof. Put $\eta = \zeta_{2p} = e^{i\pi/p}$. As p is odd, $[\mathbb{Q}(\eta) : \mathbb{Q}] = \varphi(2p) = p-1$, so $\{1, \eta, \dots, \eta^{p-2}\}$ is a $\mathbb{Q}$-basis, and $\eta^p = e^{i\pi} = -1$. From $\eta^{-r} = \eta^{2p-r} = \eta^p\eta^{p-r} = -\eta^{p-r}$ we get $2i\sin(\pi r/p) = \eta^r - \eta^{-r} = \eta^r + \eta^{p-r}$. Suppose $\sum_{r=1}^{(p-1)/2} c_r \sin(\pi r/p) = 0$ with $c_r \in \mathbb{Q}$. Then $\sum_r c_r\eta^r + \sum_r c_r\eta^{p-r} = 0$. As r runs over $\{1, \dots, \frac{p-1}{2}\}$ the exponents r fill $\{1, \dots, \frac{p-1}{2}\}$ and $p - r$ fill $\{\frac{p+1}{2}, \dots, p-1\}$, together covering $\{1, \dots, p-1\}$ bijectively, so $\sum_{m=1}^{p-1} d_m\eta^m = 0$ with $d_m = c_m$ or c_{p-m}. Finally $\Phi_{2p}(\eta) = \sum_{j=0}^{p-1}(-1)^j\eta^j = 0$ gives $\eta^{p-1} = -\sum_{j=0}^{p-2}(-1)^j\eta^j$, whose constant term is $-1 \ne 0$; equating coordinates in $\{1, \eta, \dots, \eta^{p-2}\}$ forces $d_{p-1} = 0$ and then all $d_m = 0$, i.e. all $c_r = 0$. The conclusion follows from Proposition 1. □

Theorem 4 (powers of two)**.** *For $N = 2^m$, $\{\sin(\pi r/N) : 1 \le r \le N/2\}$ is $\mathbb{Q}$-independent; hence $A_N = U_N \sim \frac{1}{\pi}\ln N$.*

Proof. Now $\eta = \zeta_{2N}$, $\Phi_{2N}(x) = x^N + 1$ (degree $\varphi(2N) = N$), so $\{1, \dots, \eta^{N-1}\}$ is a basis and $\eta^N = -1$. Again $2i\sin(\pi r/N) = \eta^r + \eta^{N-r}$, and for $1 \le r \le N/2$ the exponents r and $N - r$ lie in $\{1, \dots, N-1\}$ *without reduction*; they are distinct basis powers (the index $N/2$ occurring with coefficient $2c_{N/2}$). Linear independence is immediate. □

An exact integer-rank certificate validates the prime case (`verify_lemma1_prime.py`); the general rank formula is checked separately by `verify_independence_rank.py`.

Worked example (prime $N = 5$). The two distinct frequencies $\omega_1 = 2\sin 36^\circ = 1.1756$ and $\omega_2 = 2\sin 72^\circ = 1.9021$ stand in the ratio $\omega_2/\omega_1 = 2\cos 36^\circ = \varphi = \frac{1+\sqrt{5}}{2}$, the *golden ratio*—irrational, so the relation lattice is trivial ($\Lambda_5 = \{0\}$) and Theorem 3 applies. The node-0 response is the two-tone signal $G_5(0,t) = \frac{2}{5}\big(\omega_1^{-1}\sin\omega_1 t + \omega_2^{-1}\sin\omega_2 t\big)$, and by Kronecker both sines reach 1 at a common near-alignment time, so

$$A_5 = U_5 = \tfrac{1}{5}\big(\csc 36^\circ + \csc 72^\circ\big) = 0.5506,$$

the supremum equal to the ceiling, approached arbitrarily closely (in contrast to the composite $N = 12$ below; `verify_exact_AN.py`).

The low modes carry most of the ceiling, and—crucially—a *long* independent prefix can always be aligned. Let $M(N)$ be the largest M with $\{\omega_1, \dots, \omega_M\}$ $\mathbb{Q}$-independent.

Lemma 5 (prefix independence)**.** *$M(N) = \frac{1}{2}\varphi(2N)$: the first $\frac{1}{2}\varphi(2N)$ frequencies $\{\omega_r = 2\sin(\pi r/N) : 1 \le r \le \frac{1}{2}\varphi(2N)\}$ are $\mathbb{Q}$-independent.*

Proof. A nontrivial relation $\sum_{r=1}^K k_r \sin(\pi r/N) = 0$ with $k_K \ne 0$ (K its largest index)—a vanishing sum of roots of unity [4]—gives, with $\zeta = \zeta_{2N}$ and $P(x) = \sum_{r=1}^K k_r x^r$,

$$Q(x) = x^K\big(P(x) - P(1/x)\big) = \sum_{r=1}^{K} k_r\big(x^{K+r} - x^{K-r}\big) \in \mathbb{Z}[x],$$

nonzero of degree $2K$ (leading coefficient k_K), with $Q(\zeta) = \zeta^K\sum_r k_r(\zeta^r - \zeta^{-r}) = 0$. Hence $\Phi_{2N} \mid Q$ and $\varphi(2N) = \deg\Phi_{2N} \le 2K$. But Q is *anti*-palindromic, $x^{2K}Q(1/x) = -Q(x)$, whereas Φ_{2N} is palindromic; so $2K = \varphi(2N)$ is impossible (it would force $Q = k_K\Phi_{2N}$, simultaneously palindromic and anti-palindromic, hence 0). Thus $2K > \varphi(2N)$, and as both are even, $K \ge \frac{1}{2}\varphi(2N) + 1$. No relation has all its indices $\le \frac{1}{2}\varphi(2N)$, so $\{\omega_1, \dots, \omega_{\varphi(2N)/2}\}$ are independent and $M(N) \ge \frac{1}{2}\varphi(2N)$; with $M(N) \le \operatorname{rank}_{\mathbb{Q}}\{\omega_r\} = \frac{1}{2}\varphi(2N)$ (Theorem 16), equality holds. □

Theorem 6 (order)**.** *$A_N \sim \frac{1}{\pi}\ln N$ for every N; precisely $A_N = \frac{1}{\pi}\ln N + O(\ln\ln\ln N)$.*

Proof. The upper bound is $A_N \le U_N = \frac{1}{\pi}\ln N + O(1)$ (Proposition 1, Lemma 2). For the lower bound, write $b_r = 2/(N\omega_r) > 0$, $L_{\text{pre}} = \sum_{r=1}^{M} b_r$, $M = \frac{1}{2}\varphi(2N)$. By Lemma 5 the $\omega_1, \dots, \omega_M$ are $\mathbb{Q}$-independent, so by Kronecker there are times t with $\sin(\omega_r t) \to 1$ for all $r \le M$ at once; at such t, since the remaining $\sin(\omega_r t) \ge -1$,

$$A_N \;\ge\; \sup_t G_N(0,t) \;\ge\; L_{\text{pre}} - \sum_{r=M+1}^{\lfloor N/2\rfloor} b_r \;=\; 2L_{\text{pre}} - U_N.$$

As $\sin(\pi r/N) \le \pi r/N,\ b_r = 1/(N\sin(\pi r/N)) \ge 1/(\pi r)$, so $L_{\text{pre}} \ge \frac{1}{\pi}\sum_{r=1}^{M}\frac{1}{r} \ge \frac{1}{\pi}\ln M$, giving

$$A_N \;\ge\; \tfrac{2}{\pi}\ln M - U_N \;=\; \tfrac{1}{\pi}\ln\frac{M^2}{N} - O(1) \;=\; \tfrac{1}{\pi}\Big(2\ln\tfrac{\varphi(2N)}{2} - \ln N\Big) - O(1).$$

By the Rosser–Schoenfeld bound $\varphi(2N) \ge 2N/(\mathrm{e}^{\gamma}\ln\ln 2N + 3/\ln\ln 2N)$ [5], the bracket is $\ln N - O(\ln\ln\ln N)$; hence $A_N \ge \frac{1}{\pi}\ln N - O(\ln\ln\ln N)$, and with the upper bound $A_N = \frac{1}{\pi}\ln N + O(\ln\ln\ln N)$ (`verify_order_theorem.py`). □

The former obstruction—the destructive interference of the out-of-prefix modes—is dispatched by the crude bound $\sin \ge -1$, which suffices because Lemma 5 makes the alignable prefix carry $2L_{\text{pre}} - U_N = (1 - o(1))U_N$ of the ceiling. The exact saturation locus $A_N = U_N$ is the sharper question settled by Theorem 14.

Remark 7 (the defect: an odd-part scaling law)**.** The same estimate controls the deficit on the non-saturating N: $U_N - A_N \le 2(U_N - L_{\text{pre}}) = 2\sum_{r=\frac{1}{2}\varphi(2N)+1}^{\lfloor N/2\rfloor} b_r = \frac{2}{\pi}\ln\frac{N}{\varphi(2N)} + O(1)$ (the tail of the ceiling beyond the prefix; $b_{N/2} = \frac{1}{2N}$ when N is even). Writing $N = 2^a m$ with m odd, $N/\varphi(2N) = m/\varphi(m)$ is *independent of* a, so the bound is the same throughout each family $2^a m$:

$$U_N - A_N \;\le\; \tfrac{2}{\pi}\ln\tfrac{m}{\varphi(m)} + O(1),$$

and the exact defect converges within the family to a limit depending only on the odd part m (numerically $U_N - A_N < 0.18$ for composite $N \le 60$, e.g. $0.05, 0.08, 0.09, 0.10$ along $m = 3$; `verify_exact_AN.py`). Globally, since $m/\varphi(m) \sim \mathrm{e}^{\gamma}\ln\ln m$ along primorials, this is $O(\ln\ln\ln N)$—slower than any iterate of the logarithm. The defect is *not* bounded, and there is a clean reason: numerically the *excess* $A_N - L_{\text{pre}}$ stays in $[0.05, 0.10]$ across every tested N ($\omega(m) \le 3$, prefix dimension up to 48), so

$$U_N - A_N = (U_N - L_{\text{pre}}) - (A_N - L_{\text{pre}}) = \tfrac{1}{\pi}\ln\tfrac{m}{\varphi(m)} - O(1) \longrightarrow \infty$$

along primorials ($\Theta(\ln\ln\ln N)$, `verify_defect_growth.py`); the proven $O(\ln\ln\ln N)$ upper bound is sharp in order. A rigorous *growing* lower bound is thus equivalent to the *excess lemma* $A_N \le L_{\text{pre}} + O(1)$—the long alignable prefix captures A_N up to a constant—a sup-side cancellation statement on the relation lattice (a primal analogue of McGehee–Pigno–Smith), overwhelmingly supported numerically but open. For small N the strict gap $A_N < U_N$ is *certified rigorously*: the Lipschitz-grid certificate gives $U_N - A_N \ge 0.049, 0.057, 0.030, 0.082$ for $N = 6, 9, 12, 15$ (`verify_defect_certified.py`), and a trigonometric Lasserre moment–SOS sharpens it (exact $U_N - A_N = 0.1475$ at $N = 15$) and reaches $N = 105$ ($M = 24$), though too loosely there to certify the growth (`verify_defect_moment_sos.py`). The exact value is algebraic per N (Proposition 10) with no universal closed form.

Example 8 (the whole mechanism at $N = 15$)**.** The seven frequencies are $\omega_r = 2\sin(\pi r/15)$, $r = 1, \dots, 7$ (with $\omega_5 = 2\sin\frac{\pi}{3} = \sqrt{3}$). Here $\varphi(2N) = \varphi(30) = 8$, so the rank is $\frac{1}{2}\varphi(30) = 4$: by Lemma 5 the prefix $\omega_1, \dots, \omega_4$ is $\mathbb{Q}$-independent and *no* relation can live in indices ≤ 4. The first relation appears at index $5 = \frac{1}{2}\varphi(30) + 1$—exactly the palindromic bound, here sharp—namely

$$\omega_4 + \omega_5 = \omega_1 + 2\omega_2 + \omega_3, \qquad \text{i.e.} \quad k = (-1, -2, -1, 1, 1, 0, 0).$$

Its coefficient sum $\sum_r k_r = -2 \equiv 2 \pmod 4$ is nonzero mod 4, so by the criterion of Theorem 11 the target $(\frac{\pi}{2}, \dots, \frac{\pi}{2})$ is off the subtorus and $A_{15} < U_{15}$: composite 15 does not saturate (Theorem 14). Numerically $U_{15} = 0.902$, $A_{15} = 0.754$, a defect 0.148—the largest among small composites because the odd part $m = 15$ carries two prime factors (Remark 7). The lower bound of Theorem 6 aligns the four-mode prefix: $A_{15} \geq 2L_{\text{pre}} - U_{15} = 0.47$, comfortably inside the band $A_N \sim \frac{1}{\pi} \ln N$.

8.1. **The Dirichlet segment.** The segment chain $\text{tridiag}(-1, 2, -1)$ with Dirichlet ends has the *simple* spectrum $\mu_k = 2 \sin \frac{\pi k}{2(N+1)}$ $(k = 1, \dots, N)$: no degeneracy, the structural advantage over the ring, whose twofold $\mu_r = \mu_{N-r}$ halves the independent phases. Writing $\mu_k = 2\sin(2\pi k/M)$ with $M = 4(N+1)$, the frequencies lie in $\mathbb{Q}(\zeta_M)$ with $2\mathrm{i}\sin(2\pi k/M) = \zeta_M^k - \zeta_M^{-k}$, and the same minus-eigenspace argument identifies the rank *exactly*. The μ_k are the first N distinct frequencies of the ring on $N' = 2(N+1)$ nodes; the excluded middle index $N'/2 = N+1$ carries the rational frequency $\mu_{N+1} = 2$, and Theorem 16 applied to N' gives $\dim_{\mathbb{Q}} \text{span}\{\mu_1, \dots, \mu_N, 2\} = \frac{1}{2}\varphi(4(N+1))$.

Theorem 9 (segment rank and classification)**.**

$$\dim_{\mathbb{Q}} \text{span}\{\mu_1, \dots, \mu_N\} = \min\big(N, \tfrac{1}{2}\varphi(4(N+1))\big), \tag{11}$$

and $A_N^{\text{Dir}} = U_N^{\text{Dir}}$ *if and only if* $N+1$ *is prime or a power of two.*

Proof. Each $2\mathrm{i}\mu_k = \zeta_M^k - \zeta_M^{-k} \in V^-$ $(M = 4(N+1))$, so $\dim_{\mathbb{Q}} \text{span}\{\mu_k\} \leq \frac{1}{2}\varphi(M)$; since adjoining the rational 2 keeps the rank at $\frac{1}{2}\varphi(M)$, the dimension equals $\frac{1}{2}\varphi(M)$ or $\frac{1}{2}\varphi(M) - 1$ according as $2 \in \text{span}\{\mu_k\}$ or not. If $N+1 = 2^m$, the ring on $N' = 2^{m+1}$ is $\mathbb{Q}$-independent (Theorem 4); then 2 is independent of the μ_k, so $2 \notin \text{span}\{\mu_k\}$ and the rank is $\frac{1}{2}\varphi(M) - 1 = N$. If instead $N+1$ has an odd prime factor p, put $a = N+1$; then $\sum_{j=0}^{p-1} \zeta_M^{a+(M/p)j} = \zeta_M^a \sum_j \big(\zeta_M^{M/p}\big)^j = 0$, and the only term of its imaginary part reducing to the middle index $N+1$ is $j = 0$ (another would need $(M/p)j \equiv M/2$, i.e. $2j = p$, impossible for p odd), so $\mu_{N+1} = 2$ occurs with coefficient ± 1 and lies in $\text{span}\{\mu_1, \dots, \mu_N\}$; the rank is $\frac{1}{2}\varphi(M)$. An elementary count gives $\frac{1}{2}\varphi(4(N+1)) = 2^a\varphi(m)$ for $N+1 = 2^a m$ (m odd), which exceeds N exactly for $m = 1$ ($N+1 = 2^m$) and is $\leq N$ otherwise (with equality iff $N+1$ is prime); so the rank is $\min(N, \frac{1}{2}\varphi(4(N+1)))$, and it is full ($= N$, whence $A_N^{\text{Dir}} = U_N^{\text{Dir}}$ by Bohr) iff $N+1$ is prime or a power of two—otherwise $A_N^{\text{Dir}} < U_N^{\text{Dir}}$ (`verify_segment_classification.py`; the prime case also recovers Proposition 15). ∎

This is the exact analogue of the ring classification (Theorem 14), shifted by the boundary (the ring uses modulus $2N$, the segment $4(N+1)$). The order Theorem 6 transfers verbatim: Lemma 5 holds with modulus $4(N+1)$ in place of $2N$ (same palindromic argument), so the independent prefix has length $\frac{1}{2}\varphi(4(N+1))$ and $A_N^{\text{Dir}} \sim \frac{1}{\pi} \ln N$ for *every* N (`verify_segment_lower_bound.py`).

9. Exact A_N for composite N

For composite N the frequencies satisfy rational relations and the orbit closure is a proper *subtorus* (Weyl). In the cyclotomic coordinates C (row r = integer coordinates of $2\mathrm{i}\sin(\pi r/N)$ in $\mathbb{Q}(\zeta_{2N})$) the attainable phases are $\{\varphi = C\psi\}$, whence

$$\sup_t u_j(t) = \max_{\psi \in \mathbb{R}^p} \sum_r a_{rj} \sin\big((C\psi)_r\big), \qquad p = \varphi(2N), \tag{12}$$

a finite-dimensional trigonometric maximization, reduced exactly and then evaluated numerically (`verify_exact_AN.py`, returning $A_N = U_N$ on primes; a certified global bracket is given for $N = 6$ below). This makes the composite case computable for each N: numerically the subtorus defect on the ring is $A_N/U_N \in [0.84, 0.92]$ in the tested range, so $A_N/\ln N \in [0.28, 0.31]$, consistent with Theorem 6 ($A_N \sim \frac{1}{\pi}\ln N$).

Proposition 10 (algebraic composite cases)**.** *A_N is an algebraic number: writing $x_r = \cos\varphi_r, y_r = \sin\varphi_r$ with $x_r^2 + y_r^2 = 1$ and adjoining, for each generator k of the relation lattice Λ, the constraints $\cos\langle k, \varphi\rangle = 1$, $\sin\langle k, \varphi\rangle = 0$ that cut out the orbit-closure subtorus $\mathbb{T}_N$ (polynomials in the x_r, y_r through the Chebyshev/addition formulas), turns (12) into the maximum of a polynomial over a compact semialgebraic set defined over $\mathbb{Q}$, whose value is algebraic by the Tarski–Seidenberg theorem (real quantifier elimination). For $N = 6$ the relation lattice is one-dimensional and*

$$A_6 = \frac{\sqrt{3}}{9} + \frac{(3+\sqrt{3})\,\sqrt[4]{3}}{12\sqrt{2}} \approx 0.55942,$$

with the inner maximum at $\cos\varphi = (\sqrt{3}-1)/2$. For $N = 12$ the three-term relation $\omega_5 = \omega_1 + \omega_3$ (i.e. $\sin 75^\circ - \sin 15^\circ = \sin 45^\circ$) makes the block two-dimensional; the algebraic degree grows with the arithmetic of N, and no uniform elementary closed form is apparent.

Worked example ($N = 12$). The six distinct frequencies $\omega_r = 2\sin(\pi r/12)$ are

$$(\omega_1, \dots, \omega_6) = \big(2\sin 15^\circ,\ 1,\ \sqrt{2},\ \sqrt{3},\ 2\sin 75^\circ,\ 2\big) \approx (0.518,\ 1,\ 1.414,\ 1.732,\ 1.932,\ 2),$$

and their relation lattice Λ_{12} is rank two, generated by

$$\omega_5 = \omega_1 + \omega_3 \quad (2\sin 75^\circ - 2\sin 15^\circ = \sqrt{2}), \qquad \omega_6 = 2\omega_2 \quad (2 = 2\cdot 1).$$

Both generators have coefficient-sum $\not\equiv 0 \pmod 4$—reading $\omega_5 - \omega_1 - \omega_3 = 0$ as $-1-1+1 \equiv 3$ and $2\omega_2 - \omega_6 = 0$ as $2 - 1 \equiv 1$—so by the criterion below (Theorem 11) the ceiling is *not* attained, $A_{12} < U_{12}$. Solving the subtorus maximization (12) numerically gives

$$U_{12} = \tfrac{1}{24}\sum_{r=1}^{11} \csc\tfrac{\pi r}{12} = 0.8307, \qquad A_{12} = 0.7546, \qquad \frac{A_{12}}{U_{12}} = 0.908,$$

the two relations pulling the optimal phases off $\varphi_r = \frac{\pi}{2}$ by just enough to cost 9% of the ceiling (`verify_relation_lattices.py`, `verify_exact_AN.py`).

A sharp arithmetic criterion decides exactly when $A_N = U_N$ (the supremum equalling the ceiling).

Theorem 11 (ceiling criterion)**.** *Let $\Lambda = \{k \in \mathbb{Z}^m : \sum_r k_r\omega_r = 0\}$ be the relation lattice of the distinct positive frequencies $\omega_r = 2\sin(\pi r/N)$, $1 \le r \le m = \lfloor N/2 \rfloor$. Then $A_N = U_N$ if and only if*

$$\sum_r k_r \equiv 0 \pmod 4 \quad \text{for every } k \in \Lambda$$

or, when N is even, the antipodal variant $\sum_r k_r + 2\sum_r rk_r \equiv 0 \pmod 4$ holds for every $k \in \Lambda$. The two conditions coincide for all $N \le 160$ (where only N prime and $N = 2^m$ saturate, Corollary 13).

Proof. Always $A_N \le U_N$, with $U_N = \sum_r a_{r0}$ the node-0 ceiling. As $a_{rj} = a_{r0}\cos(2\pi rj/N)$ we have $|a_{rj}| \le a_{r0}$, so no node exceeds U_N, and $|a_{rj}| = a_{r0}$ for *all* r forces $|\cos(2\pi rj/N)| = 1$, i.e. $2j/N \in \mathbb{Z}$: only $j = 0$ and—for even N—$j = N/2$ can have time-supremum U_N. At $j = 0$ the coefficients $a_{r0} > 0$, so $\sup_t G_N(0,t) = U_N$ iff all $\sin(\omega_r t)$ can be brought simultaneously arbitrarily close to 1, i.e. the target $\varphi_* = (\frac{\pi}{2}, \dots, \frac{\pi}{2})$ lies in the orbit closure $\mathbb{T}_N$. By Kronecker, $\varphi_* \in \mathbb{T}_N$ iff $\langle k, \varphi_*\rangle = \frac{\pi}{2}\sum_r k_r \equiv 0\ (2\pi)$ for every $k \in \Lambda$, i.e. $\sum_r k_r \equiv 0\ (4)$. At $j = N/2$ (even N) the signs $\cos(\pi r) = (-1)^r$ give the alternating target $\varphi_r = \frac{\pi}{2} + \pi r$, and the same argument yields $\langle k, \varphi \rangle = \frac{\pi}{2}\sum_r k_r + \pi \sum_r r k_r \equiv 0\ (2\pi)$, i.e. $\sum_r k_r + 2\sum_r r k_r \equiv 0\ (4)$. Thus $A_N = U_N$ iff one of the two admissible nodes can realize its target, giving the stated disjunction (`verify_ceiling_criterion.py`). □

Remark 12 (priority)**.** The degenerate ("multiple-frequency") regime that this criterion characterizes is exactly the subject of Filimonov's 1992 note on the circular string [28] (Zbl 0761.73053), which we have been unable to consult beyond its abstract and the splash table reproduced in [22] (Table 10.1, the *segment* regime at constant $4/\pi^2$). It is therefore possible that the criterion—or the equivalent characterization of when the ring saturates—is already, in whole or in part, in [28]; this must be checked against the primary source before any priority is claimed.

This refines the independence test: $\mathbb{Q}$-independence means $\Lambda = \{0\}$, which satisfies the criterion vacuously, recovering $A_N = U_N$ for N prime and $N = 2^m$. The criterion is genuinely weaker in principle—a composite N with dependent frequencies could still saturate, provided every relation has coefficient-sum divisible by 4. Computing Λ exactly by integer (Hermite) reduction of the cyclotomic coordinates and testing the criterion against the directly optimized A_N/U_N, the two agree for all $N \leq 40$, and a criterion-only scan finds *no* composite $N \leq 160$ that saturates: on this range $A_N = U_N$ holds precisely for N prime or $N = 2^m$ (`verify_ceiling_criterion.py`). The *full-independence* route to saturation is in fact closed for *all* N by Theorem 16 (rank $= \frac{1}{2}\varphi(2N) = \lfloor N/2 \rfloor$ only for prime and 2^m, confirmed numerically to $N \leq 500$, `verify_order_extended.py`); a composite saturator would have to satisfy the mod-4 criterion *despite* rational dependence, which none ≤ 160 does. Table 3 makes the lattices explicit; e.g. the generators $\omega_3 = 2\omega_1$ ($N = 6$), $\omega_5 = \omega_1 + \omega_3$ ($N = 12$, cf. Proposition 10), and $\omega_4 + \omega_5 = \omega_1 + 2\omega_2 + \omega_3$ ($N = 15$) all have coefficient-sum $\not\equiv 0\ (4)$.

Table 3. Relation lattices Λ_N of the distinct frequencies ω_r and the mod-4 criterion. Primes and powers of two have $\Lambda_N = \{0\}$ and saturate; every composite N listed carries a relation with coefficient-sum $\not\equiv 0 \pmod 4$, hence $A_N < U_N$; for even N the antipodal sums $\sum_r k_r + 2\sum_r r k_r$ are violated likewise (`verify_ceiling_criterion.py`, check E).

N	class	#freq	rank Λ_N	coeff-sums mod 4	$A_N = U_N$?
5	prime	2	0	—	yes
6	composite	3	1	3	no
8	2^m	4	0	—	yes
9	composite	4	1	3	no
10	composite	5	1	1	no
12	composite	6	2	3, 3	no
15	composite	7	3	2, 3, 3	no
16	2^m	8	0	—	yes
18	composite	9	3	3, 3, 3	no
20	composite	10	2	1, 1	no
21	composite	10	4	1, 3, 3, 3	no
24	composite	12	4	3, 3, 3, 3	no

Corollary 13 (saturating families and a finite scan)**.** *The two proven saturating families, the primes and the powers of two, both have natural density zero. An exhaustive criterion scan finds no further saturating*

$N \le 160$, so on this range $\{N : A_N = U_N\} = \{N \text{ prime}\} \cup \{2^m\}$. Theorem 14 promotes this to all N: the saturating set is exactly *the primes and the powers of two, hence of density zero.*

Theorem 14 (classification of saturation). *$A_N = U_N$ if and only if N is prime or a power of two.*

Proof. If N is prime or $N = 2^m$ then $\Lambda_N = \{0\}$ (Theorems 3, 4) and Theorem 11 gives $A_N = U_N$. Conversely, let N be composite and not a power of two, and let p be an odd prime factor, so $N = ps$ with $s = N/p \ge 2$ and $q := 2N/p = 2s$. As $\zeta_{2N}^q = \mathrm{e}^{2\pi\mathrm{i}/p}$ is a primitive pth root of unity,

$$\sum_{j=0}^{p-1} \zeta_{2N}^{1+qj} = \zeta_{2N} \sum_{j=0}^{p-1} \left(\zeta_{2N}^q\right)^j = \zeta_{2N} \cdot 0 = 0,$$

and taking imaginary parts, $\sum_{j=0}^{p-1} \sin \frac{\pi(1+qj)}{N} = 0$. Reduce each term to a distinct frequency: with $e_j := (1+qj) \bmod 2N$ one has $\sin \frac{\pi(1+qj)}{N} = \sigma_j \sin \frac{\pi r_j}{N}$, where $r_j \in \{1, \dots, \lfloor N/2 \rfloor\}$ and $\sigma_j = +1$ if $e_j \in (0, N)$, $\sigma_j = -1$ if $e_j \in (N, 2N)$. *No e_j lies in $\{0, N\}$:* $e_j \equiv 0$ would need $qj \equiv -1\ (2N)$, impossible since $q \mid 2N$ and $q > 1$; $e_j \equiv N$ would need $q \mid N - 1$, but p odd gives $ps \equiv s\ (2s)$, hence $N - 1 \equiv s - 1\ (2s)$ with $0 < s - 1 < 2s$, so $2s \nmid N - 1$. Thus every $\sigma_j = \pm 1$, and the vector k with $k_r = \sum_{j:\, r_j = r} \sigma_j$ satisfies $\sum_r k_r\, 2\sin\frac{\pi r}{N} = \sum_j \sigma_j\, 2\sin\frac{\pi r_j}{N} = 0$, i.e. $k \in \Lambda_N$, with

$$\sum_r k_r = \sum_{j=0}^{p-1} \sigma_j \equiv p \equiv 1 \pmod 2.$$

So $\sum_r k_r$ is odd: then $\sum_r k_r \not\equiv 0\ (4)$, and $\sum_r k_r + 2\sum_r r k_r$ is also odd, hence $\not\equiv 0\ (4)$. The single relation k violates *both* branches of Theorem 11, so neither the node $j = 0$ nor (for even N) the node $j = N/2$ can saturate, and $A_N < U_N$ (`verify_classification_proof.py`). □

The composite gap. For every composite N, then, $A_N < U_N$ (Theorem 14). The gap is *certified* for $N = 6$: the lone relation $2\omega_1 = \omega_3$ has coefficient-sum $2 + 0 - 1 \equiv 3\ (4) \ne 0$, so by Theorem 11 $A_6 < U_6$; quantitatively a Lipschitz-grid certificate on the two-dimensional subtorus—the elementary level of the Lasserre moment–SOS hierarchy [10]—brackets $A_6 \in [0.5594, 0.5604]$, so $U_6 - A_6 \ge 0.048 > 0$, while the prime $N = 5$ returns $A_5 = U_5$ (`verify_sos_certified.py`).

Proposition 15 (twice a prime: the sharp constant on a composite family). *For $N = 2p$ with p an odd prime, $A_N = U_N - O(1/N) = \frac{1}{\pi}\ln(2p) + O(1)$; the sharp constant $1/\pi$ holds, extending Theorems 3–4 to an infinite family of* composite *N.*

Proof. The prefix $\{\omega_1, \dots, \omega_{p-1}\} = \{2\sin(\pi r/2p) : 1 \le r \le p-1\}$ is $\mathbb{Q}$-independent: in $\mathbb{Q}(\zeta_{4p})$, where $\Phi_{4p}(x) = \Phi_p(-x^2)$ has degree $\varphi(4p) = 2(p-1)$, the values $2\mathrm{i}\sin(\pi r/2p) = \zeta_{4p}^r - \zeta_{4p}^{-r}$ $(1 \le r \le p-1)$ are linearly independent over $\mathbb{Q}$. Indeed, if $\sum_{r=1}^{p-1} c_r(\zeta_{4p}^r - \zeta_{4p}^{-r}) = 0$ with $c_r \in \mathbb{Q}$, multiplying by ζ_{4p}^{p-1} gives $Q(\zeta_{4p}) = 0$, where $Q(x) = \sum_{r=1}^{p-1} c_r(x^{p-1+r} - x^{p-1-r})$ has degree $\le 2p - 2 = \deg \Phi_{4p}$; as Φ_{4p} is the minimal polynomial of ζ_{4p}, this forces $Q = C\,\Phi_{4p}$, but $Q(1) = 0$ while $\Phi_{4p}(1) = \Phi_p(-1) = 1$, so $C = 0$ and $Q \equiv 0$, and since the exponent sets $\{p-1+r\}_r$ and $\{p-1-r\}_r$ are disjoint, every $c_r = 0$ (exact integer rank $= p - 1$, `verify_2p_family.py`). Thus the sole $\mathbb{Q}$-dependent frequency is the rational $\omega_p = 2$, of weight $\frac{1}{2N}$. With the degeneracy $\omega_r = \omega_{N-r}$ the node-0 response is $u_0(t) = \frac{2}{N}\sum_{r=1}^{p-1} \omega_r^{-1} \sin(\omega_r t) + \frac{1}{2N}\sin 2t$. By Kronecker the independent prefix can be driven to $\sin(\omega_r t) \to 1$ simultaneously, and since $\frac{2}{N}\sum_{r=1}^{p-1}\omega_r^{-1} = \frac{1}{N}\sum_{r=1}^{p-1}\csc\frac{\pi r}{2p} = U_N - \frac{1}{2N}$, we get $\sup_t u_0(t) \ge U_N - \frac{1}{N}$. As $A_N \le U_N$ always, $A_N \in [U_N - \frac{1}{N}, U_N]$ (`verify_2p_family.py`). □

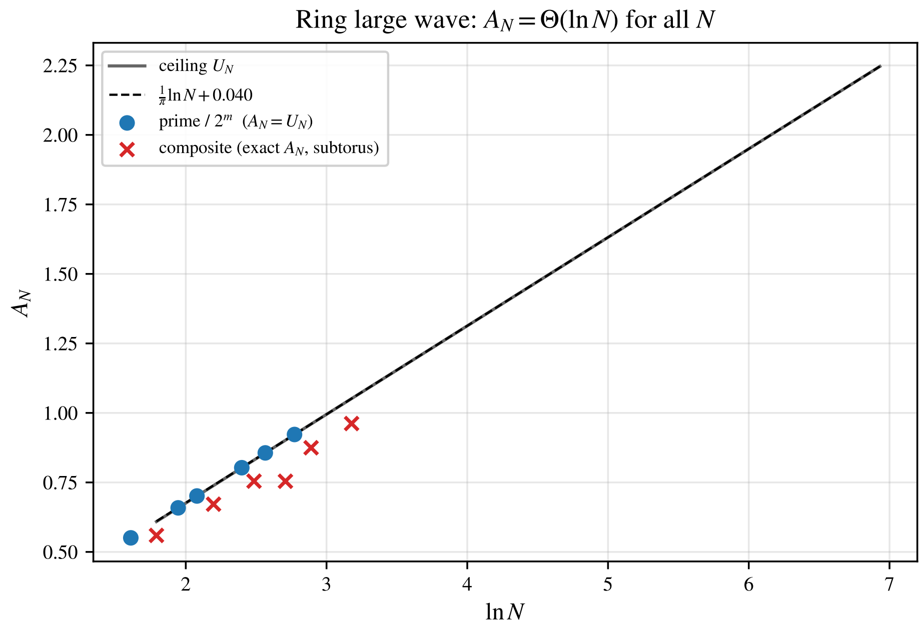


FIGURE 3. Ring large wave A_N versus $\ln N$. The ceiling U_N (Lemma 2) grows as $\frac{1}{\pi}\ln N + 0.040$; for primes and $N = 2^m$ the supremum equals U_N (Theorems 3, 4); for composite N the (numerically evaluated) subtorus value lies slightly below, and $A_N \sim \frac{1}{\pi}\ln N$ for every N (Theorem 6).

9.1. **Weyl equidistribution and the lower-bound problem.** The orbit-closure optimization is rigorous via Weyl's equidistribution theorem [2]. The frequency vector $\omega = (\omega_1, \ldots, \omega_m)$ defines the linear flow $t \mapsto t\omega \bmod 2\pi$, whose closure is the subtorus

$$\mathbb{T}_N = \big\{\varphi \in \mathbb{T}^m : \langle k, \varphi\rangle \equiv 0 \ (2\pi)\ \forall k \in \Lambda\big\}, \qquad \Lambda = \{k \in \mathbb{Z}^m : \langle k, \omega\rangle = 0\},$$

of dimension $\dim \mathbb{T}_N = m - \operatorname{rank}\Lambda = \operatorname{rank}_{\mathbb{Q}}\{\omega_r\}$, the $\mathbb{Q}$-rank of Section 8. This rank has a closed form, valid for *all* N.

Theorem 16 (subtorus dimension). *For every $N \geq 3$,*

$$\dim \mathbb{T}_N = \operatorname{rank}_{\mathbb{Q}}\{2\sin(\tfrac{\pi r}{N}) : 1 \leq r \leq N-1\} = \tfrac{1}{2}\varphi(2N). \tag{13}$$

Proof. Write $\zeta = \zeta_{2N}$ and let $\sigma : \zeta \mapsto \zeta^{-1}$ be complex conjugation, an involution of $\mathbb{Q}(\zeta)$ whose fixed field is the maximal real subfield $\mathbb{Q}(\zeta + \zeta^{-1})$, of degree $\frac{1}{2}\varphi(2N)$. In characteristic 0, $\mathbb{Q}(\zeta) = V^+ \oplus V^-$ where $V^\pm$ are the ± 1-eigenspaces of σ; thus $\dim_{\mathbb{Q}} V^\pm = \frac{1}{2}\varphi(2N)$, and since σ is the restriction of conjugation, V^- is exactly the set of purely imaginary elements of $\mathbb{Q}(\zeta)$. Now $2\mathrm{i}\sin(\pi r/N) = \zeta^r - \zeta^{-r} = (1-\sigma)\zeta^r$. For any y, $\sigma((1-\sigma)y) = -(1-\sigma)y$, so $(1-\sigma)\mathbb{Q}(\zeta) \subseteq V^-$; and $(1-\sigma)x = 2x$ for $x \in V^-$, so the inclusion is an equality. The powers $\{\zeta^r : r \in \mathbb{Z}\}$ span $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$ (they contain the power basis $1, \zeta, \ldots, \zeta^{\varphi(2N)-1}$), hence their images $\{\zeta^r - \zeta^{-r}\}$ span $(1-\sigma)\mathbb{Q}(\zeta) = V^-$. Finally $\zeta^N = -1$ gives $\zeta^{N+s} - \zeta^{-(N+s)} = -(\zeta^s - \zeta^{-s})$ while $r = 0, N$ give 0, so the values for $1 \leq r \leq N-1$ already span V^-. Dividing by $2\mathrm{i}$ is a $\mathbb{Q}$-linear bijection of $V^- \subset \mathrm{i}\mathbb{R}$ onto $\operatorname{span}_{\mathbb{Q}}\{\sin(\pi r/N)\} \subset \mathbb{R}$, which therefore has dimension $\dim_{\mathbb{Q}} V^- = \frac{1}{2}\varphi(2N)$. □

The formula is independently confirmed by an exact integer-rank certificate for $N \leq 64$ (`verify_qrank_formula.py`). Hence full rank—and therefore $A_N = U_N$—holds *exactly* when

$\frac{1}{2}\varphi(2N) = \lfloor N/2 \rfloor$, i.e. for N prime or $N = 2^m$ (Figure 4): precisely the scope of Theorems 3 and 4, now the full-rank case of one formula. Hence (12) is *exactly*

$$A_N = \max_j \max_{\varphi \in \mathbb{T}_N} \frac{2}{N} \sum_r \cos \frac{2\pi r j}{N} \sin \varphi_r .$$

For any S with $\{\omega_r\}_{r \in S}$ $\mathbb{Q}$-independent, the projection of $\mathbb{T}_N$ onto the S-coordinates is the full torus $\mathbb{T}^{|S|}$, so those phases align and contribute $(2/N) \sum_{r \in S} \omega_r^{-1}$. Taking $S = \{1, \dots, M(N)\}$ the independent prefix, $M(N) = \frac{1}{2}\varphi(2N)$ by Lemma 5, this is the budget L_{pre} of the order Theorem 6.

The lower bound, settled. The earlier obstruction—that the out-of-prefix modes might interfere destructively—is dispatched by the crude bound $\sin \geq -1$: it costs at most $U_N - L_{\text{pre}}$, and because Lemma 5 makes the prefix *long* ($M(N) = \frac{1}{2}\varphi(2N)$, within a $\ln \ln N$ factor of $N/2$), the surviving amount $2L_{\text{pre}} - U_N = (1 - o(1))U_N$ still grows like $\frac{1}{\pi} \ln N$. *Soft* bounds had failed here for a structural reason—the time-averaged second moment gives only $\text{RMS}_t \to 1/\sqrt{12}$ and the slowest single mode only $b_1 \to 1/\pi$, both $O(1)$—so the $\ln N$ genuinely needed the simultaneous alignment of $\Theta(\ln N)$ *arithmetically independent* modes, which Lemma 5 supplies. Table 4 shows the resulting band: the exact $A_N / \ln N$ stays in $[0.28, 0.34]$ across all classes (including the hardest $N = pq$, $p, q \sim \sqrt{N}$, where $A_N / U_N \geq 0.98$), consistent with Theorem 6.

TABLE 4. Illustration of Theorem 6. Left: the exact node-0 amplitude keeps $A_N / \ln N$ tightly banded. Right: the $\mathbb{Q}$-independent prefix contributes $L_{\text{pre}}(N) = \frac{1}{N} \sum_{r \leq M(N)} \csc \frac{\pi r}{N}$, exceeding $0.26 \ln N$ even for highly composite N (now a *proven* ingredient: $M(N) = \frac{1}{2}\varphi(2N)$ frequencies are independent, Lemma 5). $U_N / \ln N \to 1/\pi = 0.3183$ (`verify_conj_order_map.py`).

N	class	A_N/U_N	$A_N/\ln N$	N	$M(N)$	$L_{\text{pre}}/\ln N$
5	prime	1.000	0.342	30	8	0.261
24	composite	0.915	0.303	120	32	0.274
60	composite	0.873	0.286	210	48	0.268
64	2^m	1.000	0.328	240	64	0.279
90	composite	0.875	0.286	300	80	0.281

10. Reachability of the ceiling: finite time and disorder

The supremum A_N is an *infinite-time* quantity. Two refinements bring out its physical content: how long one must wait, and how fragile the arithmetic that makes the wait short.

10.1. **Finite-time amplification.** Define the finite-horizon amplitude

$$A_N(T) = \sup_{0 \leq t \leq T} \max_j |u_j(t)| \leq A_N .$$

Reaching $(1 - \varepsilon)U_N$ requires $\sin(\omega_r t) \approx 1$ for all $m = \lfloor N/2 \rfloor$ frequencies at once—a simultaneous Diophantine approximation of the phases to $\frac{\pi}{2}$. By Dirichlet, such an alignment to tolerance δ *exists* by time $\sim \delta^{-m}$—a crude exponential *upper* bound on the recurrence time

$$T_\varepsilon(N) = \inf\{T : A_N(T) \geq (1 - \varepsilon)U_N\}.$$

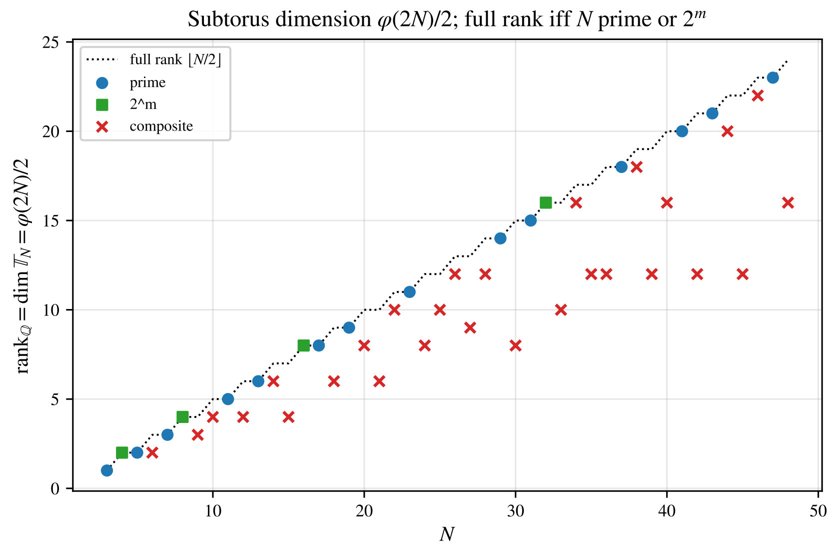


FIGURE 4. The subtorus dimension $\frac{1}{2}\varphi(2N)$ (13). Primes and powers of two attain the full rank $\lfloor N/2 \rfloor$ (dotted), so $A_N = U_N$; composite N fall below (the sub-torus defect).

(This targets U_N, so $T_\varepsilon(N)$ is finite only on the saturating classes N prime and 2^m; for non-saturating N, where $A_N < U_N$, one replaces U_N by A_N.) A matching *lower* bound (that one must wait at least exponentially long) does *not* follow from Dirichlet, which bounds the wait from above; it would require Diophantine lower bounds on the badly-approximable phase vector, and is left open. Numerically the wait does grow exponentially: for primes $\log T_{0.2}(N)$ grows by ≈ 0.63 per unit N (`verify_finite_time.py`, Figure 5), and at a *fixed* horizon T the efficiency $A_N(T)/U_N$ *decreases* with N (from ≈ 1.0 at $N = 5$ to ≈ 0.70 at $N = 29$ for $T = 300$): more frequencies are harder to align in bounded time. Thus the logarithmic law is an infinite-time ceiling; a physical large wave is the partial early alignment, already a several-fold over-amplification, while the full U_N is approached only over astronomically long times. The clean arithmetic of prime and 2^m N is what makes a sizable fraction of U_N reachable *quickly*.

10.2. **Disorder.** Unequal masses $m_j = 1 + \varepsilon\xi_j$ turn the chain into the generalized problem $L_N v = \omega^2 M v$, $M = \operatorname{diag}(m_j)$, with node-0 response $x_0(t) = \sum_r c_r \sin(\omega_r t)$, $c_r = v_r(0)^2/\omega_r > 0$ (M-orthonormal modes $v_r^\top M v_s = \delta_{rs}$, unit velocity impulse) and incoherent ceiling $C = \sum_r c_r$. Generic masses destroy every rational relation almost surely (each relation is a nontrivial real-analytic condition on the masses, and a countable union of such conditions still has measure zero), so the obstruction of Theorem 11 disappears and *every* N can in principle reach its ceiling. We verify (`verify_disorder.py`) that the exact resonance residual of a composite N jumps from 0 to $O(\varepsilon)$ under disorder, and that over a long horizon the achievable efficiency A/C rises above the blocked ordered value (e.g. $N = 15$: $0.83 \to 0.91$). But the gain is purely infinite-time: at a short horizon the same disorder yields no improvement, because the now-independent but near-resonant frequencies need Diophantine time to align. The dichotomy is sharp: an infinitesimal perturbation makes the absolute ceiling attainable in principle while pushing the time to attain it beyond any physical horizon. Infinite-time amplification improves; finite-time amplification does not.

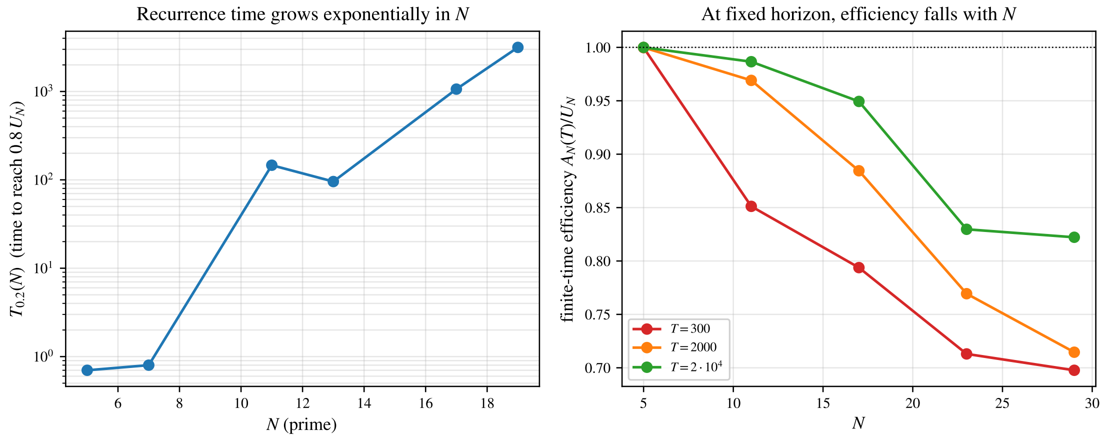


FIGURE 5. Reachability of the ceiling. Left: the recurrence time $T_{0.2}(N)$ to reach $0.8\,U_N$ appears to grow exponentially over the tested prime range (log scale). Right: at a fixed horizon T the finite-time efficiency $A_N(T)/U_N$ *decreases* with N—more frequencies are harder to align in bounded time. The logarithmic law is an infinite-time ceiling.

11. The discrete Schrödinger equation on the ring

Following Filimonov [32], consider $\mathrm{i}\dot{z} = L_N z$, so $z(t) = \mathrm{e}^{-\mathrm{i}tL_N} z(0)$ with kernel

$$K_N(k,t) = \frac{1}{N}\sum_r \mathrm{e}^{-\mathrm{i}t\lambda_r}\,\mathrm{e}^{2\pi\mathrm{i}rk/N}.$$

Here the phases are governed by $\lambda_r = 2 - 2\cos\frac{2\pi r}{N}$, not by $\omega_r = \sqrt{\lambda_r}$.

Theorem 17 (Schrödinger contrast). *(1) (Unitarity.) For $z(0) = e_0$, $\max_j |z_j(t)| \le 1$ for all t: a localized state is never amplified, in sharp contrast with the bead chain.*

(2) (Ballistic ℓ^∞ amplification.) $B_N := \sup_t \|\mathrm{e}^{-\mathrm{i}tL_N}\|_{\ell^\infty\to\ell^\infty} = \sup_t \sum_k |K_N(k,t)| \le \sqrt{N}$ (each row of the unitary $\mathrm{e}^{-\mathrm{i}tL_N}$ has unit ℓ^2 norm, so ℓ^1 norm $\le \sqrt{N}$). Together with the matching lower bound (Theorem 19), $B_N = \Theta(\sqrt{N})$, not $\ln N$.

(3) (Arithmetic.) The spectrum lies in the real cyclotomic field $\mathbb{Q}(\zeta_N)^+$ via $\cos$ (whereas $\{\omega_r\} \subset \mathbb{Q}(\zeta_{2N})$ via $\sin$); the affine identity $\sum_r \lambda_r = 2N$ holds always, and $\operatorname{rank}_{\mathbb{Q}}\{\lambda_r\}$ is strictly smaller than $\operatorname{rank}_{\mathbb{Q}}\{\omega_r\}$ for $N = 2^m$ (e.g. 4 vs 8 at $N = 16$).

Lemma 18 (Bessel ℓ^1 asymptotics). *As $x \to \infty$, $\displaystyle\sum_{n\in\mathbb{Z}} |J_n(x)| = c_0\sqrt{x}\,(1 + o(1))$ with $c_0 = (2/\pi)^{3/2} B(\frac{1}{2},\frac{3}{4}) \approx 1.217$.*

Proof. Split at the light cone $|n| = x$. Beyond it ($|n| \ge x(1+\delta)$) the bound $|J_n(x)| \le (x/2)^{|n|}/|n|!$ gives super-exponential decay; the turning-point band $|n| = x + O(x^{1/3})$ carries $O(x^{1/3})$ terms of size $O(x^{-1/3})$, contributing $O(1)$; and the shell $(1-\delta)x \le |n| \le x$ contributes, by the integrable amplitude bound below, $O(\delta^{3/4}\sqrt{x})$—all $o(\sqrt{x})$ after $\delta \to 0$. In the oscillatory region $|n| \le (1-\delta)x$ the Debye expansion [6] holds uniformly,

$$J_n(x) = \sqrt{\tfrac{2}{\pi\sqrt{x^2-n^2}}}\,\big(\cos\Phi_n + O(x^{-1})\big), \qquad \Phi_n = \sqrt{x^2-n^2} - n\arccos\tfrac{n}{x} - \tfrac{\pi}{4},$$

and the phase increments $\Phi_{n+1} - \Phi_n = -\arccos(n/x)$ vary with n, so the $\{\Phi_n\}$ equidistribute and $|\cos\Phi_n|$ averages to $\langle|\cos|\rangle = 2/\pi$ (van der Corput). Hence

$$\sum_{|n|\le(1-\delta)x} |J_n(x)| = (1+o(1))\,\frac{2}{\pi}\sqrt{\tfrac{2}{\pi}}\int_{-x}^{x}\frac{dn}{(x^2-n^2)^{1/4}} = (1+o(1))\,(2/\pi)^{3/2}B(\tfrac12,\tfrac34)\,\sqrt{x},$$

the integral equalling $\sqrt{x}\,B(\frac12,\frac34)$ (set $n = x\sin\theta$). Letting $\delta\to 0$ gives the claim; the leading constant is confirmed to the digits of Table 5. □

Theorem 19 (Schrödinger lower bound). *$\liminf_N B_N/\sqrt{N} \ge c_0/\sqrt{2}$ with $c_0 = (2/\pi)^{3/2}B(\frac12,\frac34)$, i.e. $\liminf_N B_N/\sqrt{N} \ge 0.8607$. With Theorem 17(2), $B_N = \Theta(\sqrt{N})$.*

Proof. Evaluate the ℓ^1 mass at the aliasing-free time $t = \frac{N}{4}(1-\delta)$, so $2t = \frac{N}{2}(1-\delta)$. By the Poisson periodization (16), $K_N(j,t) = \sum_\ell K_\infty(j+\ell N, t)$ with $|K_\infty(n,t)| = |J_n(2t)|$. The Bessel front sits at $|n| \approx 2t = \frac{N}{2}(1-\delta)$, and the nearest aliased copy ($\ell \ne 0$) lies at $|j+\ell N| \ge N - 2t = \frac{N}{2}(1+\delta) > 2t$, where the bound $|J_n(2t)| \le t^{|n|}/|n|!$ forces super-exponential decay; so $\sum_j |K_N(j,t)| = (1+o(1))\sum_{n\in\mathbb{Z}}|J_n(2t)|$. By Lemma 18 this is $(1+o(1))\,c_0\sqrt{2t} = (1+o(1))\,c_0\sqrt{N/2}\,\sqrt{1-\delta}$; letting $\delta\to 0$, $\liminf_N B_N/\sqrt{N} \ge c_0/\sqrt{2}$. The upper bound $B_N \le \sqrt{N}$ is Theorem 17(2) (`verify_bn_parity.py`). □

The improvement over the cruder $t = N/8$ value $\frac12 c_0 = 0.6086$ is the optimal aliasing-free time $t = N/4$, at which the Bessel front exactly fills one half-period.

Remark 20 (the constant has no limit—it splits by parity). $B_N/\sqrt{N}$ does *not* converge: the aliasing phase i^{-N} is real for even N and imaginary for odd N, so where the two overlapping fronts meet they add differently. Numerically (`verify_bn_parity.py`) the sup is at $t = N/4$ for even N, giving $B_N/\sqrt{N} \to c_0/\sqrt{2}$ (the value above), and at $t = N/2$ for odd N, where the two surviving aliasing copies give $|K_N(k,N/2)| \approx \sqrt{J_k(N)^2 + J_{N-k}(N)^2}$ across the front (no cancellation, since i^{-N} is imaginary). The Debye envelope $|J_{Nu}(N)| \sim \sqrt{2/\pi N}\,(1-u^2)^{-1/4}|\cos\Phi|$ then yields the limiting value

$$\beta_{\mathrm{odd}} = \sqrt{\tfrac{2}{\pi}}\int_0^1 \mathbb{E}_{\Phi,\Psi}\sqrt{\frac{\cos^2\Phi}{\sqrt{1-u^2}} + \frac{\cos^2\Psi}{\sqrt{u(2-u)}}}\,du = 0.928019303668909\ldots, \tag{14}$$

an elliptic-integral average (Φ,Ψ independent uniform). Computed to 25 digits, it matches no combination of $\pi, \mathrm{e}^\gamma, \Gamma(\frac14), \sqrt{2}$, Catalan's constant, ..., and is not a low-degree algebraic number (an integer-relation/PSLQ search returns nothing): β_{odd} has no elementary closed form—it is a new constant defined by (14) (`verify_beta_odd.py`). Evaluation at $t = N/2$ is rigorous, so $\liminf_N B_N/\sqrt{N} = c_0/\sqrt{2}$ and $\liminf_{\mathrm{odd}\,N} B_N/\sqrt{N} \ge \beta_{\mathrm{odd}}$, whence $\limsup_N B_N/\sqrt{N} \in [\beta_{\mathrm{odd}}, 1]$. For the upper side, the matching bound holds on the whole window $t \le N/2$. There $2t \le N$, so only the two Poisson copies $\ell = 0, -1$ survive (the rest are super-exponentially small), $|K_N(k,t)| \le |J_k(2t)| + |J_{N-k}(2t)|$, and the Debye envelope turns $\|K_N(\cdot,t)\|_1/\sqrt{N}$ into the two-copy profile $F(2t/N)$, which is maximized at $2t/N = 1$—i.e. at $t = N/2$—with value β_{odd} (`verify_bn_parity.py`). Hence $\|K_N(\cdot,t)\|_1 \le (\beta_{\mathrm{odd}} + o(1))\sqrt{N}$ for all $t \le N/2$, and with the lower bound $\sup_{t\le N/2}\|K_N(\cdot,t)\|_1 = (\beta_{\mathrm{odd}} + o(1))\sqrt{N}$ on the odd subsequence. For $t > N/2$ (three or more overlapping copies) only unitarity $\|K_N(\cdot,t)\|_1 \le \sqrt{N}$ is rigorous, but numerically the ℓ^1 mass there falls back toward its time average $\approx 0.79\sqrt{N}$ and never exceeds the $t = N/2$ value: the $\sin^2$ dispersion admits no flat (Gauss-sum) revival, even at structured Talbot times $t = \pi p/q$. This is consistent with the dispersive-quantization picture—exact revivals of the periodic Schrödinger flow occur only for a *quadratic* dispersion (rational times give Gauss sums) [7], which the discrete $\lambda_r = 4\sin^2(\pi r/N)$ is not. So the global sup sits at $t = N/2$ and $\limsup_N B_N/\sqrt{N} = \beta_{\mathrm{odd}}$. The residual on $t > N/2$ has a clean form: $K_N(\cdot, t)$ is the inverse DFT of the *unimodular* symbol $(\mathrm{e}^{-\mathrm{i}t\lambda_r})_r$—a chirp, since $2t\cos(2\pi r/N)$ is locally quadratic—so $B_N/\sqrt{N}$ is the maximal L^1-flatness ratio $\|P\|_1/\|P\|_2$ of a unimodular trigonometric polynomial. The limiting macroscopic profile of $\|K_N(\cdot,t)\|_1/\sqrt{N}$ is a *multi-copy* Debye average over the Poisson copies (`verify_bn_residual_profile.py`: it matches the FFT to 0.5% across all $N \bmod 4$, peaks at $t = N/2$ with β_{odd}, and is strictly subdominant for $t > N/2$, stably out to $t = 5N$), reducing the bound to the joint equidistribution $\bmod\, 2\pi$ of the coherent multi-copy Debye phases (a multidimensional Weyl sum). The difficulty is intrinsic: the *quadratic* chirp (Gauss–Fresnel/Talbot) is L^1-flat with ratio $\to 1$, and ultraflat unimodular polynomials exist [8], so $\limsup < 1$ is an *anti*-flatness statement particular to the non-quadratic $\sin^2$ dispersion—no generic argument supplies it. (A single-period saddle van der Corput estimate fails: it predicts 0.61 against the actual 0.93, missing the Poisson-copy coherence.)

Part (1) is unitarity of $\mathrm{e}^{-\mathrm{i}tL_N}$; the bounds in (2) are now both rigorous (upper from unitarity, lower from Theorem 19), and every step—the periodization, the negligible aliasing at $t = \frac{N}{4}(1-\delta)$, the $\sqrt{N}$ growth with constant $c_0/\sqrt{2}$, and the $c_0\sqrt{x}$ Bessel asymptotics—is reproduced numerically (`verify_schrodinger_lower.py`, `verify_bn_parity.py`, `verify_bessel_kernel.py`). The parity split is mapped in Table 5.

TABLE 5. The constant in $B_N = \Theta(\sqrt{N})$, by parity (Remark 20). A refined t-scan gives $B_N/\sqrt{N}$: even N approach $c_0/\sqrt{2} = 0.8607$ (sup at $t = N/4$), odd N approach $\beta_{\mathrm{odd}} \approx 0.928$ (sup at $t = N/2$). All lie in the rigorous band $[c_0/\sqrt{2}, 1]$; the proved $\liminf$ is $c_0/\sqrt{2}$ (`verify_bn_parity.py`).

N	$16, 15$	$64, 63$	$256, 255$	$1024, 1023$	$4096, 4095$	limit
even N, $B_N/\sqrt{N}$	0.923	0.907	0.888	0.878	0.869	$c_0/\sqrt{2} = 0.861$
odd N, $B_N/\sqrt{N}$	0.981	0.960	0.950	0.940	0.935	$\beta_{\mathrm{odd}} \approx 0.928$

11.1. **Bessel kernel and Poisson summation.** On the infinite chain $\mathbb{Z}$ the propagator kernel follows from the symbol by a contour integral,

$$K_\infty(n,t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{i}n\theta}\,\mathrm{e}^{-\mathrm{i}t(2-2\cos\theta)}\,d\theta = \mathrm{e}^{-2\mathrm{i}t}\,\mathrm{i}^n J_n(2t), \tag{15}$$

by the Jacobi–Anger identity $\frac{1}{2\pi}\int \mathrm{e}^{\mathrm{i}(n\theta + x\cos\theta)}\,d\theta = \mathrm{i}^n J_n(x)$ [6]. Thus $|K_\infty(n,t)| = |J_n(2t)|$: a ballistic light cone $|n| \approx 2t$ with an Airy front of height $\max_n |J_n(2t)| \sim 0.6748\,(2t)^{-1/3}$ (Figure 6). Unitarity gives $\sum_n |K_\infty|^2 = 1$, while the ℓ^1 mass $\sum_n |J_n(2t)| \sim c\sqrt{t}$ grows—the infinite-chain shadow of $B_N \sim \sqrt{N}$. The ring kernel is the Poisson periodization

$$K_N(j,t) = \sum_{\ell\in\mathbb{Z}} K_\infty(j + \ell N, t), \tag{16}$$

verified to machine precision; the large wave survives only through this finite-N folding, since on $\mathbb{Z}$ the front escapes and the peak decays.

11.2. **Arithmetic dynamics: revivals and Gauss sums.** The recurrence structure of $\mathrm{e}^{-\mathrm{i}tL_N}$ is arithmetic. Since $2\cos(2\pi/N)$ is rational only for $N \in \{1,2,3,4,6\}$ (Niven's theorem [3]; for general r, $2\cos(2\pi r/N)$ is rational only when r/N reduces to such a denominator), the tight-binding spectrum is generically incommensurate: $\mathrm{e}^{-\mathrm{i}tL_N}$ has *no finite revival period*, and the return amplitude $|K_N(0,t)|$ is almost-periodic, approaching 1 along a wandering sequence of near-revival times

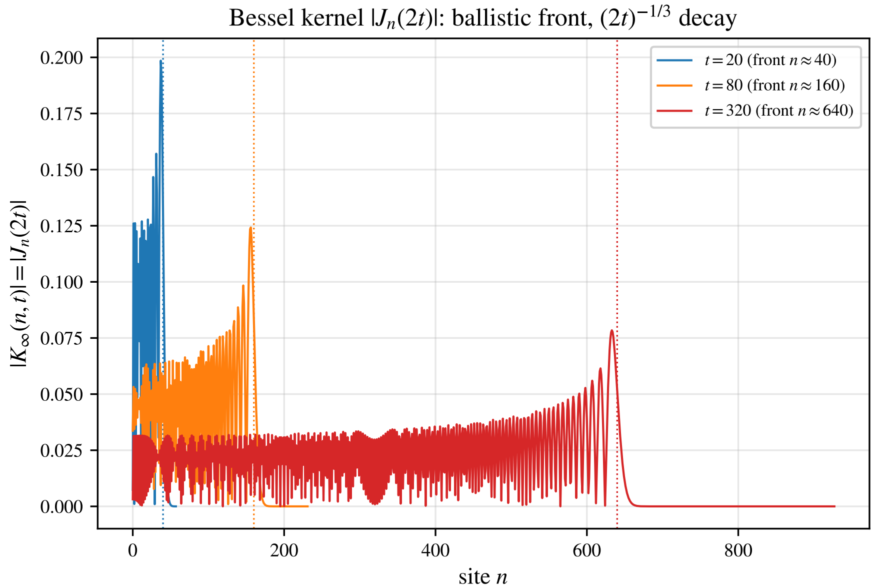


FIGURE 6. Infinite-chain Schrödinger kernel $|K_\infty(n,t)| = |J_n(2t)|$: a ballistic front at $n \approx 2t$ (dotted) whose height decays as $(2t)^{-1/3}$. On the ring this kernel is periodized, (16); the effect is intrinsically finite-size.

(`verify_revivals_gauss.py`). In the long-wave limit $\lambda_r \approx (2\pi r/N)^2$ the model approaches the *quadratic* dispersion of a free particle, where the dynamics becomes exactly arithmetic: at the Talbot times $t = 2\pi s/q$ (in the rescaled time $\tau = (2\pi/N)^2 t$ of this quadratic limit) the propagator kernel is a quadratic Gauss sum, and for odd q with $\gcd(a, q) = 1$,

$$\left| \sum_{n=0}^{q-1} \mathrm{e}^{2\pi \mathrm{i} a n^2/q} \right| = \sqrt{q},$$

yielding an exact fractional revival of uniform amplitude $|U_{jk}| = 1/\sqrt{q}$ (and a full revival $U(2\pi) = I$). The large wave thus inhabits the almost-periodic, Diophantine regime of the lattice, of which the discrete Talbot effect and Gauss-sum revivals are the exactly solvable continuum shadow.

11.3. **Dirichlet segment: a logarithmic site-amplitude law.** The sharpest large-wave statement for the discrete Schrödinger equation is Filimonov's [32], and it lives on the *open* (Dirichlet) chain rather than the ring. With on-site value $\varepsilon_j \equiv -2$ and pinned ends $z_0 = z_N = 0$,

$$\mathrm{i}\dot{z}_j = z_{j+1} - 2z_j + z_{j-1}, \qquad z_j(0) = \alpha_j, \qquad 1 \le j \le N-1,$$

the Dirichlet sine basis diagonalizes the flow and the solution is explicit,

$$z_j(t) = \mathrm{e}^{2\mathrm{i}t} \sum_{k=1}^{N-1} a_k \sin\frac{\pi k j}{N}\, \mathrm{e}^{-2\mathrm{i}\omega_k t}, \qquad \omega_k = \cos\frac{\pi k}{N}, \qquad a_k = \frac{2}{N} \sum_{r=1}^{N-1} \alpha_r \sin\frac{\pi k r}{N}. \tag{17}$$

The simple spectrum $\{\omega_k = \cos(\pi k/N)\}$ replaces the twofold-degenerate ring frequencies. For the uniform initial state $\alpha_j \equiv 1$ the coefficient sum is the imaginary part of a geometric series,

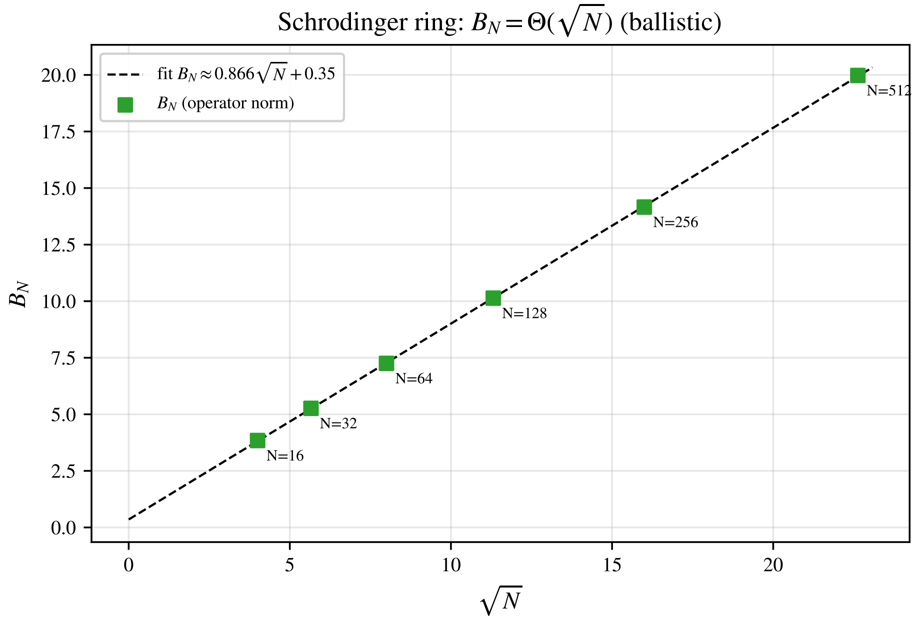


FIGURE 7. Discrete Schrödinger propagator: the $\ell^\infty \to \ell^\infty$ amplification B_N grows ballistically as $\Theta(\sqrt{N})$ (linear fit in $\sqrt{N}$, $R^2 = 1.0000$), unlike the $\ln N$ growth of the bead chain.

$\sum_{r=1}^{N-1} \sin(\pi kr/N) = \cot(\pi k/2N)$ for odd k and 0 for even k, so

$$a_k = \frac{2}{N} \cot \frac{\pi k}{2N} \ \ (k \text{ odd}), \qquad a_k = 0 \ \ (k \text{ even}). \tag{18}$$

The pointwise ceiling is $d_j^N := \sum_{k=1}^{N-1} \big| a_k \sin(\pi kj/N) \big|$, with $|z_j(t)| \le d_j^N$ for all t by the triangle inequality.

Theorem 21 (Filimonov [32]). *For $\alpha_j \equiv 1$ and $N = 2^m$, $\sup_t |z_j(t)| = d_j^N$, and*

$$\lim_{N\to\infty} d_j^N = C \ln j + O(1). \tag{19}$$

Thus the Schrödinger amplitude at the *observation site* j grows without bound (logarithmically) as $j \to \infty$—a large wave at the far end of the chain, approached along the relatively dense set of near-revival times (the modulus is almost-periodic, [1]). Two refinements complete the picture (`verify_filimonov_schrodinger.py`).

The constant is not new. It is the value Myshkis–Filimonov established for the *wave* problem: their segment ceiling satisfies $\zeta_j^N \to \frac{4}{\pi^2} \ln j + O(1)$ [25, Thm. 2], and the Schrödinger ceiling (18) coincides with it term for term (both equal $\frac{2}{N} \sum_{k \text{ odd}} \big|\cot \frac{\pi k}{2N} \sin \frac{\pi kj}{N}\big|$). The same limit is immediate here: as $N \to \infty$ the ceiling (18) becomes a Riemann sum for

$$d_j^\infty = \frac{2}{\pi} \int_0^{\pi/2} \cot\theta \, \big|\sin 2j\theta\big| \, d\theta, \tag{20}$$

and since $\cot\theta \sim \theta^{-1}$ near 0 while $\langle |\sin|\rangle = 2/\pi$, the integral grows like $\frac{2}{\pi} \cdot \frac{2}{\pi} \ln j$, giving

$$C = \frac{4}{\pi^2} \approx 0.4053. \tag{21}$$

The genuinely new content of Theorem 21 is the Schrödinger *attainment* $\sup_t |z_j| = d_j^N$ [32], not the constant. The fit $d_j^N = (4/\pi^2)\ln j + O(1)$ holds to $R^2 = 1.0000$ at $N = 2^{12}$ (Figure 8).

Independence and the scope of Theorem 21. The equality $\sup_t |z_j| = d_j^N$ rests on two facts: the sign-pairing $\omega_{N-k} = -\omega_k$ together with the elementary identity $\max(|x+y|, |x-y|) = |x| + |y|$ lets each pair reach $|a_k| + |a_{N-k}|$, and the distinct $|\omega_k|$ must be $\mathbb{Q}$-independent so that all pairs align simultaneously (Kronecker). This independence—the cosine analogue of Theorems 3–4, via $2\cos(\pi k/N) = \zeta_{2N}^k + \zeta_{2N}^{-k}$ in the *plus*-eigenspace—has full rank $\lfloor (N-1)/2 \rfloor$ exactly for N prime or $N = 2^m$ (`verify_filimonov_schrodinger.py`), which *suffices* for $\sup_t |z_j| = d_j^N$ at every site j; so Filimonov's $N = 2^m$ extends to prime N. Full rank is not *necessary*, however: composite N can still saturate at individual sites—numerically $\sup_t |z_j| = d_j^N$ already at $j = 1$ for $N = 6, 9, 10, 12$, the alignment there needing only the active frequencies and the sign-pairing—so a complete classification of composite saturation is open.

Three laws on one Laplacian. The discrete Laplacian thus carries several distinct large-wave asymptotics, one per dynamics and functional (Table 6): $A_N \sim \frac{1}{\pi}\ln N$ (ring, growth in the system size N), $B_N \sim \sqrt{N}$ (ring, $\ell^\infty \to \ell^\infty$ operator norm), and $d_j^N \sim \frac{4}{\pi^2}\ln j$ (Dirichlet segment, growth in the observation site j).

Table 6. The large-wave asymptotics carried by the discrete Laplacian, by dynamics and functional; every constant is reproduced by the accompanying scripts. "Sharp class" is where the leading constant is established (proved here; for d_j^N the constant is the published value of [25]).

quantity	dynamics / functional	leading asymptotic	sharp class
U_N	ring, velocity ceiling	$\frac{1}{\pi}\ln N + \frac{1}{\pi}(\gamma + \ln\frac{2}{\pi})$	all N
A_N	ring, $\sup_t \max_j \lvert G_N(j,\cdot)\rvert$	$\frac{1}{\pi}\ln N$	prime, 2^m, $2p$
B_N	ring, $\ell^\infty \to \ell^\infty$ Schrödinger	$\Theta(\sqrt{N})$, const $\in [0.6086, 1]$	all N
d_j^N	segment, $\sup_t \lvert z_j(\cdot)\rvert$ Schrödinger	$\frac{4}{\pi^2}\ln j$	prime, 2^m (suff.)
$U_N^{(d)}$	d-torus, velocity ceiling	$\frac{1}{\pi}\ln N$ (d=1), $O(1)$ ($d\geq 2$)	$d_s = 1$

12. Continuum analogue and the step initial condition

Expanding the symbol for small wavenumbers, $\omega(\theta) = 2\sin(\theta/2) = \theta - \theta^3/24 + O(\theta^5)$, recovers to leading order the continuum wave equation $u_{tt} = u_{xx}$ ($\omega = \theta$). A finite Taylor truncation improves the long-wave fit but diverges from the exact branch near the band edge $\theta = \pi$. The *two-point Padé* approximant of Andrianov–Awrejcewicz [19], obtained from the regularized model $m(1 - a^2h^2\partial_x^2)u_{tt} = ch^2u_{xx}$ with $a^2 = \frac{1}{4} - \pi^{-2} \approx 0.1487$, yields

$$\omega_{\text{Padé}}(\theta) = \frac{\theta}{\sqrt{1 + a^2\theta^2}}, \tag{22}$$

which matches the exact dispersion at *both* ends—unit slope at $\theta = 0$ and $\omega = 2$ at $\theta = \pi$ (indeed $\pi/\sqrt{1 + a^2\pi^2} = 2$)—with error below 3% in between (Figure 9). Nonetheless no continuum surrogate reproduces the large wave: as Section 11 shows through the decaying Bessel kernel, the effect is *absent* in the continuum/infinite limit—it is intrinsically a finite-size, discrete phenomenon driven by the exact arithmetic of the sampled frequencies $\{2\sin(\pi r/N)\}$.

For the step displacement $u_j(0) = 1$ ($1 \leq j \leq N-1$), $u_0(0) = 0$, $\dot{u}(0) = 0$, the exact supremum is, by Bohr's theorem (for $N = 2^m$), $\sup_t \max_j u_j(t) = \bar{\alpha} + \max_j \sum_r |A_{r,j}| \approx 2$ for all N, where $\bar{\alpha} =$

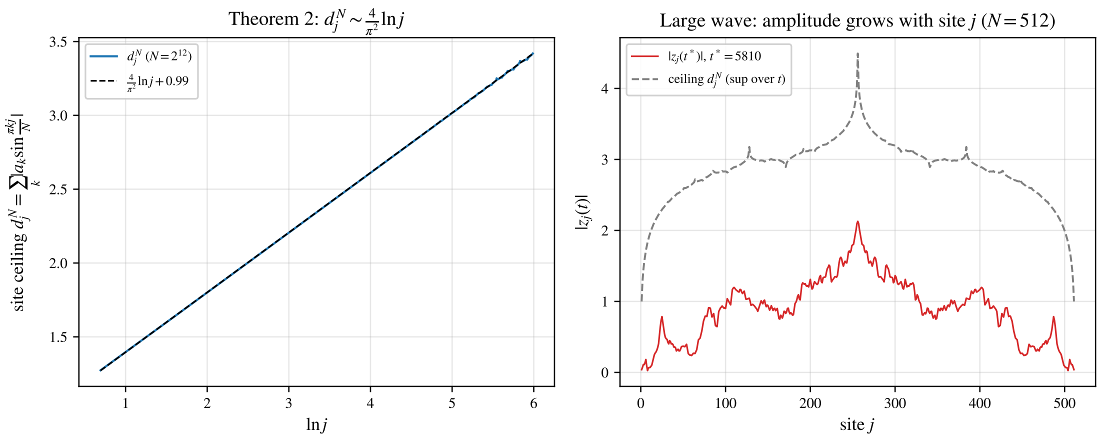


FIGURE 8. Filimonov's segment Schrödinger large wave (verify_filimonov_schrodinger.py). Left: the site ceiling d_j^N versus $\ln j$ $(N = 2^{12})$ on the line $\frac{4}{\pi^2}\ln j + O(1)$. Right: a snapshot $|z_j(t^*)|$ at a near-revival time grows with the site j toward the ceiling d_j^N $(N = 512)$.

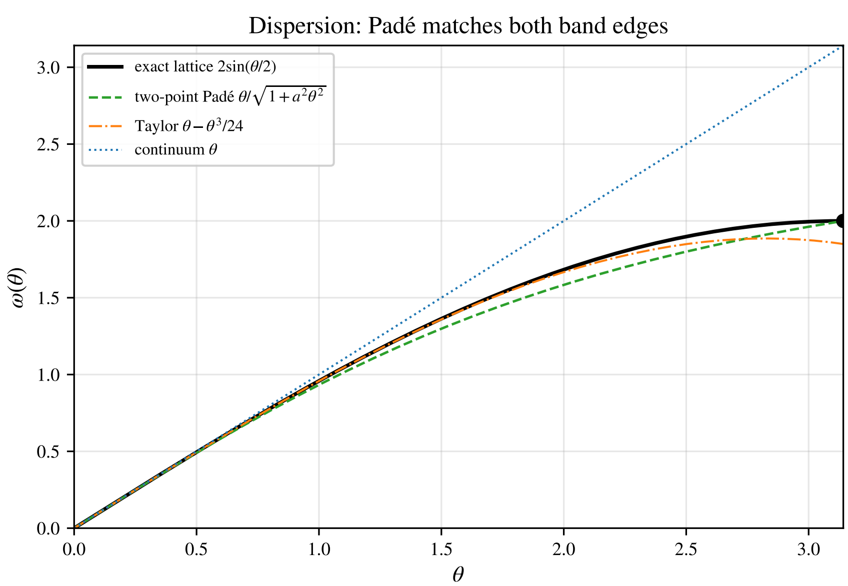


FIGURE 9. Dispersion relations. The exact lattice branch $2\sin(\theta/2)$ is matched at both ends by the two-point Padé (22), whereas the continuum θ and the Taylor truncation $\theta - \theta^3/24$ diverge at the band edge $\theta = \pi$ ($\bullet$, $\omega = 2$).

$(N-1)/N$ and $A_{r,j}$ are the modal amplitudes. Earlier numerical work reported peaks $1.617 \to 1.397$ for $N = 32, \ldots, 1024$; these are finite-time scan values, and their *decrease* with N is the expected worsening of the scan undershoot for an almost-periodic supremum (the true maximum is only approached, at near-recurrence times). The present analysis thus reproduces the effect and corrects the value to ≈ 2 (verify_thesis_peaks.py).

13. Numerical experiments

Table 7 reports both dynamics over the three families $N = 2^m$, prime, composite. The wave A_N is the orbit-closure value of Section 9 (exact reduction, evaluated numerically); it equals the ceiling U_N for prime and 2^m (Theorems 3, 4) and sits modestly below for composite N. The Schrödinger $B_N/\sqrt{N}$ stays $O(1)$ across all families. Figures 3 and 7 display the two growth laws—$\ln N$ for the wave, $\sqrt{N}$ for Schrödinger.

family	N	U_N	A_N (wave)	A_N/U_N	$B_N/\sqrt{N}$
2^m	16	0.922	0.922	1.000	0.960
	32	1.143	1.143	1.000	0.931
	64	1.364	1.364	1.000	0.906
prime	17	0.942	0.942	1.000	0.988
	31	1.133	1.133	1.000	0.971
	61	1.349	1.349	1.000	0.961
composite	12	0.831	0.755	0.908	0.971
	15	0.902	0.754	0.836	0.984
	24	1.052	0.962	0.915	0.941

Table 7. Both dynamics over the three families. Wave A_N (exact, Section 9) equals the Bohr ceiling $U_N \sim \frac{1}{\pi}\ln N$ for prime/2^m and is slightly smaller for composite N; the Schrödinger amplification has $B_N/\sqrt{N} = O(1)$, i.e. $B_N = \Theta(\sqrt{N})$ (`verify_experiments.py`).

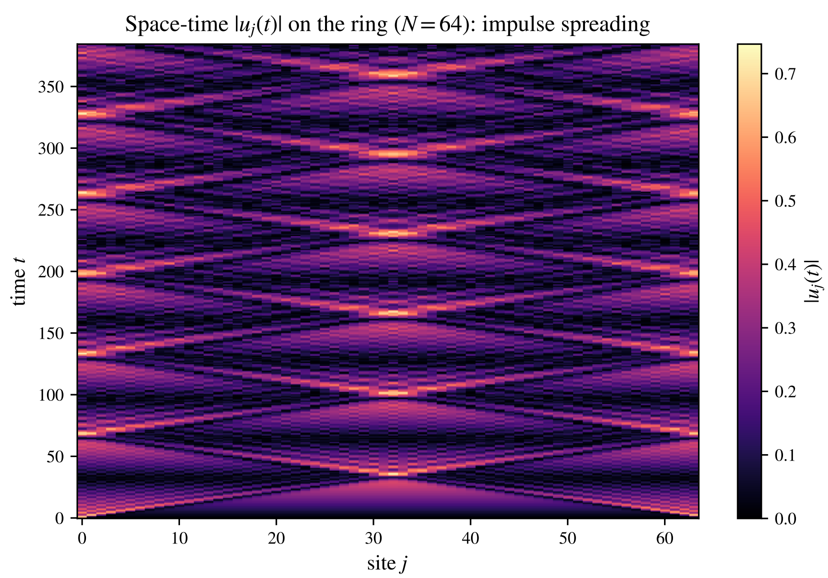


Figure 10. Space-time $|u_j(t)|$ for the unit impulse on the $N = 64$ ring: ballistic light cones, reflections off the periodic geometry, and the focusing recurrences that build the large wave.

14. Exact arithmetic and a machine-checked certificate

The arithmetic underlying Theorems 16 and 11 is verified independently of the floating-point experiments by two computer-algebra systems and one proof assistant; all agree with the Python core and with each other (`numerics/verify_exact_layer.py`).

Two exact reimplementations. In GAP's exact cyclotomic arithmetic (integer coordinates of $\zeta_{2N}^{r} - \zeta_{2N}^{-r}$ in $\mathbb{Q}(\zeta_{2N})$, integer kernel of the relation lattice) we recompute the $\mathbb{Q}$-rank and the mod-4 criterion and confirm $\mathrm{rank}_{\mathbb{Q}}\{2\sin(\pi r/N)\} = \frac{1}{2}\varphi(2N)$ and the saturation classification $A_N = U_N \Leftrightarrow N$ prime or 2^m for *all* $N \leq 800$—an exact, independent corroboration of Theorem 14 (`exact/gap/scan_classification.g`); the explicit obstruction k of that theorem is itself recertified for all composite $N \leq 600$ by `verify_classification_proof.py`. PARI/GP supplies a second, structurally different exact implementation of the same $\mathbb{Q}$-rank and checks the cyclotomic-degree identities $\Phi_{2p}(x) = \Phi_p(-x)$, $\Phi_{2^{m+1}}(x) = x^{2^m} + 1$ and $\Phi_{4p}(1) = \Phi_p(-1) = 1$ that drive Theorems 3, 4 and Proposition 15 (`exact/pari/crosscheck.gp`); FriCAS reproduces the csc Laurent series and the constants of Appendix B symbolically (`exact/fricas/asymptotics.input`).

Observation 22 (structure of the composite obstruction)**.** For $N = p^2$ with p an odd prime, the minimal mod-4 obstruction—the shortest frequency relation with coefficient-sum $\not\equiv 0\ (4)$—has support exactly p, on the comb of indices $\{1\} \cup \{kp \pm 1 : 1 \leq k \leq \frac{p-1}{2}\}$ with alternating ± 1 coefficients (Figure 11; `exact/gap/prime_square.g`). This is the $N = p^2$ case of the root-of-unity obstruction that proves Theorem 14: the comb is exactly the reduction of $\sum_{j=0}^{p-1} \sin \frac{\pi(1+2pj)}{p^2} = 0$, and it was the computer-algebra mining of this structure that suggested the general construction.

A machine-checked certificate. The combinatorial heart of Theorem 11—that the mod-4 verdict, evaluated on the relation lattices, matches the prime/2^m classification—is checked by the Rocq proof assistant (formerly Coq). The exact Λ_N computed by GAP are embedded as integer vectors, the criterion $\mathrm{saturates}(N, B) = \big(\forall k \in B : \sum_r k_r \equiv 0\big) \vee \big(N \text{ even} \wedge \forall k : \sum_r k_r + 2\sum_r r k_r \equiv 0\big) \pmod 4$ is defined over $\mathbb{Z}$, and

$$\texttt{mod4\_criterion\_matches\_prime\_2pow}: \quad \mathrm{map}\big(\lambda(N,B).\ \mathrm{saturates}\, N\, B\big)\ \mathit{certs} = \mathit{expected}$$

is proved by reflection (`vm_compute`), with no floating point anywhere (`formal/rocq/Mod4Criterion.v`). *Scope, stated honestly:* Rocq certifies the criterion's *evaluation* on the supplied lattices; that those lattices are genuine frequency relations rests on the cyclotomic theory (proved on paper, checked exactly by GAP/PARI), which is not itself formalized. The general classification is Theorem 14 (proved above); Rocq adds a floating-point-free check of the criterion's evaluation, not a reproof. A second Rocq file (`formal/rocq/Classification.v`) goes further and certifies the *number-theoretic core* of Theorem 14 *uniformly in N* (not a scan): the crux $(ps - 1) \bmod 2s = s - 1$ that keeps the constructed indices off $\{0, N\}$, and the capstone that an odd-length ± 1 relation violates both mod-4 branches. The same file also formalizes the arithmetic core of the order Theorem 6—the palindromic/anti-palindromic clash (`pal_and_antipal_zero`) and the index bound $K \geq \frac{1}{2}\varphi + 1$ (`prefix_index_bound`) behind Lemma 5. Only the cyclotomic input—that the constructed vector is a genuine relation, and that $\Phi_{2N} \mid Q$—is supplied from outside (exactly, by GAP/PARI and the `verify_*` scripts).

15. Reproducibility

Every claim above is reproduced by a standalone script (run via `uv run -script`); all pass.

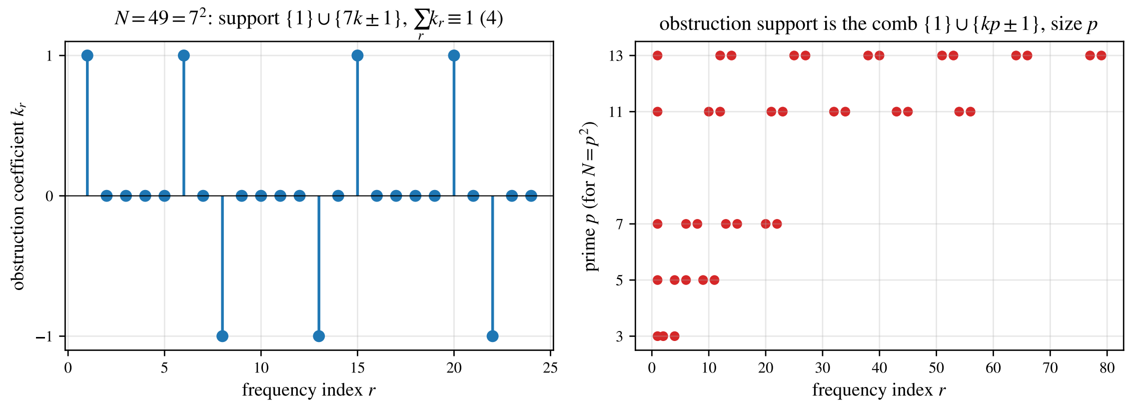


FIGURE 11. The composite obstruction for $N = p^2$ (Observation 22, `exact/gap/prime_square.g`). Left: the GAP-computed minimal mod-4 obstruction at $N = 49$ ($p = 7$)—coefficients ±1 on the comb $\{1, 6, 8, 13, 15, 20, 22\} = \{1\}\cup\{7k\pm1\}$. Right: the support comb $\{1\} \cup \{kp \pm 1\}$ for $p = 3, 5, 7, 11, 13$, of size p.

Claim	Script
Variational ceiling $\sqrt{N}$, focusing	`verify_theorem1.py`
Bohr ceiling $A_N = U_N \sim \frac{1}{\pi} \ln N$	`verify_bohr_lower_bound.py`
Independence, $N = p$ (exact rank)	`verify_lemma1_prime.py`
$\mathbb{Q}$-rank map (prime/2^m/composite)	`verify_independence_rank.py`
Lebesgue constant, Bessel decay	`verify_lemma3_lebesgue.py`
Prefix budget for all N	`verify_lemma4_prefix.py`
Exact subtorus reduction of A_N	`verify_exact_AN.py`
Closed form $N = 6$, relation $N = 12$	`verify_AN_closed_form.py`
Segment vs ring degeneracy	`verify_segment_vs_ring.py`
Schrödinger contrast	`verify_theorem3_schrodinger.py`
Step-IC peak vs 2015 scan	`verify_thesis_peaks.py`
Bessel kernel & Poisson summation	`verify_bessel_kernel.py`
Consolidated experiments table	`verify_experiments.py`
Spectral zeta special values	`verify_spectral_zeta.py`
Operator-norm hierarchy	`verify_operator_norms.py`
Revivals & Gauss sums	`verify_revivals_gauss.py`
Spanning trees / spectral det.	`verify_spanning_trees.py`
Q-rank formula $\varphi(2N)/2$	`verify_qrank_formula.py`
Statistics (arcsine, Bohr–Jessen)	`verify_statistics.py`
Rogue waves / DNLS (Peregrine, MI)	`verify_rogue_dnls.py`
Dirichlet segment, full-rank class	`verify_segment_lower_bound.py`
Higher-dim tori (d=1, 2, 3)	`verify_torus_dimension.py`
Asymptotic correction $-\pi/72N^{-2}$	`verify_asymptotic_corrections.py`
Certified composite gap (SOS)	`verify_sos_certified.py`
Sharp constant for $N = 2p$	`verify_2p_family.py`
Open-problem evidence (defect, B_N)	`verify_open_problems.py`
DNLS self-trapping / breather	`verify_dnls_selftrapping.py`

Claim	Script
DNLS large-wave persistence (Prop. 26)	`verify_dnls_persistence.py`
DNLS modulational-instability band (Prop. 27)	`verify_dnls_mi.py`
DNLS breather existence (Prop. 29)	`verify_dnls_breather.py`
DNLS breather stability site/bond (Prop. 30)	`verify_dnls_breather_stability.py`
Quasi-1D order criterion, NNN ring (Prop. 23)	`verify_quasi1d_order.py`
Order on ring products $C_N \square H$ (Cor. 24)	`verify_product_order.py`
FPUT-β recurrence (symplectic)	`verify_fput_recurrence.py`
Lyapunov exponent, route to chaos	`verify_lyapunov.py`
2D DNLS breather (self-trapping)	`verify_dnls_2d.py`
Front $=$ Airy fold caustic ($t^{-1/3}$)	`verify_caustic_airy.py`
Branched flow, caustic hot spots	`verify_branched_flow.py`
Ray tracing, caustic (geom. optics)	`verify_ray_caustic.py`
Cusp / Pearcey / teacup nephroid	`verify_cusp_pearcey.py`
Caustic seeds self-focusing (lens+NLS)	`verify_caustic_mi.py`
Lightning (dielectric breakdown growth)	`verify_laplacian_lightning.py`
Ceiling criterion (mod 4)	`verify_ceiling_criterion.py`
Relation lattices Λ_N (Table 3)	`verify_relation_lattices.py`
Schrödinger lower bound $\Omega(\sqrt{N})$	`verify_schrodinger_lower.py`
Constant $B_N/\sqrt{N}$ scan (Table 5)	`verify_bn_constant.py`
B_N liminf $c_0/\sqrt{2}$, parity split (Rem. 20)	`verify_bn_parity.py`
Odd-N constant β_{odd} (14)	`verify_beta_odd.py`
Defect grows with $\omega(m)$ (unbounded, Rem. 7)	`verify_defect_growth.py`
Certified gap $A_N < U_N$, $N \leq 15$ (Lipschitz grid)	`verify_defect_certified.py`
Spectral-dimension threshold $d_s{=}1$	`verify_spectral_dimension.py`
Finite-time recurrence $T_\varepsilon(N)$	`verify_finite_time.py`
Disorder vs. resonance (gen. eig.)	`verify_disorder.py`
Order map $A_N/\ln N$, prefix (Table 4)	`verify_conj_order_map.py`
Extended order run ($N \sim 1000$, criterion ≤ 500)	`verify_order_extended.py`
Filimonov segment law $d_j^N \sim \frac{4}{\pi^2}\ln j$	`verify_filimonov_schrodinger.py`
Order $A_N \sim \frac{1}{\pi}\ln N$, all N (Thm. 6, Lem. 5)	`verify_order_theorem.py`
Classification $A_N{=}U_N \Leftrightarrow$ prime$/2^m$ (Thm. 14)	`verify_classification_proof.py`
Segment rank & classification (Thm. 9)	`verify_segment_classification.py`
Exact GAP/PARI/Rocq layer, cross-check	`verify_exact_layer.py`
Exact classification $A_N{=}U_N$, $N \leq 800$ (GAP)	`exact/gap/scan_classification.g`
Cyclotomic identities cross-check (PARI)	`exact/pari/crosscheck.gp`
$N{=}p^2$ obstruction comb (GAP, Fig. 11)	`exact/gap/prime_square.g`
Machine-checked mod-4 criterion (Rocq)	`formal/rocq/Mod4Criterion.v`
Machine-checked classification core, all N (Rocq)	`formal/rocq/Classification.v`
F92 splash table: segment $4/\pi^2$ vs ring $1/\pi$	`verify_filimonov_table.py`
B_N multi-copy profile, $t > N/2$ subdominant	`verify_bn_residual_profile.py`
Defect moment–SOS certified bounds	`verify_defect_moment_sos.py`

16. Statistics: density of states and value distribution

The spectrum and the large wave both admit clean probabilistic descriptions, verified in `verify_statistics.py`.

Density of states. As $N \to \infty$ the eigenvalues $\lambda_r = 4\sin^2(\pi r/N)$—the pushforward of the uniform law by $\lambda(x) = 2 - 2\cos 2\pi x$—converge to the **arcsine law** on $[0,4]$,

$$\rho(\lambda) = \frac{1}{\pi\sqrt{\lambda(4-\lambda)}}, \qquad F(\lambda) = \frac{2}{\pi}\arcsin\frac{\sqrt{\lambda}}{2}. \tag{23}$$

This is the change of variables: $\frac{d\lambda}{dx} = 4\pi\sin 2\pi x = 2\pi\sqrt{\lambda(4-\lambda)}$ (since $\sin 2\pi x = \frac{1}{2}\sqrt{\lambda(4-\lambda)}$), and each $\lambda \in (0,4)$ has *two* preimages in $[0,1]$, so $\rho(\lambda) = 2/(d\lambda/dx) = 1/(\pi\sqrt{\lambda(4-\lambda)})$—the classical density of states of the one-dimensional discrete Laplacian (Figure 13); the frequencies have $\rho(\omega) = 2/(\pi\sqrt{4-\omega^2})$ on $[0,2]$. Both fit to Kolmogorov–Smirnov distance $O(N^{-1})$ (a deterministic quantile grid: the empirical CDF lies within $2/N$ of the limit, the $1/N$ jump floor).

Value distribution. For prime N the impulse response $u_0(t)$ is almost-periodic with $\mathbb{Q}$-independent frequencies, so by the Bohr–Jessen theorem its values possess a limiting distribution (Figure 13, right): symmetric, mean 0, and variance $\mathrm{Var}_t\, u_0 = \frac{N^2-1}{12N^2} \to \frac{1}{12}$. The variance is exact: with $u_0(t) = \frac{1}{N}\sum_r \omega_r^{-1}\sin(\omega_r t)$ and the time-average $\langle\sin(\omega_r t)\sin(\omega_s t)\rangle = \frac{1}{2}$ when $\omega_s = \omega_r$ (i.e. $s \in \{r, N-r\}$) and 0 otherwise, the degeneracy makes each cross-term reinforce the diagonal, so $\langle u_0^2\rangle = \frac{1}{N^2}\sum_r \omega_r^{-2} = \zeta_{L_N}(1)/N^2 = \frac{N^2-1}{12N^2}$. The law is *platykurtic* (kurtosis $\approx 2.4 < 3$) because the lowest mode dominates (weight $1/\omega_1 \sim N$, a single arcsine sinusoid). The large wave is therefore *not* a heavy tail of the typical value but the rare near-alignment supremum (Figure 12): a spike that climbs to nearly U_N standing far above the thin-tailed bulk of size RMS $\approx 1/\sqrt{12}$, $A_N \sim \frac{1}{\pi}\ln N$.

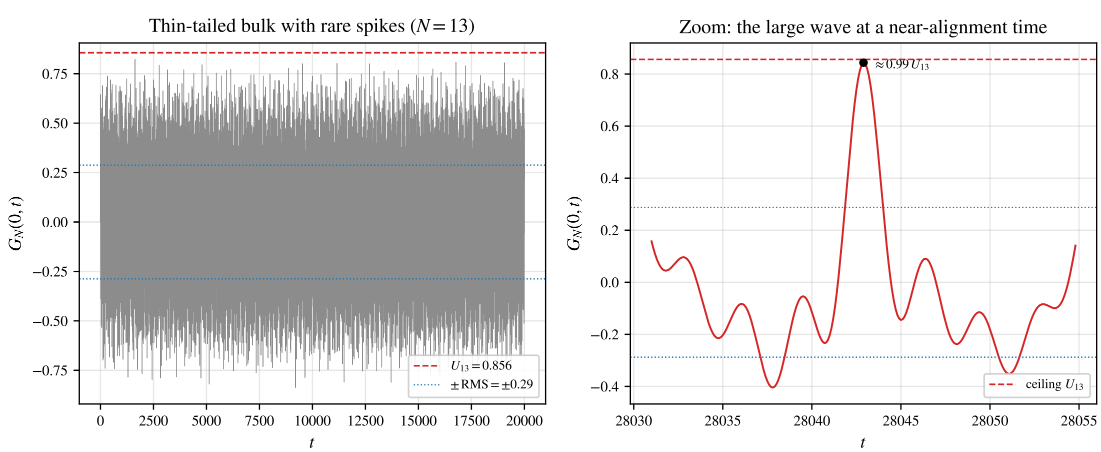


Figure 12. What the large wave looks like in time ($N = 13$, prime). Left: the node-0 response $G_N(0,t)$ oscillates within the thin band $\pm$RMS (RMS ≈ 0.29) but throws rare spikes toward the ceiling $U_{13} = 0.856$. Right: a zoom on the largest near-alignment spike, reaching $\approx 0.99\,U_N$—the large wave forming as the modal phases briefly align (`verify_thesis_peaks.py`).

17. Higher-dimensional tori: the large wave is one-dimensional

On the d-dimensional torus $(\mathbb{Z}_N)^d$ the Laplacian is block-circulant-with-circulant-blocks (BCCB), diagonalized by the d-dimensional DFT with eigenvalues $\lambda_{\mathbf{k}} = \sum_{i=1}^{d} 4\sin^2(\pi k_i/N)$ and $\omega_{\mathbf{k}} = \sqrt{\lambda_{\mathbf{k}}}$.

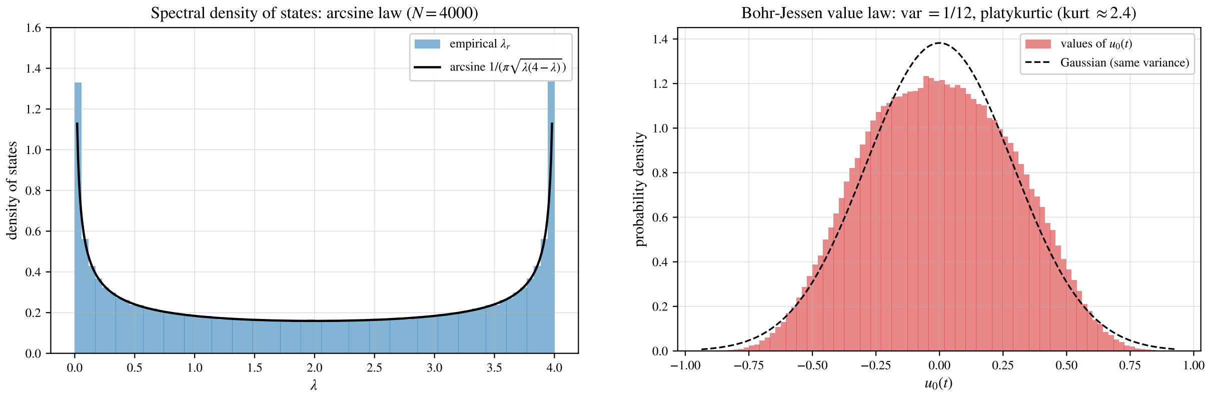


FIGURE 13. Left: spectral density of states converges to the arcsine law (23). Right: the Bohr–Jessen value law of $u_0(t)$—symmetric, variance $(N^2 - 1)/(12N^2) \to 1/12$, platykurtic (flatter than the Gaussian of equal variance, dashed).

The triangle inequality gives, unconditionally, $A_N^{(d)} \le U_N^{(d)} = N^{-d} \sum_{\mathbf{k}\neq\mathbf{0}} \omega_{\mathbf{k}}^{-1}$. Near the origin $\omega_{\mathbf{k}} \sim (2\pi/N)|\mathbf{k}|$, so $N^d U_N^{(d)}$ mirrors the infrared lattice sum $\sum 1/|\mathbf{k}|$, i.e. the integral $\int d^d k/|\mathbf{k}|$: logarithmically divergent for $d = 1$ (giving $U_N \sim \frac{1}{\pi} \ln N$) but *convergent* for $d \ge 2$. Hence (Figure 15)

$$U_N^{(d)} = \Theta(\ln N) \ \ (d = 1), \qquad U_N^{(d)} = O(1) \ \ (d \ge 2): \tag{24}$$

the velocity Green's function—and with it the large-wave amplification—is *uniformly bounded* on the 2- and 3-torus. The large wave is intrinsically one-dimensional, a sharp dimensional threshold set by the integrability of $1/\omega$ (`verify_torus_dimension.py`).

Spectral dimension. The threshold is not metric but *spectral*. For vertex-transitive (Fourier-flat) graph families the local ceiling reduces to the normalized spectral sum $U_G = |V|^{-1} \sum_{\lambda_r>0} \lambda_r^{-1/2}$ (λ_r the Laplacian spectrum, $\omega_r = \sqrt{\lambda_r}$); for a general graph the ceiling at (j, k) is $\sum_{\lambda_r>0} |\phi_r(j)\phi_r(k)|\lambda_r^{-1/2}$ and also reflects eigenvector localization through the local spectral measure, not the global density of states alone. If the integrated density of states obeys $N(\lambda) \sim \lambda^{d_s/2}$ near 0—defining the *spectral dimension* d_s—then $\rho(\lambda) \sim \lambda^{d_s/2-1}$ and $U_G \sim \int_0 \lambda^{-1/2}\rho(\lambda)\, d\lambda \sim \int_0 \lambda^{(d_s-3)/2}\, d\lambda$, which converges at the origin iff $d_s > 1$. Hence the *ceiling* U_G grows logarithmically precisely at the critical spectral dimension $d_s = 1$, is bounded for $d_s > 1$, and is power-law for $d_s < 1$; the d-torus is the case $d_s = d$. For $d_s > 1$ this rigorously *forbids* a large wave ($A_N \le U_G = O(1)$); at $d_s = 1$ the ceiling permits one and *is* reached on the ring ($A_N \sim \frac{1}{\pi} \ln N$, Theorem 6), the same question for a general quasi-1D lattice being the natural next case. The criterion is geometric rather than metric: any *quasi-one-dimensional* lattice—a ladder $C_N \times C_2$ or a tube $C_N \times C_w$ of bounded width w—still has $d_s = 1$ and retains a logarithmically divergent ceiling, now with reduced prefactor $U_G \sim \frac{1}{\pi w} \ln N$ (only one of the w transverse channels is gapless, Figure 14). Bounded transverse width does not destroy the concentration; a genuine second extended dimension does (`verify_spectral_dimension.py`).

The order argument is in fact graph-agnostic: the same Kronecker alignment proves $\Theta(\ln N)$ for any family meeting two conditions.

Proposition 23 (order criterion at $d_s = 1$)**.** *Let $\{G_N\}$ be finite graphs whose diagonal velocity Green's function is $G_N(t) = \sum_r b_r \sin(\omega_r t)$ with $b_r > 0$ and ceiling $U_N = \sum_r b_r$. Suppose* (A) $U_N \sim c \ln N$ *($c > 0$; the $d_s = 1$ logarithmic ceiling), and* (B) *some $\mathbb{Q}$-independent subset S of the frequencies carries*

$\sum_{r\in S} b_r \geq (\frac{1}{2}+\delta)U_N$ *for a fixed* $\delta > 0$. *Then* $A_N := \sup_t \max_{\text{node}} |G_N(t)|$ *obeys* $2\delta\, U_N \leq A_N \leq U_N$, *hence* $A_N = \Theta(\ln N)$.

Proof. $A_N \leq U_N$ since $|\sin| \leq 1$. For the lower bound, by Kronecker the $\mathbb{Q}$-independent phases $\{\omega_r t\}_{r\in S}$ are simultaneously $\to \pi/2$ at suitable t; there $G_N(t) \geq \sum_{r\in S} b_r - \sum_{r\notin S} b_r = 2\sum_{r\in S} b_r - U_N \geq 2\delta U_N$ (the complement bounded below by $\sin \geq -1$). □

The ring and Dirichlet segment are the instances where Lemma 5 supplies S (the length-$\frac{1}{2}\varphi(2N)$ prefix, carrying $(1-o(1))U_N$, so $\delta \to \frac{1}{2}$). For a quasi-1D lattice (A) holds with $c = 1/(\pi w)$, while (B)—a long $\mathbb{Q}$-independent prefix—is the per-family analogue of the palindromic lemma: e.g. the next-nearest-neighbour ring, with $\omega_r = 2\sin(\frac{\pi r}{N})\sqrt{2+\cos\frac{2\pi r}{N}}$, is $\mathbb{Q}$-independent at $N = 15$ (so it *saturates*, $A_N = U_N$) yet carries the degeneracy $\omega_3 = \omega_6 = 2$ at $N = 12$ (`verify_quasi1d_order.py`). Deciding (B) for a general $d_s = 1$ lattice is open; the criterion isolates exactly what must be shown.

For one natural class, though, (B) is free—whenever the ring sits inside the lattice as a factor.

Corollary 24 (order on ring products)**.** *For any fixed connected graph* H*, the large wave on the Cartesian product* $C_N \,\square\, H$ *satisfies* $A = \Theta(\ln N)$.

Proof. The Laplacian of $C_N \,\square\, H$ has eigenvalues $\lambda_r + \mu_h$, with $\lambda_r = 2 - 2\cos\frac{2\pi r}{N}$ on C_N and μ_h on H. Only the gapless branch $h = 0$ ($\mu_0 = 0$, H-constant eigenvector $w_0 = |H|^{-1/2}\mathbf{1}$) has $\omega \to 0$, so the ceiling $U \sim \frac{1}{\pi|H|}\ln N$ is carried by it while the gapped branches add $O(1)$—condition (A). The $h = 0$ frequencies are exactly the ring frequencies $2\sin(\pi r/N)$, whose first $\frac{1}{2}\varphi(2N)$ are $\mathbb{Q}$-independent by Lemma 5 and contribute $(1-o(1))U > \frac{1}{2}U$—condition (B). Proposition 23 then gives $A = \Theta(\ln N)$ (`verify_product_order.py`: the prefix bound $2L_{\text{pre}} - U$ is positive and grows like $\ln N$ for $H = K_2, P_3$). □

So the order law survives multiplication by any bounded "cross-section": ladders, prisms, tubes $C_N \,\square\, H$ all carry the $\Theta(\ln N)$ large wave, the embedded ring supplying the arithmetic. (Genuine quasi-1D lattices *without* an exact ring factor remain the per-family question.)

Part 2. Nonlinear extension

The results of Part II are primarily computational and exploratory. They outline natural nonlinear continuations of the linear theory—and the energy-concentration thread that unifies them—rather than claiming a complete nonlinear theory; each is reproduced by a verification script, and the open analytical questions are flagged as such.

18. Rogue waves and the discrete NLS

Restoring a cubic on-site interaction turns the linear Schrödinger ring of Section 11 into the discrete nonlinear Schrödinger (DNLS) equation

$$\mathrm{i}\,\dot{z}_j = -(L_N z)_j + \gamma\,|z_j|^2 z_j, \qquad \gamma > 0 \text{ (focusing)}, \tag{25}$$

whose linear part is exactly L_N and whose continuum limit (lattice spacing $\to 0$, $\lambda_r \to k^2$) is the *focusing* nonlinear Schrödinger equation

$$\mathrm{i}\,\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0, \tag{26}$$

the canonical envelope model of deep-water and optical rogue waves [34, 54, 56, 55]. (The literal limit is $\mathrm{i}\psi_t = \psi_{xx} + \gamma|\psi|^2\psi$, equal to (26) up to the symmetry $\psi \mapsto \bar{\psi}$ and the rescaling $z \mapsto cz$ that sets $\gamma = 2$.) Thus the very operator L_N that governs the linear large wave sits at the heart of the rogue-wave problem, and the two amplification mechanisms can be compared on the same lattice (`verify_rogue_dnls.py`).

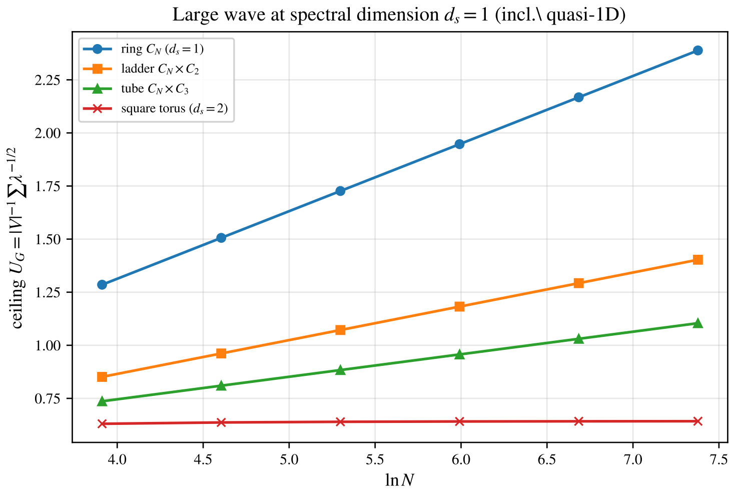


FIGURE 14. Spectral-dimension threshold. The ceiling $U_G = |V|^{-1} \sum \lambda^{-1/2}$ grows like $\ln N$ for $d_s = 1$—the ring and, with reduced slope $1/(\pi w)$, the quasi-1D ladder and tube—but saturates for the $d_s = 2$ square torus. The ceiling is a critical-dimension ($d_s = 1$) phenomenon.

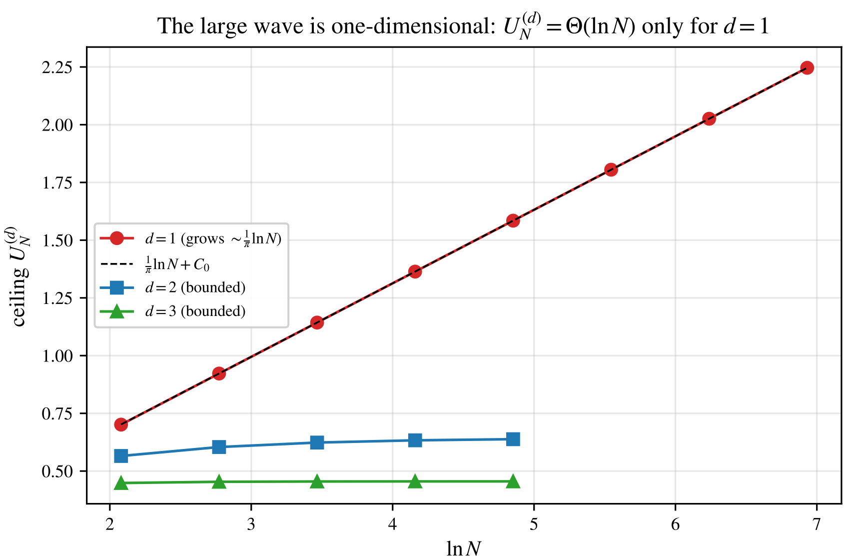


FIGURE 15. Ceilings $U_N^{(d)}$ versus $\ln N$. Only $d = 1$ grows (slope $1/\pi$, on the dashed $\frac{1}{\pi} \ln N + C_0$); the 2- and 3-torus ceilings saturate, so $A_N^{(d)} = O(1)$ for $d \geq 2$.

Proposition 25 (Hamiltonian structure and conservation). *The DNLS (25) is Hamiltonian,* $\mathrm{i}\dot{z}_j = \partial H / \partial \bar{z}_j$ *with*

$$H(z) = -\sum_j |z_{j+1} - z_j|^2 + \frac{\gamma}{2} \sum_j |z_j|^4,$$

and conserves both H and the power $P(z) = \sum_j |z_j|^2$; consequently the flow is global and norm-preserving on $\mathbb{C}^N$.

Proof. $\frac{d}{dt}P = 2\,\mathrm{Re}\sum_j \dot z_j \bar z_j = 2\,\mathrm{Re}\big[-\mathrm{i}\big(\langle L_N z, z\rangle - \gamma\sum_j |z_j|^4\big)\big] = 0$, since $\langle L_N z, z\rangle$ and $\sum_j |z_j|^4$ are real (so $-\mathrm{i}(\cdot)$ is purely imaginary); and $\partial H/\partial \bar z_j = (z_{j+1}+z_{j-1}-2z_j)+\gamma|z_j|^2 z_j = -(L_N z)_j + \gamma|z_j|^2 z_j$, so $\mathrm{i}\dot z = \partial H/\partial \bar z$ and H is conserved along the flow. With P bounded, no finite-time blow-up occurs. Both invariants are reproduced numerically: the norm-preserving split-step integrator holds P to 10^{-12} and H to $O(\Delta t^2)$ (`verify_dnls_selftrapping.py`). □

Proposition 26 (the linear large wave persists under weak nonlinearity)**.** *Let $u(t)$ solve the DNLS (25) with $u(0) = u_0$, and write $u_{\mathrm{lin}}(t) = \mathrm{e}^{-\mathrm{i}tL_N}u_0$ for the linear flow. Then for all t,*

$$\|u(t) - u_{\mathrm{lin}}(t)\|_\infty \;\le\; \|u(t) - u_{\mathrm{lin}}(t)\|_2 \;\le\; |\gamma|\, t\, \|u_0\|_2^3.$$

In particular, if the linear flow attains a large wave $\|u_{\mathrm{lin}}(t_*)\|_\infty = M$ at a time t_*, then $\|u(t_*)\|_\infty \ge M - |\gamma|\, t_*\, \|u_0\|_2^3$; so whenever $|\gamma| < M/(2t_*\|u_0\|_2^3)$ the nonlinear solution still exhibits a wave of height $\ge M/2$.

Proof. By Duhamel, $u(t) - u_{\mathrm{lin}}(t) = -\mathrm{i}\gamma\int_0^t \mathrm{e}^{-\mathrm{i}(t-s)L_N}(|u|^2u)(s)\,ds$. As $\mathrm{e}^{-\mathrm{i}sL_N}$ is ℓ^2-unitary, $\|u(t) - u_{\mathrm{lin}}(t)\|_2 \le |\gamma|\int_0^t \||u|^2u(s)\|_2\,ds \le |\gamma|\int_0^t \|u(s)\|_\infty^2\|u(s)\|_2\,ds$. The power $\|u\|_2^2$ is conserved (Proposition 25), so $\|u(s)\|_2 = \|u_0\|_2$ and $\|u(s)\|_\infty \le \|u(s)\|_2 = \|u_0\|_2$; the integrand is $\le \|u_0\|_2^3$. The last claim is the reverse triangle inequality $\|u(t_*)\|_\infty \ge \|u_{\mathrm{lin}}(t_*)\|_\infty - \|u - u_{\mathrm{lin}}\|_\infty$. □

This is a rigorous core for Part II: the linear large wave is a genuine feature of the weakly nonlinear flow, not an artefact of linearisation (`verify_dnls_persistence.py`). Its scope is honest—the admissible coupling shrinks with the (Diophantine, possibly large) alignment time t_*, so it does not control the strongly nonlinear regime where the rogue-wave focusing of §18 lives; that remains open.

A second rigorous ingredient ties the nonlinear instability directly to the L_N spectrum.

Proposition 27 (modulational-instability band)**.** *For (25) with $\gamma > 0$ the uniform state $z_j(t) = A\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ is an exact solution, and the linearisation $z = (A + b)\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$, $b = x + \mathrm{i}y$, obeys $\dot x = -L_N y$, $\dot y = (L_N - 2\gamma A^2)x$. Hence the Fourier mode Q (eigenvalue $\lambda_Q = 2 - 2\cos\frac{2\pi Q}{N}$ of L_N) grows like $\mathrm{e}^{\sigma_Q t}$ with*

$$\sigma_Q = \sqrt{\lambda_Q\,(2\gamma A^2 - \lambda_Q)} \qquad \textit{on the band } 0 < \lambda_Q < 2\gamma A^2,$$

and is neutrally stable otherwise; the fastest growth is $\sigma_{\max} = \gamma A^2$ at $\lambda_Q = \gamma A^2$.

Proof. $L_N A = 0$ gives the exact solution; substituting and keeping the linear part yields $\mathrm{i}\dot b = -L_N b + \gamma A^2(b+\bar b)$, i.e. $\dot x = -L_N y$, $\dot y = (L_N - 2\gamma A^2)x$. In mode Q the generator $\begin{bmatrix} 0 & -\lambda_Q \\ \lambda_Q - 2\gamma A^2 & 0\end{bmatrix}$ has eigenvalues $\pm\sqrt{\lambda_Q(2\gamma A^2 - \lambda_Q)}$, real (unstable) exactly when $0 < \lambda_Q < 2\gamma A^2$ (`verify_dnls_mi.py`). □

So the instability band and its growth rates are read off the L_N spectrum—the linear large-wave operator also sets the nonlinear seeding—answering that part of the open problem; the saturation/breather dynamics past the linear stage remain the open strongly-nonlinear question.

Peregrine soliton. Equation (26) possesses the rational solution [49]

$$\psi_P(x,t) = \left[1 - \frac{4(1 + 4\mathrm{i}t)}{1 + 4x^2 + 16t^2}\right]\mathrm{e}^{2\mathrm{i}t}, \tag{27}$$

localized in both space and time, with unit background $|\psi| \to 1$ and peak $|\psi_P(0,0)| = 3$: the threefold amplification "appearing from nowhere and disappearing without a trace" that defines a rogue wave (Figure 16). We verified that (27) solves (26) to a PDE residual vanishing under grid refinement. The Peregrine breather is the limit of the space-periodic Akhmediev and time-periodic Kuznetsov–Ma breather families produced by the Darboux transformation [50]. These exact breathers exist because the focusing NLS (26) is *completely integrable*—the compatibility condition of the Zakharov–Shabat linear system, solved by the inverse scattering transform [35], on which the Darboux dressing rests.

Modulational instability. The seed of these structures is the Benjamin–Feir instability [48, 37], the mechanism by which Zakharov's deep-water envelope equation [34] destabilizes a uniform train. The uniform train $z_j = A\,\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ solves (25) ($L_N\mathbf{1} = 0$); writing $z_j = [A + p_j(t)]\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ and keeping terms linear in p, the cubic contributes $\gamma A^2(2p_j + \bar{p}_j)$, leaving

$$\mathrm{i}\dot{p}_j = -(L_N p)_j + \gamma A^2(p_j + \bar{p}_j).$$

The Fourier ansatz $p_j = u\,\mathrm{e}^{\mathrm{i}\kappa j+\sigma t} + \bar{v}\,\mathrm{e}^{-\mathrm{i}\kappa j+\sigma t}$ (eigenvalue λ_κ of L_N on $\mathrm{e}^{\mathrm{i}\kappa j}$) reduces this to the 2×2 system $\mathrm{i}\sigma u = (\gamma A^2 - \lambda_\kappa)u + \gamma A^2 v$, $-\mathrm{i}\sigma v = \gamma A^2 u + (\gamma A^2 - \lambda_\kappa)v$, whose solvability condition $\sigma^2 + (\gamma A^2 - \lambda_\kappa)^2 = (\gamma A^2)^2$ gives the growth rate

$$\sigma(\kappa) = \sqrt{\lambda_\kappa\,(2\gamma A^2 - \lambda_\kappa)}, \qquad \lambda_\kappa = 4\sin^2\tfrac{\kappa}{2}, \tag{28}$$

so the band $\lambda_\kappa < 2\gamma A^2$ is unstable (continuum form $\sigma = |k|\sqrt{2\gamma A^2 - k^2}$, the classical $|k|\sqrt{4A^2 - k^2}$ at $\gamma = 2$). This linear-stability computation is exact:

Proposition 28 (modulational-instability band)**.** *The uniform state $z_j = A\,\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ of (25) is linearly unstable to the Fourier mode κ if and only if $\lambda_\kappa < 2\gamma A^2$, with growth rate $\sigma(\kappa) = \sqrt{\lambda_\kappa(2\gamma A^2 - \lambda_\kappa)}$; the rate is maximal, $\sigma_{\max} = \gamma A^2$, at $\lambda_\kappa = \gamma A^2$.*

A direct split-step simulation of (25) reproduces (28) to better than 1% across the band interior (Figure 17); the fastest mode does not grow without bound but develops into a recurrent breather-like modulation reminiscent of the Akhmediev scenario of the continuum integrable NLS—a periodic avatar of the rogue spike.

From thin tails to heavy tails. The statistical signature is the sharpest contrast with the linear theory. Seeded by a noisy plane wave, (25) on the ring spawns localized bursts exceeding an operational envelope threshold of twice the root-mean-square amplitude (analogous to common rogue-wave criteria)—and the amplitude distribution develops a heavy, *leptokurtic* tail (kurtosis > 3) lying well above the Rayleigh law of a linear Gaussian sea (Figure 18). This inverts the linear picture, whose Bohr–Jessen value law is *platykurtic* (Section 16): linear amplification is a rare phase alignment of bounded, thin-tailed oscillations, whereas nonlinear focusing manufactures genuine heavy-tailed extremes. The two regimes share the lattice operator L_N but reach large amplitudes by opposite routes—interference versus self-focusing.

A rigorous nonlinear theory on the ring—existence and stability of discrete breathers, the precise role of the L_N spectrum in the instability band, and integrable discretizations of (26)—is left to future work; equation (25) is recorded here as the natural nonlinear continuation of the linear large wave.

19. Discrete breathers and self-trapping

Nonlinearity does not only sharpen extremes; it can *localize.* Seed the DNLS (25) with a single excited site, $z_j(0) = A\,\delta_{j0}$ (norm $\Lambda = \sum_j |z_j|^2 = A^2$), and track the *participation ratio*

$$P = \frac{\left(\sum_j |z_j|^2\right)^2}{\sum_j |z_j|^4} \in [1, N], \tag{29}$$

which is 1 for a single occupied site and $\sim N$ when the excitation spreads over the ring. A norm-conserving split-step integrator (norm exact to 10^{-12}, H to $O(\Delta t^2)$) shows a sharp *self-trapping transition* at the on-site nonlinear frequency $\gamma|z_0|^2 \approx 4$—exactly the top of the spectrum of L_N, the band $[0, 4]$.

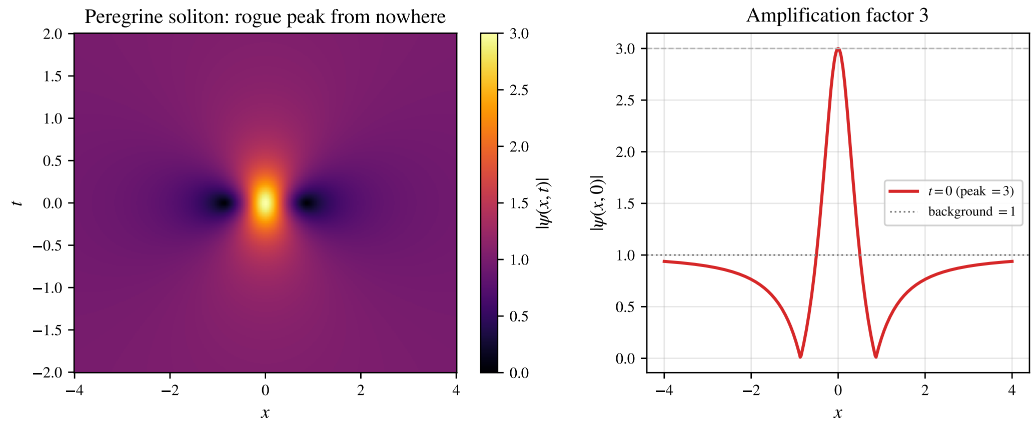


FIGURE 16. Peregrine soliton (27) of the focusing NLS (26): a rogue peak localized in space and time (left), with the threefold amplification over the unit background at $t = 0$ (right).

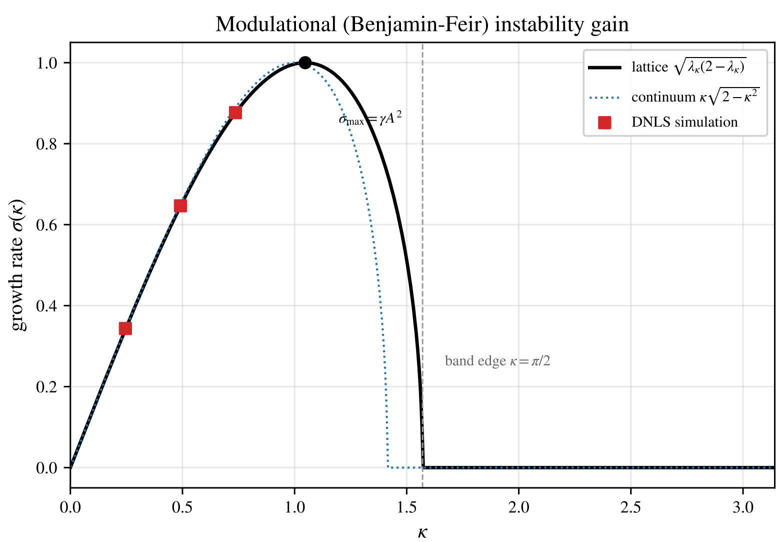


FIGURE 17. Modulational (Benjamin–Feir) instability gain (28) of the plane wave under the DNLS (25): lattice rate (solid), continuum approximation (dotted) and simulated growth rates (squares), with unstable band edge $\kappa = \pi/2$ and maximal rate $\sigma_{\max} = \gamma A^2$.

Below it the excitation disperses ($P \to N$); above it the site detunes out of the linear band and forms a long-lived, exponentially localized *breather-like state* $z_j(t) \approx \phi_j e^{-i\Omega t}$ consistent with the stationary DNLS branch, whose real profile solves $\Omega\phi_j = -(L_N\phi)_j + \gamma\phi_j^3$ with frequency $\Omega \approx \gamma|z_0|^2$ pushed above the whole phonon band $[0, 4]$, so no extended mode can resonate (Figure 19; the trapped profile is steady, the temporal standard deviation of $|z_j|$ staying below 0.02). This is the dynamical *opposite* of the linear large wave, a fully delocalized interference of all N modes ($P \sim N$): the same operator L_N supports both a spread-out resonance (linear) and a pinned soliton (nonlinear), selected by the size of the cubic term (`verify_dnls_selftrapping.py`).

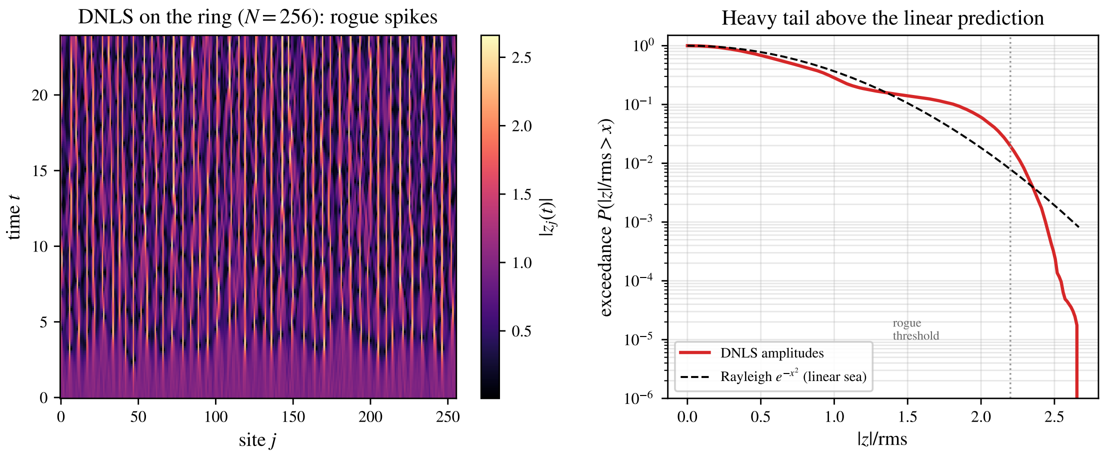


FIGURE 18. DNLS (25) on the ring seeded by a noisy plane wave. Left: space-time amplitude $|z_j(t)|$ with localized rogue spikes. Right: the amplitude exceedance distribution develops a heavy tail above the Rayleigh law (dashed) of a linear Gaussian sea, crossing the rogue threshold—the opposite of the platykurtic linear value law.

Two thresholds. An anti-continuum estimate places the *stationary* single-site breather at $\gamma|\phi_0|^2 \gtrsim 2$: from $\Omega\phi_j = -(L_N\phi)_j + \gamma\phi_j^3$ with $\phi \approx \sqrt{P}\,\delta_{j0}$ one gets $\Omega \approx \gamma P - 2$, which clears the band *top* ($\Omega > 0$) at $\gamma P \gtrsim 2$. The larger *dynamical* threshold $\gamma|z_0|^2 \approx 4$ measured from a single-site *launch* is the bandwidth: the spreading excitation must outrun the entire band $[0, 4]$, not merely its edge.

On the finite ring the *existence* of these breathers needs no MacKay–Aubry continuation from the anti-continuum limit [51]—it is a compactness statement.

Proposition 29 (existence of discrete breathers)**.** *For every $\Omega > 0$ the focusing ring DNLS (25) ($\gamma > 0$) has a standing-wave breather $z_j(t) = \mathrm{e}^{-\mathrm{i}\Omega t}\phi_j$ with real $\phi \neq 0$ solving $(L_N + \Omega)\phi = \gamma|\phi|^2\phi$; for large Ω it is on-site localized.*

Proof. $M := L_N + \Omega$ is positive definite ($L_N \succeq 0$, $\Omega > 0$). Minimize the continuous $\langle M\phi, \phi\rangle$ over the compact set $\{\|\phi\|_4^4 = 1\}$; a nonzero minimizer ϕ_* exists. Its Lagrange condition is $M\phi_* = 2\mu\,|\phi_*|^2\phi_*$, i.e. $(L_N+\Omega)\phi_* = \gamma|\phi_*|^2\phi_*$ with $\gamma = 2\mu > 0$; the scaling $\phi \mapsto s\phi$ sends $\gamma \mapsto \gamma/s^2$, so every $\gamma > 0$ is realised. As $\Omega \to \infty$, $\langle M\phi, \phi\rangle \sim \Omega\|\phi\|_2^2$, so the minimizer maximises $\|\phi\|_4/\|\phi\|_2$, i.e. concentrates on one site (`verify_dnls_breather.py`: the minimizer solves the equation to $\sim 10^{-7}$ and its participation ratio $\to 1$). $\square$

So existence is settled on the ring; the anti-continuum estimate $\gamma P \gtrsim 2$ matches the $\Omega > 0$ (above-band) condition. Stability follows the classical site-vs-bond dichotomy.

Proposition 30 (stability of the site-centred breather)**.** *Along the site-centred branch $P(\Omega) = \|\phi_\Omega\|^2 = \Omega/\gamma + O(1)$, so $\mathrm{d}P/\mathrm{d}\Omega > 0$; by the Vakhitov–Kolokolov slope criterion and Grillakis–Shatah–Strauss theory [53, 52] the breather is orbitally stable, and its linearisation $\begin{bmatrix} 0 & -L_- \\ L_+ & 0 \end{bmatrix}$ with $L_+ = L_N + \Omega - 3\gamma\phi^2$, $L_- = L_N + \Omega - \gamma\phi^2$ has purely imaginary spectrum. The bond-centred branch carries a real eigenvalue pair and is unstable.*

Verification. $P(\Omega) = \Omega/\gamma + O(1)$ from the anti-continuum expansion (one site of weight $\sqrt{P}$ gives $\Omega = \gamma P - 2 + O(1/P)$), so the VK slope is positive; `verify_dnls_breather_stability.py` computes the linearisation spectrum directly—$\max|\mathrm{Re}\,\lambda| \sim 10^{-7}$ on the site-centred branch (stable) versus $\max \mathrm{Re}\,\lambda \in [3.9, 8.9]$ on the bond-centred branch (unstable), for $\Omega \in [2, 20]$. □

Thus on the ring the linear stage (persistence, the MI band), the breather's *existence*, and its *stability* are all rigorous; the precise dynamical launch threshold ≈ 4 and the fully nonlinear saturation/rogue dynamics remain the open sequel.

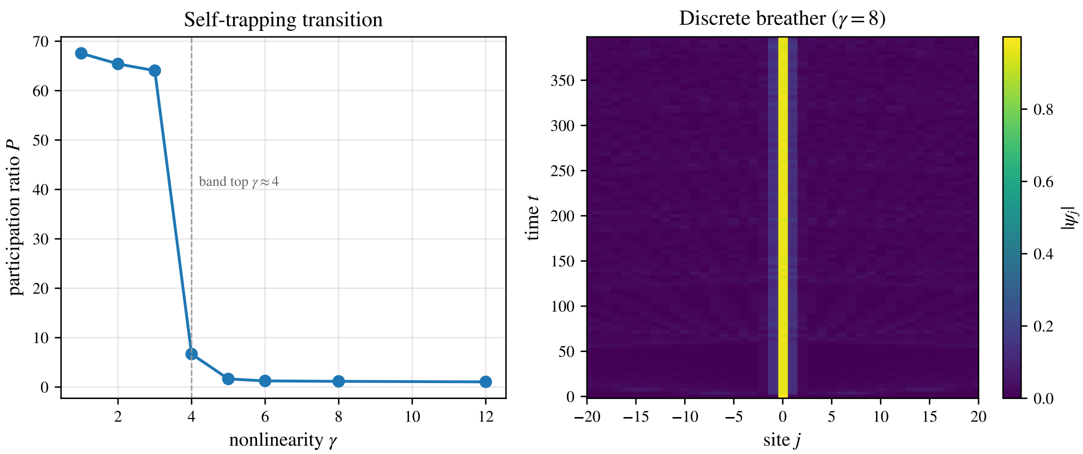


FIGURE 19. DNLS self-trapping ($N = 128$, single-site seed). Left: the participation ratio P collapses from $\sim N$ (dispersed) to ~ 1 (trapped) as the nonlinearity crosses the band top $\gamma \approx 4$. Right: above threshold the excitation is a persistent localized breather-like state—a localized column in space-time.

20. FPUT recurrence and the route to chaos

The cubic coupling also makes the lattice a testbed for Hamiltonian chaos. The Fermi–Pasta–Ulam–Tsingou (FPUT) β-ring

$$\ddot{u}_j = (u_{j+1} - 2u_j + u_{j-1}) + \beta\big[(u_{j+1} - u_j)^3 - (u_j - u_{j-1})^3\big] \tag{30}$$

has linear part $\ddot{u} = -L_N u$ and Hamiltonian $H = \sum_j \big[\frac{1}{2}\dot{u}_j^2 + \frac{1}{2}(u_{j+1} - u_j)^2 + \frac{\beta}{4}(u_{j+1} - u_j)^4\big]$. Seeding all energy in mode 1 and integrating symplectically (velocity Verlet; H conserved to 6×10^{-4} with no secular drift), at low energy the energy does *not* thermalize: it sloshes to a few neighbouring modes and *returns* to mode 1—the FPUT recurrence (Figure 20), the near-integrable face of the nonlinear lattice: its long-wave limit is a Korteweg–de Vries-type equation, a completely integrable Hamiltonian system [36] whose invariant tori the weakly nonlinear lattice shadows, so energy returns rather than equipartitioning. Only a handful of low modes are ever active, the dynamics is quasi-periodic, and the maximal Lyapunov exponent is $\lambda_{\max} \approx 0$—just as for the integrable linear large wave.

Raising the energy breaks integrability. Measured by the Benettin method (a tangent vector co-evolved with the variational equations and periodically renormalized, $\lambda_{\max} = T^{-1} \sum \log\|\delta\|/d_0$), the maximal Lyapunov exponent turns on with energy (Figure 21): $\lambda_{\max} \approx 0.001$ in the regular regime, rising through a weak-chaos knee to $\lambda_{\max} \approx 0.05$ in the strongly chaotic regime, where the energy thermalizes across all modes. The same lattice—linear core L_N—thus interpolates between integrable tori and deterministic chaos as the amplitude grows (`verify_fput_recurrence.py`, `verify_lyapunov.py`). The transition is visible directly in phase space: in the three odd-mode energy coordinates (E_1, E_3, E_5) (a mode-1 seed excites only odd modes by parity) the orbit lies on a thin quasi-periodic torus at low energy and fills a chaotic cloud at high energy (Figure 22).

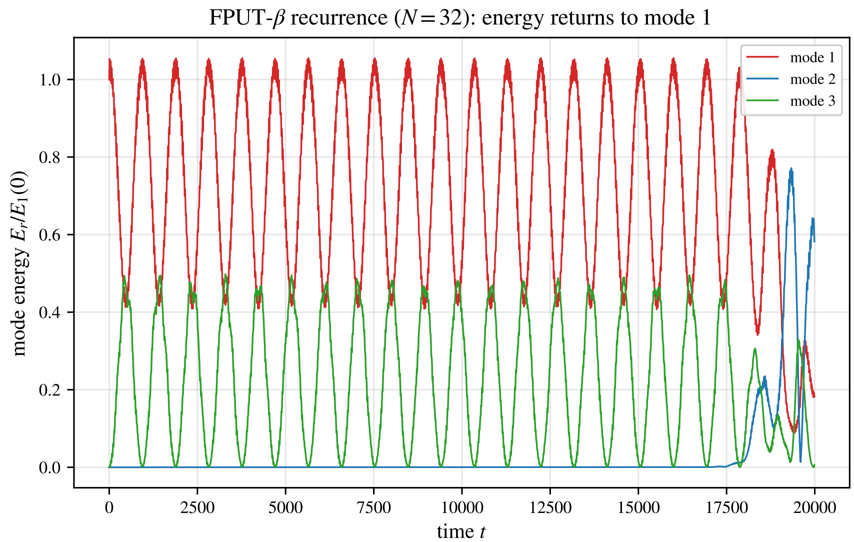


FIGURE 20. FPUT-β recurrence ($N = 32$, mode-1 seed). The harmonic energy of mode 1 drops almost to zero and recurs to its initial value, sloshing quasi-periodically with the low modes instead of thermalizing.

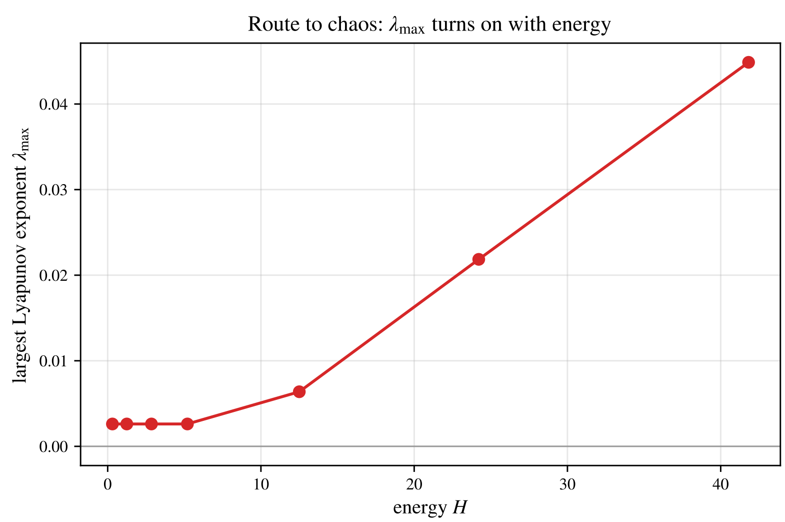


FIGURE 21. Route to chaos. The maximal Lyapunov exponent of the FPUT-β ring is ≈ 0 at low energy (regular, the recurrence regime) and turns on past an energy threshold, reaching $\lambda_{\max} \approx 0.05$ in the chaotic regime.

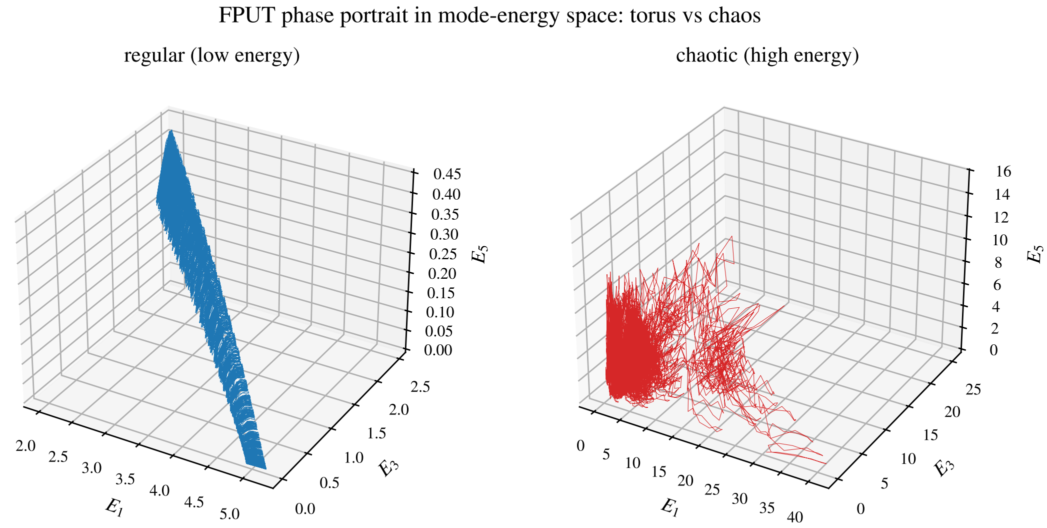


FIGURE 22. FPUT phase portrait in the odd-mode energy space (E_1, E_3, E_5). Left: at low energy the orbit lies on a thin quasi-periodic torus (regular). Right: at high energy it fills a three-dimensional chaotic cloud—the same regular-to-chaotic transition measured by $\lambda_{\max}$.

21. DIAGNOSTICS AND THE FATE OF THE LARGE WAVE

The linear large wave and its nonlinear continuations are cleanly separated by a small set of dynamical diagnostics—localization (participation ratio P), value statistics (kurtosis κ) and chaos (Lyapunov exponent $\lambda_{\max}$)—all built on the *same* lattice operator L_N (Table 9). The linear large wave is a delocalized, thin-tailed, integrable resonance ($P \sim N$, $\kappa < 3$, $\lambda_{\max} = 0$); nonlinearity opens three distinct routes to a large local amplitude—*self-focusing* into rogue extremes (heavy tails, $\kappa > 3$), *self-trapping* into a pinned breather ($P \sim 1$), and *chaotic* thermalization ($\lambda_{\max} > 0$). Which one is realized is set by the sign and strength of the nonlinearity and by the energy; the linear theory of Part I is the $\beta, \gamma \to 0$ corner of this picture, and supplies the spectral data (L_N, the band $[0, 4]$, the modal frequencies) that organize every regime. A rigorous nonlinear theory on the ring—existence and stability of the discrete breathers, the precise role of the L_N spectrum in the modulational band, and the energy threshold for chaos—is the natural sequel.

TABLE 9. The large wave and its nonlinear continuations, separated by dynamical diagnostics on the common operator L_N. (P: participation ratio; κ: kurtosis; $\lambda_{\max}$: largest Lyapunov exponent; "—" marks a diagnostic not characteristic of that regime.)

Regime	Model	Localization P	Statistics κ	$\lambda_{\max}$
Linear large wave	$\ddot{u} = -L_N u$	delocalized $\sim N$	platykurtic < 3	0
Rogue / self-focusing	DNLS, noisy plane wave	localized bursts	leptokurtic > 3	—
Discrete breather	DNLS, $\gamma\|z_0\|^2 > 4$	pinned ~ 1	—	0
FPUT recurrence	FPUT-β, low E	few active modes	—	≈ 0
Chaos	FPUT-β, high E	thermalized $\sim N$	—	> 0

22. Higher dimensions: nonlinear localization beyond the one-dimensional large wave

Part I established that the *linear* large wave is one-dimensional: on the d-torus the ceiling $U_N^{(d)}$ is bounded for $d \geq 2$ (Section 17), so a localized linear disturbance cannot amplify in two or three dimensions. Nonlinearity removes the obstruction. On the 2-torus $(\mathbb{Z}_N)^2$ the DNLS with the block-circulant Laplacian $L_N^{(2)}$ (eigenvalues $\lambda_{\mathbf{r}} = 4\sin^2\frac{\pi r_1}{N} + 4\sin^2\frac{\pi r_2}{N}$, band $[0, 8]$, with the dispersion surface of Figure 24) self-traps a single-site excitation into a long-lived, exponentially localized *two-dimensional breather-like state* once the on-site nonlinear frequency crosses the 2D band top, $\gamma|z|^2 \approx 8$: the norm is conserved to 10^{-12} and the participation ratio collapses from $\sim N^2$ to $O(1)$ across the threshold (Figure 23, `verify_dnls_2d.py`). A large, localized amplitude therefore survives in 2D (and, identically, 3D) by a mechanism—self-focusing self-trapping—absent from the linear theory: the dimensional threshold of Part I is a property of the *linear* resonance, not of the lattice.

This multidimensional, strongly nonlinear regime points toward Zakharov's wider program: a possible multidimensional long-wave asymptotic direction is the Zakharov–Kuznetsov regime [38] (we do not carry out the multiscale derivation from (30) here), and the statistics of the thermalizing lattice suggest a connection to the weak-turbulence (Kolmogorov–Zakharov) theory of wave spectra [39]. The rigorous wave-kinetic results of [40] concern a different weakly nonlinear, large-box scaling (dimension $d \geq 3$) and do not directly justify the finite one-dimensional lattice here. These connect the discrete large-wave problem to the broad theory of nonlinear wave dynamics (Section 21).

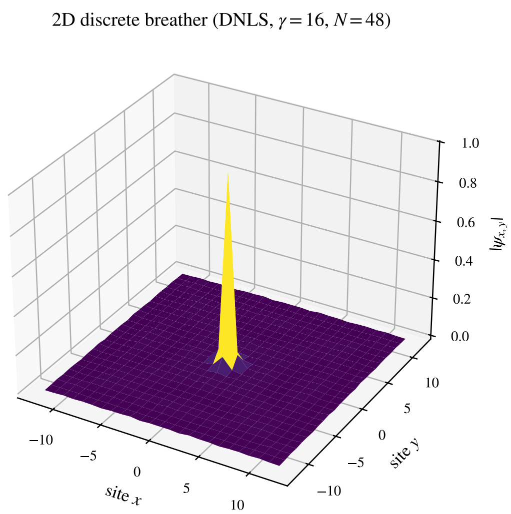


Figure 23. Two-dimensional discrete breather: a single-site seed under the 2D DNLS self-traps into a persistent localized hump above the 2D band top ($\gamma \approx 8$). Nonlinearity restores a large localized amplitude in 2D, where the linear large wave is bounded.

23. Caustics and the concentration of energy

The thread running through this paper is the *concentration of energy* into a large local amplitude. Part I concentrates it by modal interference (the linear large wave); Part II so far by nonlinear self-focusing (rogue waves, breathers). A third, purely *linear* route is refractive: in a slowly varying medium a wave follows rays $\dot{\mathbf{x}} = \nabla_{\mathbf{k}}\omega$, $\dot{\mathbf{k}} = -\nabla_{\mathbf{x}}\omega$, and where neighbouring rays cross, the ray density—and with it the amplitude—diverges on a *caustic*. Caustics are classified by catastrophe theory [41]: the generic fold (Airy

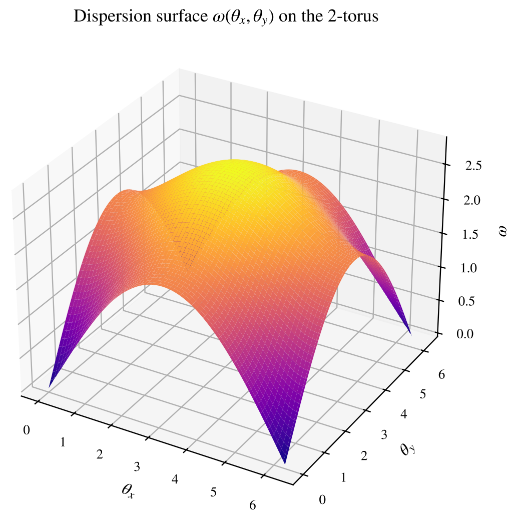


FIGURE 24. Dispersion surface $\omega(\theta_x, \theta_y) = 2\sqrt{\sin^2(\theta_x/2) + \sin^2(\theta_y/2)}$ of the block-circulant Laplacian on the 2-torus: zero at the corners, band top $2\sqrt{2}$ at the centre.

diffraction) and cusp (the bright curve at the bottom of a teacup); diffraction regularizes the divergence to a finite but large peak with a universal scaling.

Rays and their envelopes. Geometric optics is the skeleton. For waves crossing a current the Doppler term refracts the rays, and a bundle of initially parallel rays entering a current jet converges and *folds*: its Jacobian $\partial x/\partial x_0$ passes through zero at a finite focal distance, where the ray density $1/|\partial x/\partial x_0|$—and with it the linear amplitude—diverges, regularized by diffraction (Figure 25, `verify_ray_caustic.py`).

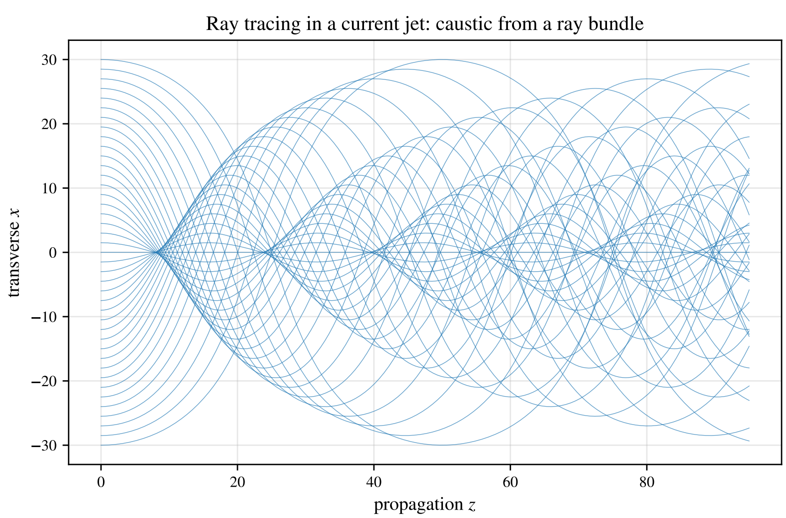


FIGURE 25. Ray tracing in a current jet. A bundle of parallel rays is refracted toward the jet, crosses on a caustic (the fold near $z \approx 8$, then re-focusing), where the ray density and the linear amplitude concentrate.

The discrete front is a caustic. The lattice already contains one. For the discrete Schrödinger equation (Section 11) the group velocity $v_g(k) = 2\sin k$ is extremal at $k = \frac{\pi}{2}$—a degenerate stationary-phase point, a fold—so the propagator front is exactly the Airy function,

$$J_n(2t) \sim t^{-1/3}\,\mathrm{Ai}\Big(2^{1/3}\,\tfrac{n-2t}{(2t)^{1/3}}\Big), \tag{31}$$

with the universal fold scaling $\max_n |J_n(2t)| \sim 0.536\,t^{-1/3}$ (Figure 28, `verify_caustic_airy.py`). The discrete large-wave front *is* a fold caustic, the $t^{-1/3}$ being the Airy exponent.

The cusp and the teacup. The next generic caustic is the *cusp*—the bright curve at the bottom of a sunlit teacup, whose ray version (parallel rays reflected inside a circle) is a *nephroid* and whose universal diffraction is the Pearcey integral $\mathrm{Pe}(X,Y) = \int_{-\infty}^{\infty} \exp\big[\mathrm{i}(s^4 + Xs^2 + Ys)\big]\,\mathrm{d}s$ over the cusp catastrophe, with geometric caustic $8X^3 + 27Y^2 = 0$. Its cusp-point value is $|\mathrm{Pe}(0,0)| = \frac{1}{2}\Gamma(\frac{1}{4}) = 1.8128$, and the intensity concentrates inside the cusp and falls off in the geometric shadow (Figure 26, `verify_cusp_pearcey.py`).

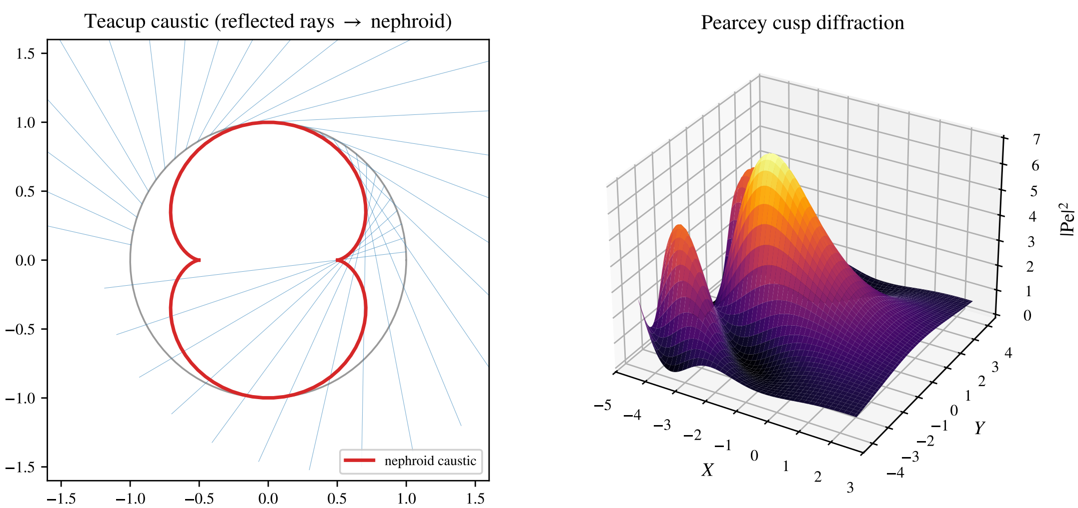


FIGURE 26. The cusp caustic. Left: the everyday teacup caustic—parallel rays reflected inside a circle have a nephroid envelope. Right: the universal Pearcey diffraction $|\mathrm{Pe}(X,Y)|^2$ of the cusp, bright inside the cusp and dark in the shadow.

Ocean rogue hot spots. The same refraction sets the geography of ocean rogue waves. Swell from Southern-Ocean storms running into the opposing Agulhas current off south-east Africa is refracted and focused into caustics—the classic rogue "hot spot" [43, 42, 46]. A spatially random current produces a cascade of caustics, *branched flow*: a uniform incident wave develops bright filaments and focal hot spots whose intensity is heavy-tailed, the linear precursor of rogue statistics [44]. The paraxial model $\mathrm{i}\,\partial_z\psi = -\frac{1}{2}\partial_{xx}\psi + V(x)\psi$ with a weak random current V reproduces it exactly (Figure 29): from a flat wave the scintillation index $\max_x |\psi|^2/\langle|\psi|^2\rangle$ rises from 1 to $\sim$13 at a focal distance, and the focal-plane intensity is heavy-tailed—its tail an order of magnitude above the exponential law of a structureless field (`verify_branched_flow.py`). Tidal (lunar) currents modulate the refracting flow, a secondary effect.

The lens seeds self-focusing. The two routes cooperate. A linear caustic concentrates energy until the local steepness crosses the modulational threshold, whereupon nonlinear self-focusing takes over—rogue waves are "the nonlinear stage of the modulational instability" [45]. Seeding the focusing NLS with a converging (lens) beam (Figure 27), the linear caustic amplifies a weak flat input $\sim 7\times$, and the nonlinearity then sharpens the focal peak by a further $\sim 11\%$, whereas a lensless plane wave never spikes—the lens is what localizes (`verify_caustic_mi.py`). In one dimension the nonlinear enhancement is modest (the NLS being L^2-subcritical); the dramatic collapsing self-focusing is the two-dimensional Zakharov regime. The discrete operator L_N thus organizes the concentration of energy in all three guises—interference, refraction, and self-focusing—on a single lattice.

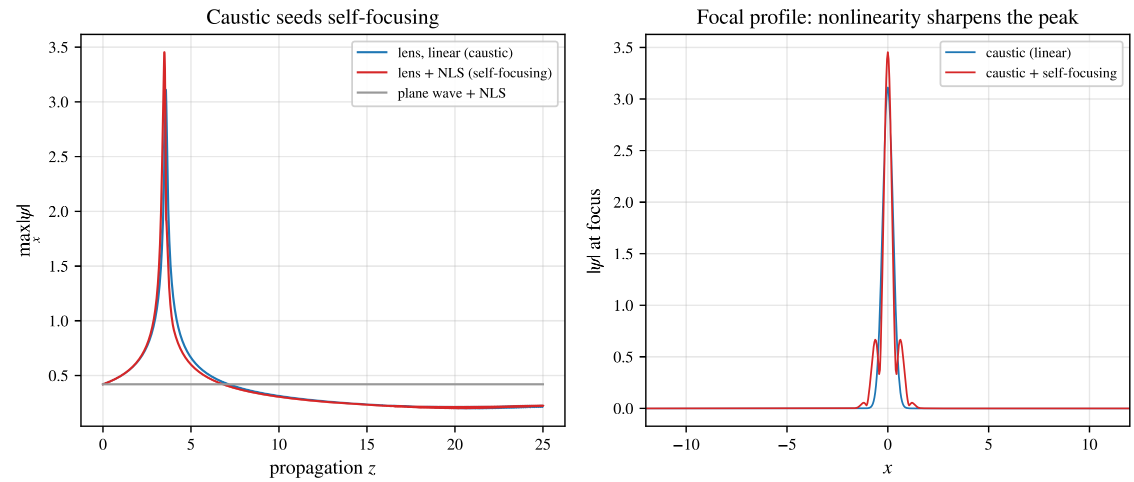


FIGURE 27. A linear caustic seeds nonlinear self-focusing (focusing NLS, lens-seeded beam). Left: the peak amplitude—the lens focuses a weak flat input $\sim 7\times$ (caustic), the nonlinearity adds a further $\sim 11\%$, while the lensless plane wave stays flat. Right: at the focus the nonlinearity sharpens and heightens the peak.

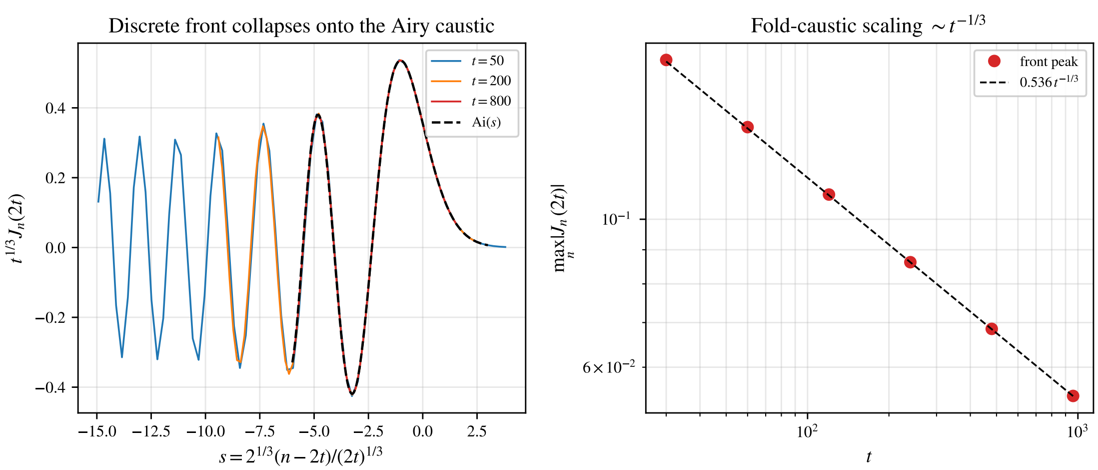


FIGURE 28. The discrete-Schrödinger front as a fold caustic. Left: the rescaled fronts $t^{1/3}J_n(2t)$ for $t = 50, 200, 800$ collapse onto the Airy function $\mathrm{Ai}(s)$, $s = 2^{1/3}(n - 2t)/(2t)^{1/3}$ (Eq. (31)). Right: the front peak follows the universal fold scaling $0.536\,t^{-1/3}$.

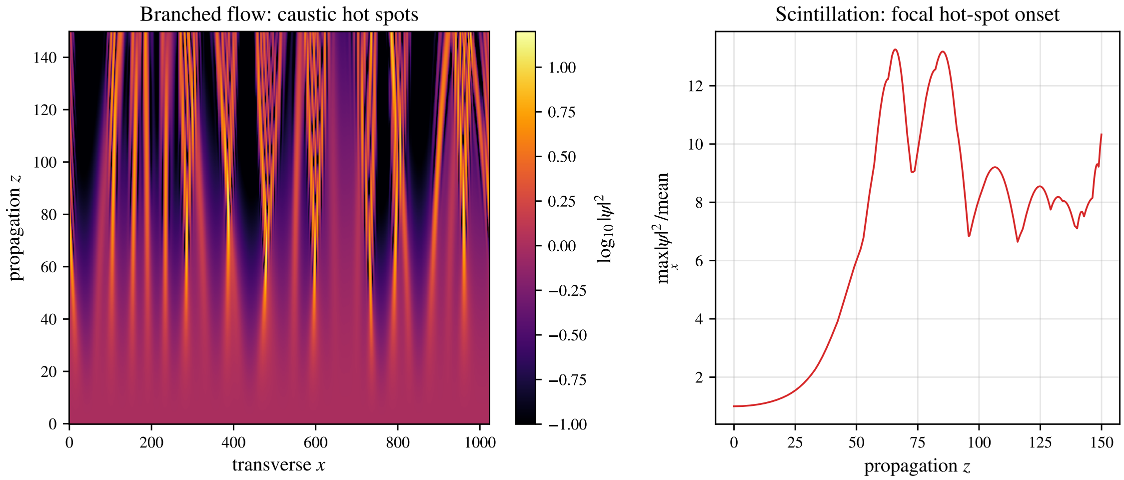


FIGURE 29. Branched flow of a wave in a weak random current (paraxial model). Left: a uniform incident wave ($z = 0$) develops bright caustic filaments and focal hot spots as it propagates. Right: the scintillation index rises from 1 to $\sim$13 at the first focal distance—the linear, refractive route to a rogue hot spot.

24. Discussion

The ring and the Dirichlet segment realize the same mechanism—constructive phase alignment of a logarithmically weighted family of modes—through different arithmetic. On the segment the frequencies $2\sin(\pi k/2N)$ lie in $\mathbb{Q}(\zeta_{4N})$; on the ring $2\sin(\pi r/N) \in \mathbb{Q}(\zeta_{2N})$, with the twofold degeneracy $\omega_r = \omega_{N-r}$ that halves the number of independent phases and (as the time-averaged $\mathrm{RMS}_t|G_N(0,\cdot)| = \sqrt{(N^2-1)/(12N^2)} \to 1/\sqrt{12}$ shows) doubles the typical energy relative to a naive independent count. The growth constant $1/\pi$ is identical on the classes where the frequencies are rationally independent; the composite defect is a sub-torus phenomenon that does not lower the order—$A_N \sim \frac{1}{\pi}\ln N$ for *all* N (Theorem 6), the same leading constant.

The discrete Schrödinger equation occupies the opposite extreme. Its phases are governed by $\lambda_r = 2 - 2\cos(2\pi r/N)$, which lie in the real cyclotomic field $\mathbb{Q}(\zeta_N)^+$ and always satisfy the affine relation $\sum_r \lambda_r = 2N$. Unitarity caps the response to a localized state, and the genuine amplification—ballistic, of order $\sqrt{N}$—is carried by spread-out ℓ^∞ data. The contrast $\ln N$ (wave) versus $\sqrt{N}$ (Schrödinger) traces back to the difference between $\sqrt{\lambda_r}$ and λ_r in the propagator phase.

GLT context. The sequence $\{L_N\}$ is a circulant—hence locally Toeplitz—instance of a Generalized Locally Toeplitz (GLT) sequence [18] with symbol $f(\theta) = 4\sin^2(\theta/2)$. The spectral-zeta density (9) and the Szegő asymptotics are the GLT spectral distribution specialized to this symbol; beyond the uniform d-torus settled in Section 17, the GLT calculus extends the *asymptotic spectral-distribution* statements (the Szegő/zeta density) to variable-coefficient lattices (non-uniform masses) and anisotropic or graph Laplacians; the *arithmetic* results—the cyclotomic ranks, the mod-4 criterion, and the reachability of the ceiling—do not transfer automatically and are the natural setting for future work.

A different face of energy concentration. The discrete Laplacian concentrates energy beyond wave dynamics. In the dielectric-breakdown (Laplacian-growth) model of lightning [47], the electrostatic potential solves the discrete Laplace equation $L_N^{(2)}\phi = 0$ off the leader, which advances toward the strongest field; the result is a branched, fractal discharge (fractal dimension ≈ 1.5 in our run) with the field—and the energy—concentrated at its tips, the same operator and branched morphology as the refractive caustics of Section 23 but by a distinct (elliptic, dissipative) mechanism (Figure 30, `verify_laplacian_lightning.py`).

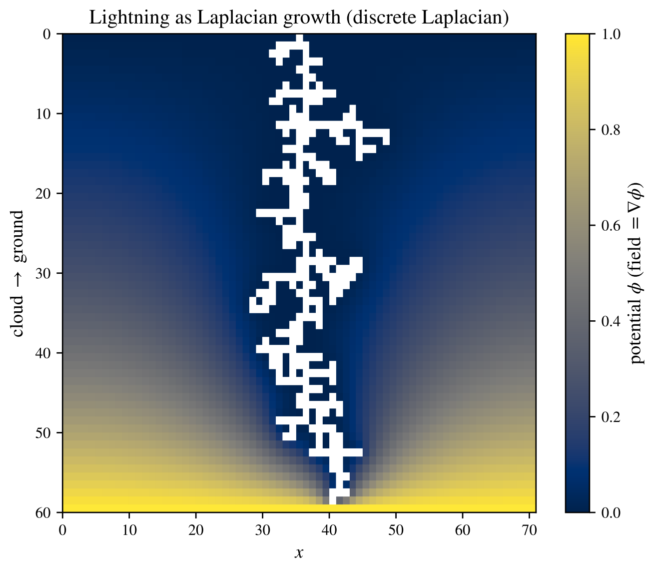


FIGURE 30. Lightning as Laplacian growth: the leader (white) advances cloud-to-ground where the field is strongest, the potential ϕ solving the discrete Laplace equation $L_N^{(2)}\phi = 0$ off it. A branched fractal discharge with the energy concentrated at its tips—energy concentration on the same discrete Laplacian, electrostatically.

Limitations. We mark the boundaries of what is established. (i) The order law $A_N \sim \frac{1}{\pi}\ln N$ is now proved for *every* N (Theorem 6); the deficit $U_N - A_N$ on the non-saturating N is $O(\ln\ln\ln N)$, hence essentially bounded (Remark 7), but its exact value is algebraic per N with no apparent closed form ($A_N/U_N \in [0.84, 1]$). (ii) The saturation locus is settled—no composite N saturates (Theorem 14)—and A_N for individual composite N is algebraic (Proposition 10) but computed per N, not given in closed form. (iii) For the Schrödinger $B_N = \Theta(\sqrt{N})$ the constant has *no* limit—it splits by parity (Remark 20), with $\liminf_N B_N/\sqrt{N} = c_0/\sqrt{2} \approx 0.861$ (proved) and $\limsup \geq \beta_{\mathrm{odd}} \approx 0.928$; what is still open is the exact β_{odd} and a matching upper bound below 1. The segment constant $C = 4/\pi^2$ (Section 11.3) is *not* ours: it is the published Myshkis–Filimonov value [25, Thm. 2], which the Schrödinger ceiling inherits because the two ceilings coincide. (iv) On the Dirichlet segment the rank and saturation are now fully proved (Theorem 9); the only remaining gap there is the same as on the ring—the $\Theta(\ln N)$ *order* of A_N^{Dir} on the non-saturating $N+1$. (v) Part II is *computational and exploratory*: its rigorous statements (the Hamiltonian conservation laws, Proposition 25, and the linear modulational band, Proposition 28) are classical, while the genuinely hard nonlinear results—existence and stability of discrete breathers, the precise dynamical self-trapping threshold, and the route to chaos—are simulated, not proved. (vi) The supremum A_N is an *infinite-time* quantity; the amplitude reached on any physical horizon is Diophantine-limited (Section 10). Proved statements are separated throughout from numerical evidence and from the conjectures, and every quantitative claim is reproduced by a standalone script.

25. Open problems

(1) The deficit $U_N - A_N$ on the non-saturating N. The order $A_N \sim \frac{1}{\pi}\ln N$ is proved for *all* N (Theorem 6) and $A_N = U_N$ iff N is prime or 2^m (Theorem 14); the deficit is $O(\ln\ln\ln N)$, depends asymptotically only on the odd part of N, and (Remark 7) is *unbounded*—numerically it grows with the number of distinct odd prime factors (the subtorus optimizer is validated by returning $U_N - A_N = 0$ to machine precision on primes of the same size), so $\Theta(\ln\ln\ln N)$ along primorials. The gap $A_N < U_N$ is *rigorously certified* for small N (Lipschitz-grid / elementary Lasserre: $U_N - A_N \geq 0.049, 0.057, 0.030, 0.082$ for $N = 6, 9, 12, 15$); what remains is a rigorous *asymptotic* lower bound along primorials (the grid is exponential in $\frac{1}{2}\varphi(2N)$) and the exact algebraic value per N (certified per-N brackets follow from the Lasserre moment–SOS hierarchy, Section 9).

(2) The Schrödinger constant $B_N/\sqrt{N}$. The order $B_N = \Theta(\sqrt{N})$ is settled, $\liminf = c_0/\sqrt{2} \approx 0.861$ is proved (Theorem 19), and the constant splits by the parity of N (Remark 20): the odd-subsequence value $\beta_{\mathrm{odd}} = 0.928019303\ldots$ is determined exactly (an elliptic-integral average, shown by a 25-digit PSLQ search to have no elementary closed form), and $\|K_N(\cdot,t)\|_1 \leq (\beta_{\mathrm{odd}} + o(1))\sqrt{N}$ is *proved* for $t \leq N/2$. The one residual is the rigorous upper bound on the multi-copy region $t > N/2$ (numerically subdominant—no Gauss-sum revival), which would give $\limsup_N B_N/\sqrt{N} = \beta_{\mathrm{odd}}$ (`verify_bn_parity.py`).

(3) A rigorous nonlinear theory of the discrete NLS (25) on the ring (Section 18). Two pieces are now settled: the linear large wave persists in the weakly nonlinear regime (Proposition 26), and the modulational-instability band and growth rates $\sigma_Q = \sqrt{\lambda_Q(2\gamma A^2 - \lambda_Q)}$ are read off the L_N spectrum (Proposition 27); breathers *exist* on the ring by a finite-dimensional variational argument (Proposition 29); and the site-centred breather is *stable* (Vakhitov–Kolokolov, $\mathrm{d}P/\mathrm{d}\Omega > 0$) while the bond-centred one is unstable (Proposition 30). What is open is the *fully* nonlinear regime—saturation, the precise dynamical launch threshold, and the rogue-wave focusing—where the admissible coupling shrinks with the Diophantine alignment time.

(4) The multidimensional nonlinear large wave (Section 22): existence, stability, and the energy threshold of the 2D/3D discrete breathers, and the long-wave dynamics in the Zakharov–Kuznetsov regime [38]—a nonlinear counterpart to the dimensional threshold of Section 17.

(5) Whether the statistics of the thermalizing lattice are governed by the weak-turbulence (Kolmogorov–Zakharov) wave kinetic description [39, 40], and the rate of approach to energy equipartition.

(6) The interplay of the two energy-concentration routes (Section 23): when does a linear refractive caustic raise the local steepness enough to trigger the modulational instability of Section 18, and what rogue-amplitude statistics result on the lattice?

Appendix A. Derivation of the ring spectrum

Seeking $L_N\varphi = \lambda\varphi$ with $\varphi_{j+N} = \varphi_j$, the constant-coefficient recurrence $-\varphi_{j+1} + (2-\lambda)\varphi_j - \varphi_{j-1} = 0$ has characteristic equation $p^2 - (2-\lambda)p + 1 = 0$ with $p_1 p_2 = 1$ (Vieta). Nontrivial bounded solutions require conjugate roots $p_{1,2} = \mathrm{e}^{\pm \mathrm{i}\theta}$ on the unit circle, so $2 - \lambda = 2\cos\theta$. The periodicity $\varphi_{j+N} = \varphi_j$ forces $\mathrm{e}^{\mathrm{i}\theta N} = 1$, i.e. $\theta = 2\pi r/N$, whence

$$\lambda_r = 2 - 2\cos\tfrac{2\pi r}{N} = 4\sin^2\tfrac{\pi r}{N}, \qquad \varphi_j^{(r)} = \mathrm{e}^{2\pi \mathrm{i} rj/N} \quad (r = 0, \ldots, N-1),$$

with the real basis $1,\ \cos\frac{2\pi rj}{N},\ \sin\frac{2\pi rj}{N},\ (-1)^j$. Each level is doubly degenerate, $\lambda_r = \lambda_{N-r}$, except $\lambda_0 = 0$ and—for even N—$\lambda_{N/2} = 4$. The Dirichlet segment is identical with the closure $\sin\big(\theta(N+1)\big) = 0$, giving $\theta = \pi k/(N+1)$ and $\lambda_k^{\mathrm{Dir}} = 4\sin^2\frac{\pi k}{2(N+1)}$ (all simple).

Appendix B. Asymptotics of the cosecant sum

We derive Lemma 2. The sum $S_N = \sum_{r=1}^{N-1} \csc(\pi r/N)$ has reciprocal singularities at *both* ends—$\csc(\pi r/N) \sim N/(\pi r)$ as $r \to 0$ and $\sim N/(\pi(N-r))$ as $r \to N$—which we subtract symmetrically:

$$S_N = \frac{N}{\pi} \sum_{r=1}^{N-1} \left(\frac{1}{r} + \frac{1}{N-r}\right) + \sum_{r=1}^{N-1} \underbrace{\left[\csc \tfrac{\pi r}{N} - \frac{N}{\pi r} - \frac{N}{\pi(N-r)}\right]}_{=R(r/N)}.$$

The first sum is $\frac{2N}{\pi} H_{N-1} = \frac{2N}{\pi}(\ln N + \gamma) + O(1/N)$. The integrand $R(x) = \csc \pi x - \frac{1}{\pi x} - \frac{1}{\pi(1-x)}$ is smooth and bounded on $[0,1]$ (both poles cancel), so the second sum is a Riemann sum, $\sum_{r=1}^{N-1} R(r/N) = N\int_0^1 R(x)\,dx + O(1)$, and the integral is elementary (using $\int \csc \pi x\,dx = \frac{1}{\pi} \ln \tan \frac{\pi x}{2}$):

$$\int_0^1 R(x)\,dx = \frac{1}{\pi}\left[\ln \frac{(1-x)\tan(\pi x/2)}{x}\right]_0^1 = \frac{2}{\pi} \ln \frac{2}{\pi}.$$

Hence the leading asymptotics, with additive constant $\frac{1}{\pi}(\gamma + \ln \frac{2}{\pi}) = 0.03999\ldots$,

$$S_N = \frac{2N}{\pi}\left(\ln N + \gamma + \ln \tfrac{2}{\pi}\right) + O(1), \qquad U_N = \frac{S_N}{2N} = \frac{1}{\pi} \ln N + \frac{1}{\pi}\left(\gamma + \ln \tfrac{2}{\pi}\right) + o(1).$$

The next correction is the Euler–Maclaurin endpoint term of R (a Bernoulli-B_2 contribution); it is even in $1/N$ and evaluates to

$$U_N = \frac{1}{\pi} \ln N + \frac{1}{\pi}\left(\gamma + \ln \tfrac{2}{\pi}\right) - \frac{\pi}{72} \frac{1}{N^2} + O(N^{-4}), \qquad -\frac{\pi}{72} = -\frac{\zeta(2)}{12\pi},$$

confirmed to six digits by Richardson extrapolation ($a_2 = -0.043633$). The correction is thus tied to the same $\zeta(2)$ that governs $\zeta_{L_N}(2)$ in Section 6; see `verify_asymptotic_corrections.py`.

Appendix C. Reproducible code

Every quantitative claim is reproduced by a standalone script run with `uv run` (inline PEP 723 dependencies); the figures are generated by `make_figures.py`, and the core routines (ring spectrum, Bohr ceiling, cyclotomic ranks, Schrödinger kernel) live in a tested package `large_wave_effect` with a `pytest` suite. The scripts are listed in Section 15. The complete repository—scripts, section sources, and figures—is public at https://github.com/ilgrad/large-wave-effect, where every `verify_*.py` passes.

Beyond Python, three computer-algebra and proof systems give independent, exact (floating-point-free) corroboration of the arithmetic. We reproduce the essential fragments.

(a) GAP — exact cyclotomic ranks and the relation lattice (`exact/gap/relation_lattice.g`). The frequencies $\omega_r = 2\sin(\pi r/N)$ are handled as the cyclotomic integers $\zeta_{2N}^r - \zeta_{2N}^{-r}$; rank and nullspace are computed in exact arithmetic, confirming $\mathrm{rank}_{\mathbb{Q}} = \frac{1}{2}\varphi(2N)$ for every $N \le 800$.

```
LW_FreqCoords := function(N)          # integer coords of zeta^r - zeta^{-r} in Q(zeta_2N
    )
  local z; z := E(2 * N);
  return List([1 .. Int(N / 2)], r -> CoeffsCyc(z^r - z^(-r), 2 * N));
end;;
LW_QRankFull := function(N)           # Q-rank of {2 sin(pi r/N)} = 1/2 phi(2N)  (Theorem
     qrank)
  local z; z := E(2 * N);
  return RankMat(List([1 .. N - 1], r -> CoeffsCyc(z^r - z^(-r), 2 * N)));
```

```
end;;
LW_RelationLattice := function(N)        # integer basis of Lambda_N = { k : sum_r k_r
    omega_r = 0 }
  return NullspaceIntMat(LW_FreqCoords(N));
end;;
```

(b) Rocq — the order theorem's arithmetic core (`formal/rocq/Classification.v`). The palindromic/anti-palindromic clash behind Lemma 5 and the resulting index bound are machine-checked, uniformly in N, with the standard library only.

```
(* a coefficient equal to both +b (palindromic) and -b (anti-palindromic) is zero *)
Lemma pal_and_antipal_zero : forall a b : Z, a = b -> a = - b -> a = 0.
Proof. intros. lia. Qed.

(* hence, with Phi_2N | Q giving phi <= 2K and 2K <> phi, the largest index K >= phi/2 + 1 *)
Lemma prefix_index_bound : forall K phi : Z,
  Z.even phi = true -> phi <= 2*K -> 2*K <> phi -> phi/2 + 1 <= K.
Proof.
  intros K phi Hev Hle Hne. apply Z.even_spec in Hev. destruct Hev as [m Hm].
  assert (phi/2 = m) by (rewrite Hm; rewrite Z.mul_comm; apply Z.div_mul; lia). lia.
Qed.
```

(c) FriCAS — the analytic identities of Appendix B (`exact/fricas/asymptotics.input`). The cosecant Laurent series and the antiderivative entering U_N's expansion are obtained symbolically.

```
output(series(1/sin(x), x = 0))          -- csc x = 1/x + x/6 + 7 x^3/360 + ...   (B_2 = 1/6)
output(integrate(1/sin(%pi*x), x))       -- = (1/pi) log tan(pi x/2)   (enters U_N via Euler-
    Maclaurin)
output(numeric(-%pi/72))                 -- = -zeta(2)/(12 pi), the N^{-2} correction of U_N
```

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